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Time-varying elite sand cat optimization algorithms for engineering design and feature selection

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ABSTRACT

The Sand Cat Optimization Algorithm (SCSO) is well known for its efficient search capability. However, SCSO has limitations in solving complex, high-dimensional problems, particularly in balancing exploration and exploitation and maintaining population diversity. It is prone to falling into local optimal solutions. This paper addresses these limitations by improving SCSO's search strategies, enhancing its ability to solve complex high-dimensional problems and avoiding local optima. Here, we propose a Time-Varying Elite Sand Cat Optimization Algorithm (TVESCSO). This algorithm achieves a multi-strategy search by dividing the sand cat population into a new and traditional population and further dynamically dividing the traditional population into an optimal and free population. The optimal population focuses on global exploration and local exploitation guided by improved adaptive factors and elite individuals. In contrast, the free population conducts extensive searches guided by the elite pool, expanding the search scope and balancing exploration and exploitation dynamically. The new population's diversity is further improved by merging the two sub-populations, information exchange and new population ordering. Finally, the elite individuals in the new population are mutated using the Logistic Chaos method, accelerating the algorithm's convergence and helping to jump out of local optimal solutions. Experimental results show that TVESCSO exhibits faster convergence and higher accuracy in the benchmark functions of the Congress on Evolutionary Computation (CEC2017 and CEC2022), outperforming a wide range of state-of-the-art and well-known optimization algorithms. The Friedman Test , the Nemenyi and Wilcoxon Rank-Sum tests further validates the effectiveness and robustness of TVESCSO. The application of TVESCSO to four real-world engineering problems and twelve feature selection datasets further confirms its excellent performance. TVESCSO significantly improves SCSO performance and demonstrates significant advantages in complex numerical optimization, engineering optimization, and high-dimensional feature selection problems. This improvement addresses the issue of SCSO being prone to local optima and provides new insights for SCSO algorithm research, advancing the field further.

1. Introduction

In real life, design problems in different engineering domains can be transformed into optimization problems. As the complexity of engineering increases, these optimization problems face more and more challenges, mainly in terms of large scale, high dimensionality, and multimodality (Cheng & Sholeh, 2023). Optimization methods (Rostami et al., 2021) mainly include deterministic and meta-heuristic algorithms. Deterministic methods are based on rigorous mathematical analysis theory, such as the Lagrange multiplier, conjugate gradient, and interior point linear programming methods. Although these methods are effective in dealing with small-scale, superficial problems, they face the challenges of high computational cost and a tendency to fall into local optima in large-scale, complex optimization problems (Abdel-Basset et al., 2023). In contrast, meta-heuristic algorithms search by simulating natural phenomena and biological behaviors, such as Grey Wolf Optimizer (Mirjalili et al., 2014), Sine Cosine Algorithm (Mirjalili, 2016), and Equilibrium optimizer (Faramarzi et al., 2020), these algorithms do not depend on initial conditions and apply to a broader range of problem types. Thus, they have become a hot topic of research and application in the past decades due to their advantages in dealing with complex problems (Deng & Liu, 2024; Fu et al., 2024).

Meta-heuristic algorithms (MA), such as Partial Reinforcement Optimizer (Taheri et al., 2024) and Newton Raphson Optimizer (Sowmya et al., 2024), rely on the evolutionary information of elite individuals to

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guide the in-depth exploration of improving the quality of the solution. These algorithms generate new solutions based on the current solution state and predefined evolutionary rules and continuously optimize them through an iterative process to approximate the global optimal solution (Mohammadi & Sheikholeslam, 2023). However, in the face of complex problems, MA often requires a large number of iterations to reach a satisfactory solution set, which challenges the efficiency of the algorithms (Jia et al., 2024). In MA, the balance between exploitation and exploration is critical to understanding their core (Zhang & Chen, 2025). Precisely finding the optimal balance between these two and dynamically adjusting this ratio according to the algorithm's progress is a central issue in optimizing the performance of metaheuristic algorithms (Ahmadianfar et al., 2021). Therefore, designing algorithms that adjust the exploration-to-exploitation ratio in response to the problem's complexity and the algorithm's evolutionary state is a hot topic in current research (Hussien, Gharehchopogh et al., 2024; Zhao et al., 2023).

On the one hand, existing MA mostly use the whole population (Civicioglu & Besdok, 2024) or a fixed population (Deng & Liu, 2023) as the evolutionary unit, relying on elite individuals to guide the evolutionary direction and improve the performance of the population. However, this reliance on elite individuals may lead to the population being too homogeneous and lacking diversity during the iteration process, thus limiting the algorithm's exploration capability. Especially when dealing with high-dimensional complex optimization problems, highly homogeneous populations need help performing effective global searches and are prone to falling into local optimal solutions, thus making it difficult to find the global optimal solution (Zhao et al., 2024). According to the 'no free lunch' theorem (Wolpert & Macready, 1997), no algorithm can perform best on all problems. Therefore, to solve specific optimization problems more efficiently, algorithms for these problems need to be developed to improve their performance and competitiveness.

In 2023, Seyyedabbasi et al. proposed Sand Cat Swarm Optimization (SCSO) (Seyyedabbasi & Kiani, 2023), which effectively balances exploration and exploitation by simulating the survival behavior of sand cats, thus avoiding falling into the local optimal solution. However, SCSO, as a single-population elite-guided intelligent algorithm, is prone to triggering a high degree of intra-population similarity when dealing with high-dimensional complex optimization tasks, which hinders the exploration of global optimal solutions. Time-varying Elite Sand Cat Swarm Optimization (TVESCSO) is introduced in this study to overcome this limitation. TVESCSO avoids population homogenization by dynamically adjusting the population diversity and adopting a multi-strategy search approach. Moreover, it improves the exploration and exploitation of the algorithm. At the beginning of the iteration, TVESCSO uses global elite individuals and elite pools to expand the exploration of the solution space and dynamically balances global exploration and local exploitation. In the late iteration, through the elite grouping mechanism and Logistic Chaotic Learning, TVESCSO can effectively avoid falling into local optima. This strategy combines global and local optimization information, significantly improves the efficiency of the evolutionary process, and gives full play to the advantages of each unit, thus improving the algorithm's overall performance. The main innovations of this study are as follows.

- TVESCSO dynamically divides traditional populations into optimal and free populations through an adaptive time-varying method and adjusts the ratio and composition of the two to adapt to the optimization needs of different iteration stages.
- Nonlinear adaptive factors r_G and optimal elite individuals are introduced into the optimal population to enhance the depth of global exploration and local exploitation. Meanwhile, the free population uses the elite pool and the dynamic exponential term parameter CF to adaptively regulate the effectiveness of global exploration and local exploitation and improve the exploration capability of the solution space.
- Increase the diversity of new populations through information interaction mechanisms and sorting.
- Adopt the elite grouping mechanism of Logistic Chaos Learning to enhance the diversity of local exploitation, avoid the algorithm falling into the local optimum, and provide a broader search space and higher global optimization capability.
- The robustness and stability of the TVESCSO algorithm are verified on the CEC2017 and CEC2022 benchmark functions, four real-world engineering problems, and 12 feature selection datasets.
- The superiority of the TVESCSO algorithm in terms of efficiency and robustness is further confirmed by Friedman's sequencing test, Nemenyi's test and Wilcoxon's rank sum test.

The paper is organized as follows. Section 2 provides an in-depth discussion of strategies for balanced exploration and exploitation of swarm intelligence algorithms and their variants. Section 3 presents the mathematical model of SCSO. Section 4 concentrates on the TVESCSO algorithm proposed in this study. In Section 5, we systematically evaluate the stability of the TVESCSO algorithm on the CEC2017 and CEC2022 benchmark functions and provide a comprehensive performance comparison with the current leading optimization algorithms, thus validating its superiority on theoretical benchmarks. Section 6 extends the study to practical applications by evaluating the effectiveness of TVESCSO on four engineering problems and comparing its performance with well-known optimization algorithms, demonstrating its potential for real-world problem-solving. Section 7 further explores the high-dimensional optimization performance of TVESCSO on 12 feature selection datasets and comprehensively validates the high-dimensional optimization capabilities of TVESCSO through exhaustive comparisons with SCSO variants and other well-known algorithms. Finally, Section 8 summarizes this study's key findings and suggests future research directions.

2. Related work

In MA, exploration and exploitation are two key concepts (Yang et al., 2021) that affect the algorithm's performance and efficiency. Exploration searches for new, under-examined regions in the solution space by Increasing the diversity of solutions, avoiding local optima, exploiting deeper searches in the vicinity of known solutions and improving the quality of the solutions. In summary, MA increases the likelihood of finding a globally optimal solution by finding a balance between exploring new regions and exploiting known regions. Common approaches to balance exploration and exploitation (Ma et al., 2023) are population diversity, multi-strategy fusion and hybrid algorithms.

2.1. Population diversity

Zhang et al. proposed an improved Group Teaching Optimization Algorithm (GTOA) called ISGTOA (Zhang & Chi, 2023) (to enhance performance. ISGTOA effectively facilitates the exchange of information between the excellent and average groups by reusing two population individuals in the archive, thus enhancing the algorithm's global search capability. Experimental results show that ISGTOA exhibits significant advantages in dealing with high-dimensional expensive optimization problems, especially in avoiding local optima and enhancing convergence efficiency. Since ISGTOA does not introduce additional control parameters, its population size and termination conditions still need to be adjusted according to the specific problems, which increases the complexity in practical applications. Turgut et al. proposed a Diversity-Enhanced Equilibrium Optimisation Algorithm (DEQUIL) (Turgut & Turgut, 2024), designed for unconstrained optimization problems. The algorithm integrates the operational equations of tangent search optimization with the symbiotic phase of the symbiotic search algorithm to enhance the diversity of the individual solutions in the population.

However, as the dimensionality of the optimization problem increases, it becomes more difficult to maintain diversity and ensure efficient exploration. Yang proposed a dual-population cooperative algorithm (DPDCA) (Yang et al., 2024), which maintains two populations and an archive and can dynamically adjust the search strategy according to the evolutionary stage. In the early stage of evolution, the two populations are responsible for different tasks, promote fast convergence by exchanging solutions, and save promising solutions in the archive. Different local search strategies are used in the later stages of evolution to obtain a high-quality set of feasible solutions, depending on the diversity and convergence of the central populations. To prevent the algorithm from falling into local optimums in the later stages, the archive is utilized to enhance the population diversity and to explore the significant populations in depth to finally obtain a set of solutions located in the frontiers of the Pareto. However, the archive update strategy directly affects the algorithm's diversity and global search capability in later iterations. Suppose the archive is not properly managed or updated with a lag. In that case, the quality of the new solutions will be negatively affected, which in turn weakens the algorithm's ability to find globally optimal solutions. Liang proposed a GSK framework for enhanced population diversity (EPD-GSK) (Liang & Wang, 2024). The algorithm utilizes a low-discrete Sobol sequence for population initialization, which helps to promote diversity among the initial solutions. A Cauchy mutation strategy is employed during the primary stage to perturb individuals and expand the search space. Moreover, the framework incorporates oppositional learning to improve convergence. Nevertheless, a potential issue with decreasing population diversity could limit the algorithm's effectiveness. Wang proposed a co-evolutionary algorithm (CdEA-SCPD) (Wang et al., 2024) aimed at exploring interpretable constrained optimization problems. The dynamic archiving strategy and a shared replacement mechanism enhance the diversity of the population. Meanwhile, weights are assigned to different constraints according to the severity of constraint violation to accelerate the convergence to the global optimal solution. However, the CdEA-SCPD algorithm needs to adjust the search strategy at different evolutionary stages dynamically. Otherwise, it may lead to insufficient search efficiency or premature convergence.

2.2. Multi-strategy fusion

Zhou et al. proposed the Random Follower Ant Colony Optimization (RFACO) (Zhou et al., 2023) to address the limitations of ACO algorithms in high-dimensional optimization problems. The core innovation of this algorithm is introducing a random following strategy, which enhances information exchange between search agents. However, in complex optimization problems, adaptive adjustment mechanisms can lead to under-exploration, potentially hindering the convergence speed. Xu proposed a diversity-driven opposites fusion strategy (SQOBL) (Xu et al., 2023), which constructs extended opposites and quasi-opposites variants by exploiting the principle of refractive imaging to balance the population's exploratory and exploitative capabilities. However, at a later stage, the exploration capability may weaken, causing the algorithm to fall into local optimal solutions. Xu proposed an improved artificial bee colony algorithm (ABC-MES) (Xu et al., 2024). The employed bee stage introduces a multi-strategy evolutionary approach to improving the population's quality. A decision support database is constructed in the learning stage. The UCB algorithm is used in the bystander bee stage to combine prior and new knowledge to evaluate and screen evolutionary strategies, ensuring efficient evolution. Updating old nectar sources through T-distribution and oppositional learning mechanisms in the scout bee stage prevents prematurely falling into local optimal solutions. This staged strategy design enhances the global search capability and comprehensively explores and exploits high-quality solutions. However, the conflict between diversity and convergence may weaken exploration ability, thus increasing the risk of falling into local optimal solutions. Additionally, computational complexity issues may affect the algorithm's overall performance. Xu proposed a multi-strategy improved multi-objective particle swarm optimization algorithm (CMOPSO-MSI) (Xu et al., 2024). The algorithm ensures the uniformity and diversity of the population by introducing an adaptive grid congestion distance, a roulette wheel selection strategy, and an improved dynamic nonlinear weighting strategy, enhancing the algorithm's global search capability and convergence performance. However, the adaptive grid congestion distance parameter sensitivity and roulette wheel selection strategy may affect the balance between exploration and exploitation. Xiao et al. (2024) improved the efficiency of the SAO algorithm through a good point set initialization strategy, a greedy selection method, a differential evolution strategy, and a dynamic lens opposition learning strategy. However, uncertainty arises in the dynamic adjustment of the k-value in the DLOBL strategy concerning the fitness of the optimization stage. This lack of fitness interferes with the algorithm's exploration and development balance and prevents it from jumping out of the local optimum. Zamani proposed an Evolutionary Crow Search Algorithm (ECSA) (Zamani & Nadimi-Shahraki, 2024) for optimizing the hyperparameters of Multilayer Perceptron (MLP). ECSA incorporates an evolutionary search strategy, adaptive flight length, and interactive memory mechanisms to overcome the limitations of standard Crow search algorithms. However, it remains a challenge to dynamically adjust the balance between adaptive flight length and interactive memory mechanisms at different stages. Hu proposed an enhanced differential evolutionary algorithm named DHRDE (Hu et al., 2024), which combines a two-population hybrid update and a random population replacement strategy. Before the update phase, DHRDE constructs an elite rival population based on the initial population to explore more potential search regions. The algorithm also integrates a perturbation mechanism into the DE/rand/2 method, significantly improving search efficiency. Two updating models are established through the co-guidance of random and locally optimal individuals, and a two-population hybrid updating strategy is adopted to achieve a comprehensive and multi-angle search. Although this strategy aids in multi-perspective exploration, the collaboration and updating methods between the populations need dynamic adjustment at different stages of the algorithm. If not properly adjusted, it may result in an imbalance between exploration and exploitation.

2.3. Hybrid algorithms

Braik et al. proposed an improved hybrid chameleon swarm algorithm (HMICSA) (Braik et al., 2023), which precisely balances global search and local development needs through adaptive parameters and fuzzy adaptive inertia weights. On this basis, HMICSA incorporates the individual memory mechanism of the AFT algorithm, which significantly improves the search's depth and accuracy, enhances the algorithm's efficiency in utilizing the past valuable information, and improves the performance of the feature selection. However, over-reliance on historical information may fall into a local optimum, limiting the exploration of a broader solution space, thus negatively affecting the balance between exploration and exploitation, especially when dealing with high-dimensional complex problems. Duan and Yu (2023) significantly enhanced the exploration capabilities of the Grey Wolf optimization algorithm by integrating the Social Class Model and the Trigonometric Function Transformation of the Sine Cosine Algorithm (SCA). However, introducing SCA and its multiple control parameters increases the computational complexity, which may lead to higher computational costs when dealing with complex and high-dimensional optimization problems. Ouyang proposes a hybrid cuckoo search algorithm based on linearly decreasing populations (LHCS) (Ouyang et al., 2024), which aims to optimize the local search and accelerate the convergence speed. The algorithm incorporates the solution update strategy of the Grey Wolf Optimiser. It uses a linearly decreasing rule to dynamically adjust the proportion of calls to the strategy to balance exploration and exploitation. However, the linear reduction

rule of LHCS may be too simple to adapt to complex and variable optimization problems, which may lead to premature a local optimum. Ji proposed a new particle swarm optimization combined with a whale optimization algorithm (HPWA) (Ji et al., 2024) for identifying irregular shapes of defects in the inner walls. Although HPWA combines the advantages of particle swarm optimization and whale optimization algorithms, its exploratory capability may be insufficient to fully cover the solution space when dealing with high-dimensional complex defective morphologies. Kumar proposed a hybrid Whale Ant Optimization Algorithm (WAOA) (Kumar et al., 2024) for energy-efficient routing in WSNs. In WAOA, the whale algorithm is utilized to find a suitable cluster head in a predefined search space, while the ant colony algorithm searches for the best route from the source cluster sensor to the cluster head in its predefined space. However, this synergistic mechanism increases the computational complexity, affecting the overall convergence of the algorithm.

2.4. Optimization algorithm for variant sand cat

Kiani proposed the SCSO with a multidisciplinary principle inspired by the Political system (PSCSO) algorithm (Kiani et al., 2023), which aims to improve the optimization algorithm's global search capability by mimicking the mechanism of multi-party competition and cooperation in the political system and the convergence speed. Although by introducing the concept of the political system, SCSO enhances the diversity and collaboration ability of the algorithm in its theoretical design, the complexity of its computational model also increases significantly when dealing with complex engineering scenarios, resulting in a longer computation time. Muqeet et al. proposed a Q-learning-based SCSO (QSCSO) algorithm (Muqeet et al., 2023), which uses the SCSO algorithm to optimize reinforcement learning and thus improve the surface quality of turning machining. However, when dealing with high-dimensional complex problems, Q-learning may encounter the problem of long convergence time. Specifically, each state–action pair needs to be fully sampled to ensure accurate updating of Q-values, which may be impractical in practical applications. Niu et al. proposed an Enhanced SCSO (ESCSO) algorithm (Niu et al., 2024). The algorithm first determines the position of an individual in the current population through a social ranking mechanism, then uses an adaptive social neighborhood search mechanism to construct a dynamic neighborhood for the sand cat, and updates the position of the sand cat during the predation phase using a Lévy flight strategy. However, the ESCSO algorithm's performance is highly reliant on the parameter settings in the social neighborhood search mechanism and the Lévy flight strategy, and this dependence may lead to large fluctuations in the algorithm's performance in practical applications. Li et al. proposed Chaotic Dynamic Disturbance SCSO (CDD-SCSO) (Li, Xiao, Zhang et al., 2024). Firstly, the algorithm initializes the sand cat swarm using cubic chaotic mapping to enhance the randomness and diversity of the population. Secondly, the dynamic perturbation factor is calculated by sine and cosine functions to help the algorithm eliminate the non-optimal value trap. However, the CDD-SCSO algorithm contains several key parameters. Finding the optimal combination of these parameters requires much time for fine-tuning. Li et al. proposed a virtual force-directed improved SCSO (VF-ISCSO) (Li, Zhao, Wang, & Wen, 2024). First, a nonlinear convergence strategy for sand cat sensitivity was designed to better balance global search and local exploitation. Secondly, the algorithm utilizes the fitness values of individual sand cats to balance their local and global search capabilities dynamically. However, the VF-ISCSO algorithm is not designed and optimized for node heterogeneity, which makes it difficult for sensor nodes to be effectively guided to the optimal location in heterogeneous network environments, thus affecting the effectiveness of network coverage performance improvement.

Despite the remarkable achievements of the above algorithms, they still face multiple challenges. Firstly, most of these algorithms are single-population elite-guided augmented MA, leading to a high degree of homogeneity within the population, thus hindering the search for the globally optimal solution (e.g., DEQUIL, PSCSO, and HMICSA). Secondly, some algorithm variants adopt a single search strategy, which limits their effectiveness in solving high-dimensional complex problems (e.g., HMICSA, LHCS, and ABC-MES). Finally, the large number of control parameters in the swarm intelligence algorithm increases the algorithm's complexity and the difficulty of parameter tuning (e.g., WAOA, CDD-SCSO, and QSCSO).

To improve the applicability of the algorithm and its ability to solve high-dimensional complex problems, future research should focus on the following key aspects. Firstly, local optimal solutions should be avoided by improving population diversity to ensure the algorithm can explore the search space more comprehensively. Second, diversified learning mechanisms should be explored to balance exploration and exploitation effectively to enhance the algorithm's ability to cope with large-scale and complex optimization problems. Finally, the updating strategy of the algorithm should be simplified to reduce the computational cost and improve the overall operational efficiency.

3. Sand cat optimization algorithm

The SCSO algorithm, a swarm intelligence optimization algorithm that mimics the survival behavior of sand cats in nature, enhances the flexibility and robustness of the optimization process by incorporating regulation parameters. Next, the mathematical formulation of the algorithm's parameters is described.

$$r_G = 2 \cdot \left(1 - \frac{t}{t_{\max}}\right) \tag{1}$$

$$R = 2 \cdot r_G \cdot rand - r_G \tag{2}$$

$$r = r_G \cdot rand \tag{3}$$

In Eqs. (1), t represents the current iteration count, while $t_{\max}$ denotes the maximum number of iterations. As the iteration count grows, the value of r_G linearly decreases from 2 to 0. The parameter r is crucial for deciding between search and attack behaviors. It determines the sensitivity range for each sand cat. The term $rand \in [0, 1]$.

3.1. Attacking prey (exploitation phase)

The sand cat will attack the prey when the absolute value of R is less than or equal to 1. The attack process first generates a random position X_{rand} through the best position X_{best} and the current position X_c. A circle can represent the sensitive range of a sand cat, and a random angle θ is chosen for each sand cat using a roulette wheel. Using random positions for the sand cats ensures they are very close to their prey, thus increasing the probability of successful hunting. Choosing a random angle for each sand cat avoids falling into a local optimal solution.

$$X_{rand} = \left| rand \cdot X_{best} - X_c \right| \tag{4}$$

$$X_c(t+1) = X_{best}(t) + r \cdot X_{rand}(t) \cdot \cos(\theta) \tag{5}$$

As evident from Eqs. (5), this hunting strategy allows the sand cat to track the target and optimize its hunting behavior effectively. Specifically, the random positions X_{rand} are derived from the optimal positions X_{best} and current positions X_c, and the angle of attack is determined by the roulette wheel selection, ensuring proximity to the prey while evading local optima.

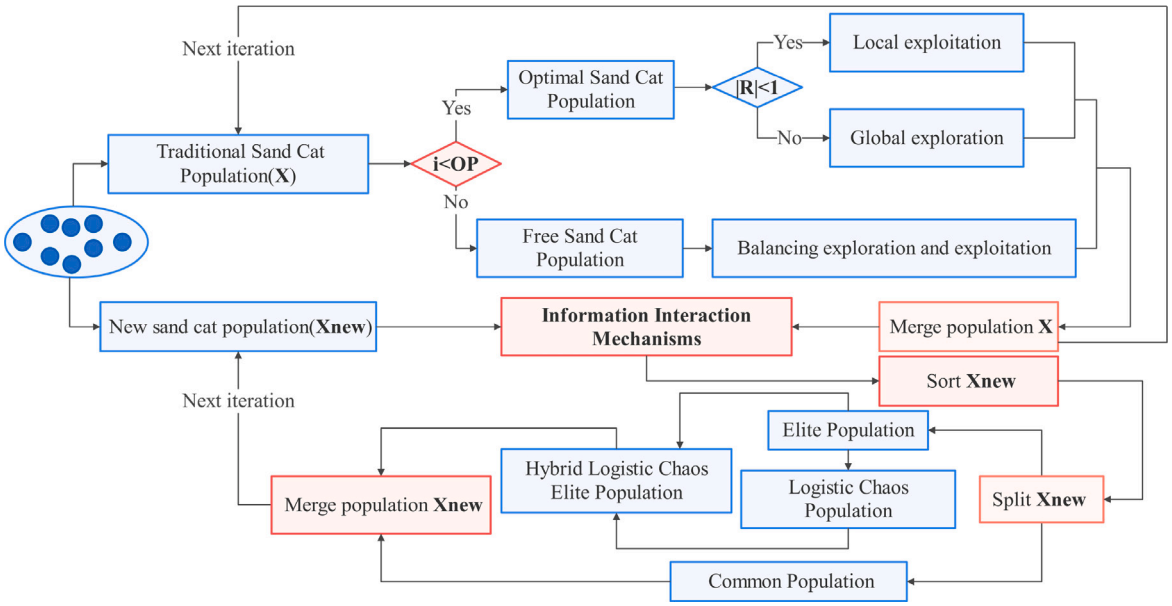


Fig. 1. TVESCSO algorithm framework.

3.2. Search for prey(exploration phase)

If the absolute value of R is more significant than one, the sand cat engages in hunting for prey. The sand cat adjusts its location by considering the random position X_r and the current position X_c, as well as the random number r_1 and the sensitivity r. Here, r_1 falls within the range of $[0, 1]$, following a particular mathematical calculation outlined in Eqs. (6).

$$X_c(t+1) = r \cdot \left(X_r - r_1 \cdot X_c(t)\right) \tag{6}$$

From the Eqs. (5) and (6), one can observe that the parameter R in the SCSO algorithm determines the movement range for the sand cat population. When $|R|$ falls within the interval $[-1, 1]$, the sand cat population aggressively targets the prey, leading to rapid convergence.

4. TVESCSO algorithm

From Section 3, it can be seen that the SCSO algorithm uses a single evaluation mechanism when searching for a globally optimal solution, which leads to homogenization of the population search strategy, thus increasing the risk of the algorithm falling into a locally optimal solution (Rakhshani & Rahati, 2017). Section 4.1 details the motivation for proposing the new variant of the SCSO algorithm and highlights the need for improvement. Section 4.2 presents the design principles and specific design details of the new variant SCSO algorithm. Section 4.3 describes the flowchart and pseudo-code of TVESCSO. Section 4.4 Describes the algorithmic complexity analysis of TVESCSO.

4.1. Motivation of the TVESCSO algorithm

The SCSO algorithm utilizes $|R|$ to achieve an equilibrium relationship between exploitation and exploration, which directly leads to the homogenization of the search strategies, thus increasing the risk of the algorithm falling into a locally optimal solution. In swarm intelligence optimization algorithms, fixing the grouping strategy and introducing diverse search strategies can improve the optimization efficiency of the algorithm (Zhao et al., 2023). For example, the Snake Optimizer (SO) algorithm (Hashim & Hussien, 2022) improves the optimization performance by dividing the population into two sub-populations of fixed size, males and females, with each sub-population adopting the search strategy of elite individuals. In Role-oriented Binary Grey Wolf Optimizer (BFLGWO) (Wang et al., 2023), the top grey wolf adopts a multi-strategy approach, such as foraging and following, improving convergence performance and enhancing the population's diversity.

However, fixed grouping and diverse search strategies can enhance the algorithm's global search and local exploitation (Zhang, 2023). However, such single/dual elite bootstrap evolutionary information for large-scale populations needs more inheritance and can easily lead to high similarity or homology (Zhao et al., 2024). Therefore, when solving complex optimization problems, these methods still suffer from global search bias and poor local extreme avoidance.

TVESCSO employs an adaptive time-varying method to dynamically divide the traditional population into the optimal population(OP) and the free population(FP) to adjust the composition of both flexibly. In the OP, nonlinear adaptive factors and optimal elite individuals are introduced to improve the depth of global exploration and local exploitation. In the FP, the elite pool and the dynamic exponential term parameter CF adaptively regulate the effectiveness of global exploration and local exploitation. In addition, the diversity of new populations is increased through information interaction mechanisms and sorting. Finally, the elite grouping mechanism of Logistic Chaotic Learning is utilized to prevent the algorithm from falling into the local optimum.

4.2. TVESCSO algorithm framework

Fig. 1 illustrates the framework of the TVESCSO algorithm, which consists of three main parts: first, the division of the traditional group dynamics into two sub-populations; second, the exchange of information between the traditional and new populations; and lastly, the differentiation of the new populations into a mixed Logistic Chaotic Elite and an Ordinary population using a segmentation mechanism.

4.2.1. Divide dynamic populations

According to Fig. 1, in the iterative process, the traditional population is divided into the optimal population and the free population through the linear dynamic division method to effectively adjust the ratio of the two populations. The formula for the linear dynamic division Eqs. (7) is as follows.

$$OP = ceil\left(N \cdot \frac{i}{t_{\max} + 1}\right) \tag{7}$$

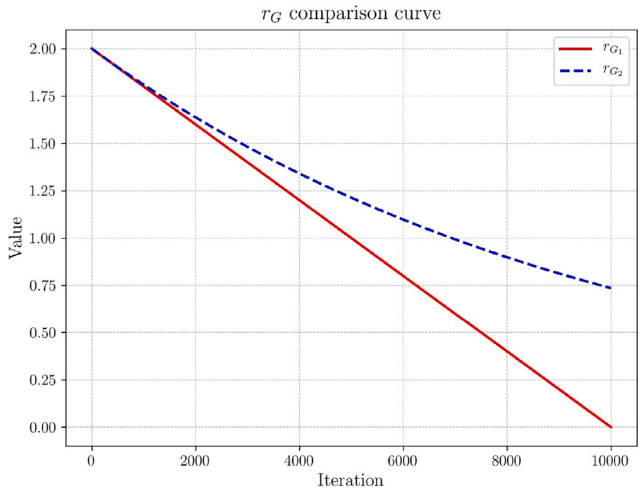


Fig. 2. r_G Contrast curve.

Eqs. (7) elucidates the correlation between the number of elite sand felines and the total number of iterations. Additionally, x equals the rounded-up value of $ceil(x)$. When $i < OP$, the i sand cat is in the optimal population. When $i > OP$, the i sand cat is in the free population. After Eqs. (8) is as follows.

$$FP = N - OP \tag{8}$$

Eqs. (7) and (8), it is possible to partition the population into the optimal and free population. The optimal population consists of sand cat individuals as $X_{OP} = \left(X_1^{OP}, X_2^{OP}, \ldots, X_i^{OP}\right)$ with elite individuals $X_i^{OP} = \left(x_{i,1}^{OP}, x_{i,2}^{OP}, \ldots, x_{i,j}^{OP}\right)$ $(i = 1, \ldots, OP; j = 1, \ldots, D)$. The free population, consisting of FP individuals in population X, is represented as $X_{FP} = \left(X_1^{FP}, X_2^{FP}, \ldots, X_i^{FP}\right)$, with ordinary individuals $X_i^{FP} = \left(x_{i,1}^{FP}, x_{i,2}^{FP}, \ldots, x_{i,D}^{FP}\right)$ $(i = 1, 2, \ldots, FP)$.

4.2.2. The optimal population

As shown in Fig. 1, compared with the SCSO algorithm, in the optimal population, the improvement strategy is mainly reflected in the introduction of the nonlinear adaptive factor r_G and the optimal elite individuals. As seen from Eqs. (1), the parameter is a regulating factor r_G for balancing the algorithm's global search and local exploitation capabilities. However, the parameter decreases linearly during the iteration process, which fails to adequately simulate the sand cat's pattern of extensive exploration followed by accelerated aggregation to share food in nature. Therefore, this study proposes an improved nonlinear adaptive parameter factor with the following mathematical formula.

$$r_G = 2 \cdot \exp\left(-\frac{t}{t_{\max}}\right) \tag{9}$$

Fig. 2 illustrates the comparative analysis between the original parameter r_{G1} and its improved counterpart r_{G2}. The enhanced parameter r_{G2} exhibits higher initial values followed by a gradual decrease. At the beginning of the iteration, this method maintains a more extensive exploration capability, while at the later stages of the iteration, the accuracy of the local search is enhanced. Its advantages help the algorithm to converge to the optimal solution more efficiently while avoiding falling into local optima. Therefore, Eqs. (9) effectively balances exploration and exploitation capabilities.

From Eqs. (6), it can be seen that the new solutions are mainly generated by linear combinations of X_r and X_c, which may lead to many similar solutions in the population, thus reducing the diversity of the population. The reduction of population diversity affects the robustness and exploration ability of the algorithm, especially when the initial solutions are of low quality or dispersed, and the generated new solutions may need to be able to guide the algorithm towards the optimal solution effectively. Therefore, the optimal elite individual guidance method is proposed with the following mathematical formula to enhance the search diversity and increase the convergence stability.

$$X_c(t+1) = X_{best}(t) + r \cdot \left(X_r(t) - r_1 \cdot X_c(t)\right) \tag{10}$$

According to Eqs. (10), through the guidance of the optimal elite individuals, the oscillation of the solution during the search process can be reduced, so that the algorithm is more stable towards the optimal solution. Even if the initial solution quality is not high or the distribution is not ideal, this steering mechanism can help the algorithm gradually improve the solution quality and enhance its robustness.

4.2.3. The free population

In the free population, the improvement strategy is mainly embodied in three components: constructing the elite pool, defining the index term parameter CF, and dynamically balancing the exploration and exploitation positions.

An elite pool consists of four sand cats with relatively optimal fitness values from the traditional population and an average sand cat, defined as follows.

$$Elite(t) \in \left[E_{first}(t), E_{second}(t), E_{third}(t), E_{fourth}(t), E_{mean}(t)\right] \tag{11}$$

In Eqs. (11), $E_{first}(t)$, $E_{second}(t)$, $E_{third}(t)$, $E_{fourth}(t)$, represent the current best solution, the second best individual, the third best individual and the fourth best individual, respectively. $E_{mean}(t)$ represents the average position of all sand cat individuals, mathematically calculated as follows.

$$E_{mean}(t) = \frac{1}{N}\sum_{i=1}^{N} X_i(t) \tag{12}$$

During each iteration, a random selection is made from the elite pool consisting of $E_{first}(t)$, $E_{second}(t)$, $E_{third}(t)$, $E_{fourth}(t)$ and $E_{mean}(t)$, which is mathematically modeled by the following equation.

$$X_{eq} = randi\{Elite(t)\} \tag{13}$$

CF is an essential parameter for balancing exploration and development capacity and is mathematically described below.

$$CF = 2 \cdot sign\left(r_1 - 0.5\right) \cdot \left[e^{z \cdot r_1} - 1\right] \tag{14}$$

z is a coefficient updated with the number of iterations, calculated as follows.

$$z = \left(1 - \frac{t}{t_{\max}}\right)^{\left(2 - \frac{t}{t_{\max}}\right)} \tag{15}$$

An elite pool guides the updating of sand cat individuals, and the parameter CF is combined with multiple search mechanisms, calculated as follows.

$$X_c(t+1) = X_{eq} + \underbrace{CF \cdot \left(X_c(t) - X_{eq}\right)}_{Global exploration} + \underbrace{\frac{CF \cdot (1 - CF)}{2r_1} \cdot \left(X_c(t) - r_1 \cdot X_{eq}\right) \cdot GP}_{Local exploitation} \tag{16}$$

From Eqs. (16), it can be seen that at the beginning of the iteration when the value of parameter CF is large, the sand cat individuals can move away from or close to X_{eq} for extensive global exploration. As the value of parameter CF is small, sand cat individuals prefer to perform fine-grained search near the current elite pool X_{eq} for local exploration. This mechanism enables the search strategy to be dynamically adjusted at different iteration stages, thus balancing global exploration and local exploitation. Where $r_2 \in [0, 1]$, GP is calculated as follows.

$$GP = \begin{cases} 0.5 \cdot r_1 & r_2 > 0 \\ 0 & r_2 \leqslant 0 \end{cases} \tag{17}$$

4.2.4. Information interaction mechanisms

When both the optimal and free population have completed their iterations, they are recombined into the traditional population and compared with the new population. The final new population is determined by the information interaction method with the following formula:

$$Xnew_i(t+1) = \begin{cases} Xnew_i(t) & f\left(Xnew_i(t)\right) < f\left(X_i(t)\right) \\ X_i(t) & f\left(Xnew_i(t)\right) > f\left(X_i(t)\right) \end{cases} \tag{18}$$

Through Eqs. (18), it can be seen that if the fitness value of the new population $Xnew_i$ individuals is greater than the fitness value of the traditional population X_i, then it indicates that the position of the new population individuals $Xnew_i$ needs to be updated to the traditional population X_i in such a way that the diversity of the new population is increased.

4.2.5. Updating new populations

Updating new populations consists of three main parts: dividing elite and everyday populations, generating Logistically chaotic elite populations, and merging new populations.

First, sort the $Xnew$ population in ascending order of fitness values, with the individual with the smallest fitness value at the front and the top $M = rp \cdot N$ individuals, where rp ranges from 0 to 1, are chosen as the elite population. This elite population, denoted as $Xnew_E = \left(Xnew_1^e, Xnew_2^e, \ldots, Xnew_i^e\right)$, consists of individuals represented by $Xnew_i^e = \left(xnew_{i,1}^e, xnew_{i,2}^e, \ldots, xnew_{i,j}^e\right)$, with D being the dimension count. Simultaneously, the remaining standard population of $(N-M)$ individuals in population $Xnew$ is labeled as $Xnew_c = \left(Xnew_1^c, Xnew_2^c, \ldots, Xnew_i^c\right)$, while the regular individuals $Xnew_i^c = \left(xnew_{i,1}^c, xnew_{i,2}^c, \ldots, xnew_{i,j}^c\right)$ $(i = 1, 2, \ldots, (N-M))$. Secondly, Logstic chaotic iterations are performed on $C_{i,j}^e$ using Eqs. (19) to obtain a chaotic sequence.

$$C_{i,j}^e(t+1) = 4 \cdot Xnew_{i,j}^e(t)\left(1 - Xnew_{i,j}^e(t)\right) \tag{19}$$

Finally, the chaotic sequence $C_{i,j}^e$ is used to perform chaotic elite mutation on $xnew_{i,j}^e$ elite individuals to generate chaotic elite individuals XC_i^e, and then, the composition of chaotic elite populations as $XC_E = \left(XC_1^e, XC_2^e, \ldots, XC_i^e\right)$, $(i = 1, 2, \ldots, M)$. XC_i^e the mathematical model is shown in Eqs. (20).

$$XC_i^e = \lambda Xnew_i^e + (1-\lambda)C_i^e \tag{20}$$

Where λ is the shrinkage factor, which is calculated as follows

$$\lambda = \frac{t_{\max} - t}{t_{\max}} \tag{21}$$

In Eqs. (21), the influence of the elite individual $xnew_{i,j}^e$ on the chaotic elite individual XC_i^e decreases as λ decreases. Simultaneously, the chaotic individual C_i^e significantly impacts XC_i^e due to its higher randomness and diversity. This approach guides the algorithm to progressively confine the search area of elite individuals near the desired solution throughout the iteration process, thereby enhancing the local search efficiency.

4.2.6. Filtering mechanism

A screening mechanism determines the mutation's success in the chaotic elite individual XC_i^e. This mechanism involves comparing the fitness value $f\left(XC_i^e\right)$ of the mutated chaotic elite individual with the fitness value $f\left(Xnew_i^e\right)$ of the corresponding elite individual $Xnew_i^e$. Based on this comparison, an excellent individual is selected according to the mathematical model presented in Eqs. (22).

$$X_i^e = \begin{cases} XC_i^e, if & f\left(XC_i^e\right) < f\left(Xnew_i^e\right) \\ Xnew_i^e, & otherwise \end{cases} \tag{22}$$

Suppose the fitness value of the mutated chaotic elite individual XC_i^e is lower than that of the elite individual $Xnew_i^e$ indicated. The mutation is deemed successful, and the mixed chaotic elite population is updated accordingly.

This screening mechanism ensures the retention of only the fittest individuals within the population. The algorithm drives evolutionary progress towards optimal solutions by favoring individuals with lower fitness values. This approach not only enhances the efficiency of the search process but also facilitates compelling exploration of the solution space, thereby improving the algorithm's overall performance.

4.3. Flowchart and pseudo code for TVESCSO

According to the details in Section 4.2, TVESCSO significantly enhances the algorithm's ability to search the solution space globally by dynamically dividing the populations, facilitating the exchange of information between the traditional and new populations, and updating the new populations based on the traditional SCSO. In the early stage of the algorithm, the optimal and free populations are utilized to improve the sand cat population's extensive exploration of the solution space and dynamically balance the relationship between exploration and exploitation. In the middle stage, the population diversity is increased through the information interaction mechanism. In the late iteration, the chaotic elite strategy is used to narrow the search range gradually and enhance the intensive search for the excellent solution region. In conclusion, TVESCSO enhances the algorithm's overall performance in solving complex optimization problems by dynamically combining global exploration and local exploitation to enhance the deep exploration of qualitatively superior solutions while maintaining the breadth of solution space exploration. The implementation steps are as follows.

Step 1, initialize the populations, which are the traditional population X and the new population $Xnew$. Subsequently, the fitness values of each individual $i = 1, 2, \ldots, N$ in these two populations are calculated as $f\left(X_i\right)$ and $f\left(Xnew_i\right)$, respectively.

Step 2, calculate the values of OP, r_G according to Eqs. (7) and (9).

Step 3, divides the traditional population into the optimal and free populations using the values of OP.

Step 4, in the optimal population, if $|R| \leqslant 1$, then according to Eqs. (5), the individual sand cat enters the exploitation phase. if $|R| > 1$, then according to (10), the individual sand cat enters the exploration phase.

Step 5, in the free population, according to Eqs. (16), The sand cat elite pool guides the sand cat individuals in balancing the relationship between exploration and exploitation dynamically.

Step 6, the optimal and free populations were recombined into the traditional population X.

Step 7, recalculate the new population $Xnew$ according to Eqs. (18).

Step 8, Sort $Xnew$ and select the top M sand cats to form an elite sand cat population. According to Eqs. (20) to form a Logistic chaotic elite population.

Step 9, According to Eqs. (22), The logistic chaotic elite population is screened.

Step 10, M and $N-M$ are re-performed to merge into a new population $Xnew$.

Step 11, update X_i, $Xnew_i$, $f\left(X_i\right)$ and $f\left(Xnew_i\right)$ respectively. Continue the loop if t is less than t_{max}. Otherwise, jump out of the loop, find the optimal solution (X_{best}) from the new population $Xnew$, and output it.

Detailed information on implementing this strategy can be found in Flowchart 3 and Algorithm 1.

From Algorithm 1, TVESCSO has the following advantages.

- Dynamically adjust the ratio between the optimal and free populations through parameter OP and adopt different global search and local exploitation strategies in different populations, fully reflecting the diversity of search strategies.
- Enhance the diversity of new populations by using the information interaction mechanism and sorting.

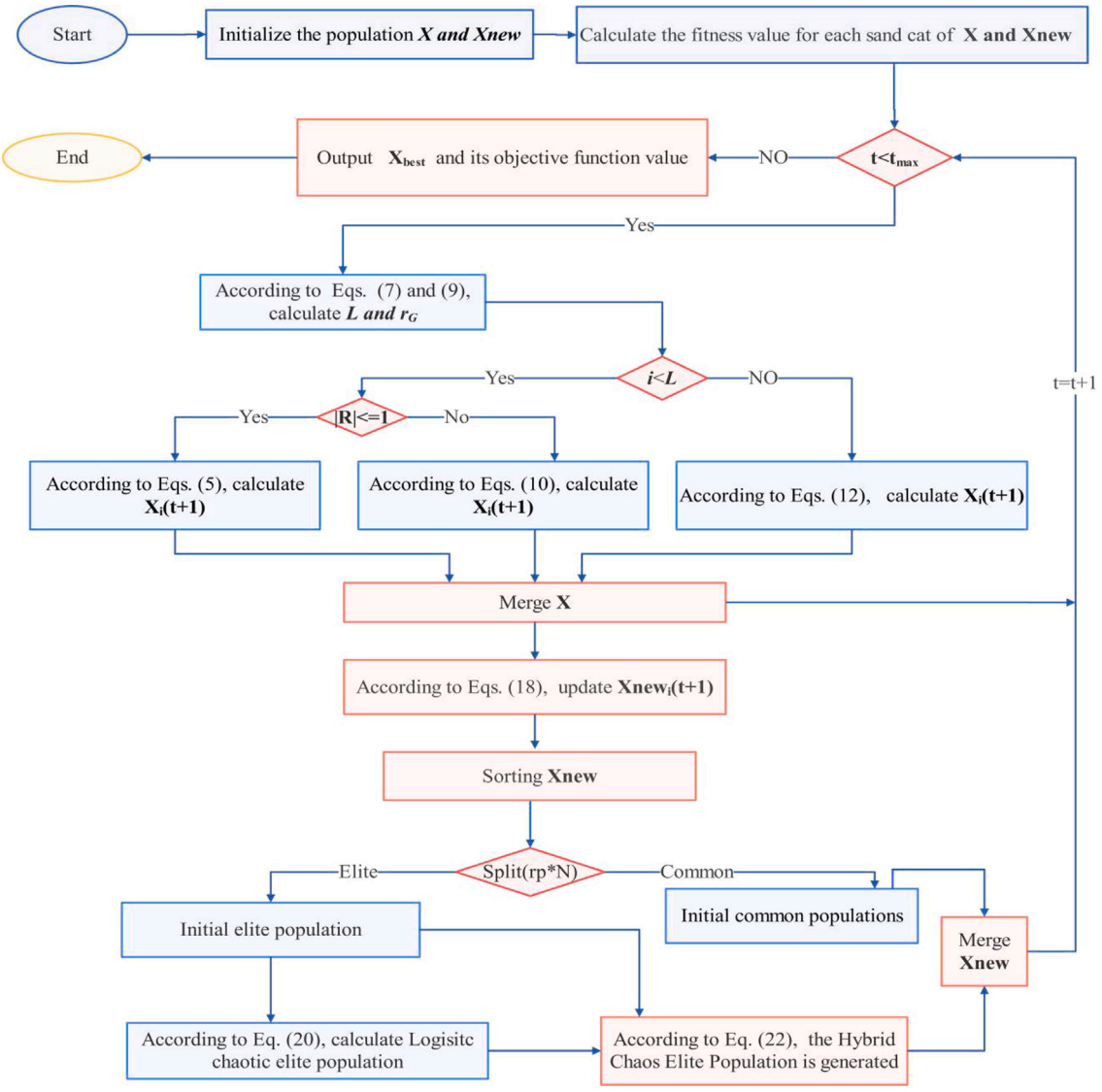


Fig. 3. Flowchart of TVESCSO Algorithm.

- Using the elite selection ratio rp to select elite individuals and enhancing its mutation searchability with the help of chaos theory makes the search process more flexible and unpredictable. This helps maintain the diversity of the populations and improve the algorithm's global search performance.

In summary, the TVESCSO algorithm can effectively concentrate the search resources and gradually reduce the search scope by dynamically adjusting the population structure and enhancing multiple search strategies. This allows it to efficiently solve complex optimization problems while maintaining the diversity of the solution space.

4.4. Algorithm complexity analysis of TVESCSO

The time complexity of TVESCSO can be evaluated by analyzing the frequency of its core operations and the time required for each operation. The critical components of the TVESCSO algorithm that contribute to its time complexity are as follows.

- At the start of the algorithm, the population is initialized with N populations and D dimensions. This operation is typically performed only once, resulting in a time complexity of $O(N \cdot D)$.
- In each iteration, for dynamically dividing the population, calculations such as fitness and updating the position of individuals need to be performed for each individual, so the time complexity for each iteration is $O(N \cdot D)$.
- Information about the interactions between the new and traditional populations, the time complexity of each iteration is $O(N \cdot D)$.
- The use of chaotic elite strategies to renew new populations is usually also $O(N \cdot D)$.

Should the algorithm execute t_{max} iterations, the overall time complexity will be $O(t_{max} \cdot N \cdot D)$. The time complexity of the TVESCSO algorithm matches that of the conventional SCSO algorithm.

5. Numerical optimization simulation experiment and analysis

To comprehensively evaluate the optimizing performance of our proposed TVESCSO, this section presents the challenging CEC2017 experimental sequence test sets. TVESCSO is compared to ten well-known meta-heuristic algorithms. For all algorithms, a consistent population size of $N = 50$ and a maximum number of iterations of $t_{\max} = 1000$ is maintained to ensure comparison fairness. Each algorithm independently is considered thirty times(runs) on each benchmark function. All

Algorithm 1: Pseudo Code of TVESCSO Algorithm

Input: the population size N; the maximum iterations t_{max}; the dimensional search space D
Output: the best fitness value $f(X_{best})$; the optimal solution X_{best}

```
Initialize the population X and Xnew ;
The fitness values f(X_i) and f(Xnew_i), i = 1, 2, ⋯, N , and the corresponding positions X_i and Xnew_i are calculated for each individual.;
while t ≤ t_max do
    for i ≤ N do
        According to Eqs. (7) and (9), Calculate the values of OP, r_G;
        if i < OP then
            if |R| ⩽ 1 then
                According to Eqs. (5), update the position of the individual X_i(t + 1) ;
            end
            else if |R| > 1 then
                According to Eqs. (10), update the position of the individual X_i(t + 1) ;
            end
        end
        else if i ⩾ OP then
            According to Eqs. (13) and (14), Calculate X_eq and CF ;
            According to Eqs. (16), update the position of the individual X_i(t + 1) ;
        end
    end
    Merge X ;
    According to Eqs. (18) , Calculate Xnew ;
    Sort Xnew;
    Selection of the M individuals.;
    The remaining N − M individuals remain unchanged and form the population .;
    for i ≤ M do
        According to Eqs. (20), Calculating XC_i^e(t + 1).;
        According to Eqs. (22) , Calculating Xnew_i^e(t + 1).;
    end
    Merge population M and N − M.;
    The fitness values f(X_i) and f(Xnew_i), and the corresponding positions X_i and Xnew_i are calculated for each individual.;
    t = t + 1 ;
end
```

experiments use Python software on an Intel(R) Core(TM) i7-9750HF 2.59 GHz CPU with 16 GB RAM and a 64-bit Windows 10 operating system.

The optimization performance of the proposed TVESCSO algorithm will be analyzed in this section, which mainly covers three aspects: General optimizing performance analysis, scalability analysis, and convergence analysis. The experimental results and the analysis are described in detail below.

5.1. Selected benchmark suites and performance measures

The TVESCSO algorithm is evaluated using 29 benchmark problems from the CEC2017 test set described in Table 1. The issues are categorized into Unimodal functions (F1-F2), Simple Multimodal Functions (F3-F8), Hybrid (F9-F19), and composite (F20-F29) functions. Unimodal Functions have a single global optimal solution. Therefore, they are suitable for assessing the algorithm's exploitation tendency and convergence speed. Simple Multimodal Functions evaluate the algorithm's ability to explore the global optimal solution with multiple locally optimal solutions. Hybrid functions combine features of essential functions to simulate natural search spaces. They test the algorithm's ability to solve practical optimization problems. Composite functions, a mixture of hybrid functions, balance global and local optimal solutions. They are often used to evaluate the overall search performance of algorithms.

The use of the CEC2022 test set (Table 2) allows an in-depth exploration of the searchability and robustness of TVESCSO when dealing with more competitive numerical optimization objects. The test set includes unimodal functions (f1), multimodal functions (f2–f5), hybrid functions (f6–f8), and composition functions (f9–f12). The test dimensions were set to Dim = 10 and Dim = 20.

In the experiments, two evaluation metrics are used to compare and analyze each method's optimization performance intuitively.

- The average value (Mean). The average fitness value usually measures the performance of an algorithm over many runs. The following formula can precisely determine the average.

$$Mean = \frac{1}{runs}\sum_{i=1}^{runs} fitness_i \tag{23}$$

  where $fitness_i$ denotes the result of the i-th run of the algorithm.
- The standard deviation (Std). The standard deviation serves as a metric for the spread of the optimization results, with a smaller value suggesting enhanced calculation stability of the algorithm. The formula for standard deviation is as follows.

$$Std = \sqrt{\frac{1}{runs-1}\sum_{i=1}^{runs}\left(fitness_i - Mean\right)^2} \tag{24}$$

From a statistical point of view, two robust non-parametric techniques (the Wilcoxon rank sum test, the Friedman rank test, and the Nemenyi Test) were used to detect any significant differences between the performance of the TVESCSO and other algorithms. For the Wilcoxon rank sum test, the threshold for statistical significance was set at the conventional level of 0.05. If the p-value derived from a test scenario is below this threshold, TVESCSO performs better than the algorithm. On the other hand, if the p-value is more significant than 0.05, it indicates that TVESCSO lags behind the algorithm. Furthermore, 'NaN' denotes a scenario where TVESCSO and the other optimization methods have statistically indistinguishable performance levels. The Friedman Rank test, on the other hand, evaluates the average rank of an optimizer over the entire set of test functions. A lower rank indicates a more favorable result.

Combining TVESCSO with Artificial Protozoa Optimizer (APO) (Wang et al., 2024), Multi-Strategy Grey Wolf Optimizer (MSGWO) (Chang et al., 2024), Snow Ablation Optimizer (SAO) (Deng & Liu, 2023), Physical phenomenon of rime-ice (RIME) (Su et al., 2023), adaptive hybrid dandelion optimizer (DETDO) (Hu et al., 2023), Sand Cat Swarm Optimization (SCSO) (Seyyedabbasi & Kiani, 2023), weIghted meaN oF vectOrs (INFO) (Ahmadianfar et al., 2022), Beluga whale optimization (BWO) (Zhong et al., 2022), adaptive LSHADE algorithm named ALSHADE (Li et al., 2022), Aquila Optimizer (AO) (Abualigah et al., 2021), Chimp optimization algorithm (CHoA) (Khishe & Mosavi, 2020), Ensemble Sinusoidal Differential Covariance Matrix Adaptation with Euclidean Neighborhood (LSHADE-cnEpSin) (Salgotra et al.) and other algorithms are compared. The parameter settings of each algorithm are shown in Table 3.

5.2. CEC2017 benchmark function results comparison and analysis

The experiments begin with parameter sensitivity (rp) analyses of the chaotic elite and ordinary populations. Then, the effectiveness of the three improvement strategies is verified. Finally, TVESCSO and

Table 1
Details of the CEC2017.

	No.	Functions	Fmin
Unimodal Functions	1	Shifted and Rotated Bent Cigar Function	100
	2	Shifted and Rotated Zakharov Function	200
Simple Multimodal Functions	3	Shifted and Rotated Rosenbrock's Function	300
	4	Shifted and Rotated Rastrigin's Function	400
	5	Shifted and Rotated Expanded Scaffer's F6 Function	500
	6	Shifted and Rotated Lunacek Bi_Rastrigin F6 Function	600
	7	Shifted and Rotated Non-Continuous Rastrigin Function	700
	8	Shifted and Rotated Levy Function	800
Hybrid Functions	9	Shifted and Rotated Schwefel's Function	900
	10	Hybrid Function 1 (N=3)	1000
	11	Hybrid Function 2 (N=3)	1100
	12	Hybrid Function 3 (N=3)	1200
	13	Hybrid Function 4 (N=4)	1300
	14	Hybrid Function 5 (N=4)	1400
	15	Hybrid Function 6 (N=4)	1500
	16	Hybrid Function 6 (N=5)	1600
	17	Hybrid Function 6 (N=5)	1700
	18	Hybrid Function 6 (N=5)	1800
	19	Hybrid Function 6 (N=6)	1900
Composition Functions	20	Composition Function 1 (N=3)	2000
	21	Composition Function 2 (N=3)	2100
	22	Composition Function 3 (N=4)	2200
	23	Composition Function 4 (N=4)	2300
	24	Composition Function 5 (N=5)	2400
	25	Composition Function 6 (N=5)	2500
	26	Composition Function 7 (N=6)	2600
	27	Composition Function 8 (N=6)	2700
	28	Composition Function 9 (N=3)	2800
	29	Composition Function 10 (N=3)	2900
Search Range: $[-100, 100]^D$			

Table 2
Details of the CEC2022.

Function	NO.	Functions	Fmin
Unimodal Function	1	Shifted and full Rotated Zakharov Function	300
Basic Functions	2	Shifted and full Rotated Rosenbrock's Function	400
	3	Shifted and full Rotated Expanded Schaffer's f6 Function	600
	4	Shifted and full Rotated Non-Continuous Rastrigin's Function	800
	5	Shifted and full Rotated Levy Function	900
Hybrid Functions	6	Hybrid Function 1 (N=3)	1800
	7	Hybrid Function 2 (N=6)	2000
	8	Hybrid Function 3 (N=5)	2200
Compostion Functions	9	Compostion Functions 1 (N=5)	2300
	10	Compostion Functions 2 (N=4)	2400
	11	Compostion Functions 3 (N=5)	2600
	12	Compostion Functions 4 (N=6)	2700
Search Range: $[-100, 100]^D$			

Table 3
Parameter configurations for TVESCSO other algorithms.

Algorithm	Year	Parameter settings
TVESCSO	–	$r_G \in [2, 0]$, $R \in \left[-2r_G, 2r_G\right]$
APO	2024	$theta1 = [1, 0],\ F = 0.5$
MSGWO	2024	$alpha = [2, 0]$
SAO	2023	–
DETDO	2023	$F = 0.9,\ a = [0, 1],\ k = [0, 1]$
RIME	2023	$W = 5$
SCSO	2023	$a = [2, 0]$
HBA	2022	$HA = 6,\ c = 2$
INFO	2022	$c = 2,\ d = 4$
BWO	2022	$w_f = [0.1, 0.05]$
ALSHADE	2022	$p = 0.11,\ H = 6,\ e = 0.5$
GJO	2022	$pa = [0.5, 2], pc = [0.5, 1]$
AO	2021	$G_1 = [-1, 1], G_2 = [0, 2]$
CHoA	2020	$f = [-1, 1]$
LSHADE-cnEpSin	2019	$H = 5, NPmin = 4, Pbest = 0.11, Arcrate = 2, pc = 0.4, ps = 0.5$

Table 4
Experimental results for different rp values on CEC2017.

Functions	Metric	$rp = 0.1$	$rp = 0.2$	$rp = 0.3$	rp = 0.4	$rp = 0.5$	Functions	$rp = 0.1$	$rp = 0.2$	$rp = 0.3$	$rp = 0.4$	$rp = 0.5$
f1	mean	2.17E+03	2.91E+03	7.12E+03	**1.48E+03**	7.19E+03	f16	**2.96E+03**	3.31E+03	3.29E+03	7.08E+03	3.00E+03
	std	6.84E+02	1.97E+03	2.82E+03	1.19E+03	2.80E+03		2.98E+02	2.16E+02	2.67E+02	5.59E+03	6.65E+02
f2	mean	2.00E+02	2.00E+02	2.00E+02	**2.00E+02**	2.00E+02	f17	3.18E+05	1.98E+05	2.22E+05	**1.54E+05**	1.83E+05
	std	1.68E−03	1.53E−03	1.33E−04	1.42E−07	6.43E−07		1.05E+05	8.29E+04	9.03E+04	2.61E+04	2.34E+04
f3	mean	4.02E+02	**3.26E+02**	3.66E+02	3.32E+02	3.80E+02	f18	5.14E+04	2.93E+04	3.99E+04	**1.46E+04**	1.97E+04
	std	6.82E+01	8.46E+00	4.88E+01	7.14E−01	4.00E+01		3.60E+04	2.08E+04	2.63E+04	5.31E+03	2.09E+04
f4	mean	5.37E+02	5.13E+02	5.66E+02	**4.69E+02**	5.10E+02	f19	2.23E+03	2.34E+03	2.31E+03	2.26E+03	**2.09E+03**
	std	4.35E+01	2.11E+01	1.11E+02	1.89E+01	1.07E+01		1.90E+02	1.41E+02	1.19E+02	8.21E+01	1.18E+02
f5	mean	5.00E+02	5.00E+02	5.00E+02	**5.00E+02**	5.00E+02	f20	2.25E+03	**2.10E+03**	2.24E+03	2.33E+03	2.32E+03
	std	5.17E−04	2.20E−05	1.89E−04	5.21E−04	2.27E−04		1.20E+02	7.04E−05	1.02E+02	2.53E+01	1.70E+01
f6	mean	7.94E+02	7.86E+02	**7.34E+02**	7.37E+02	8.37E+02	f21	2.28E+03	**2.27E+03**	2.28E+03	2.28E+03	2.28E+03
	std	3.06E+01	6.75E+01	1.29E+01	3.33E+01	4.86E+01		1.26E+01	6.28E+00	5.79E+00	6.85E+00	4.26E+00
f7	mean	9.57E+02	9.18E+02	9.47E+02	**8.56E+02**	9.33E+02	f22	2.53E+03	2.40E+03	2.40E+03	**2.40E+03**	2.40E+03
	std	7.01E+01	6.41E+01	5.80E+01	4.86E+01	2.01E+01		1.79E+02	1.29E−04	1.46E−04	1.55E−06	3.58E−05
f8	mean	8.04E+02	**8.03E+02**	8.03E+02	8.04E+02	8.03E+02	f23	2.50E+03	2.50E+03	2.55E+03	**2.50E+03**	2.50E+03
	std	2.08E+00	3.64E+00	7.78E−01	7.78E−01	2.59E+00		1.67E−03	2.55E−04	7.41E+01	6.39E−06	3.54E−06
f9	mean	5.13E+03	4.94E+03	**4.48E+03**	5.19E+03	4.33E+03	f24	2.83E+03	2.82E+03	2.82E+03	**2.82E+03**	2.82E+03
	std	3.10E+02	1.06E+03	2.59E+02	3.05E+02	7.45E+02		5.63E+00	2.31E−01	1.75E−01	3.11E−01	3.19E−01
f10	mean	1.81E+04	1.59E+04	1.48E+04	**1.34E+04**	4.45E+04	f25	3.37E+03	3.36E+03	3.35E+03	**3.35E+03**	3.38E+03
	std	1.08E+04	5.71E+03	4.95E+03	9.64E+03	2.32E+04		2.79E+01	2.62E+01	2.39E+01	2.61E+01	2.25E−02
f11	mean	6.56E+04	**3.20E+04**	3.78E+05	1.15E+05	1.20E+05	f26	3.13E+03	**3.11E+03**	3.14E+03	3.13E+03	3.14E+03
	std	1.83E+04	5.84E+03	4.62E+05	7.12E+04	1.60E+04		1.77E+01	1.19E+01	2.29E+01	2.06E+01	2.43E+01
f12	mean	1.08E+05	**7.97E+03**	1.22E+04	1.99E+04	1.17E+04	f27	2.70E+03	2.76E+03	2.70E+03	**2.70E+03**	2.71E+03
	std	1.41E+05	3.18E+03	1.06E+04	2.06E+04	5.06E+03		7.54E−03	9.05E+01	3.39E−02	9.19E−05	7.30E+00
f13	mean	2.39E+05	3.01E+05	2.38E+05	**1.52E+05**	3.06E+05	f28	2.30E+04	4.04E+04	1.54E+04	**8.47E+03**	1.48E+04
	std	1.22E+05	1.75E+04	3.30E+04	2.75E+04	2.15E+04		1.05E+04	3.58E+04	1.43E+03	1.39E+03	8.42E+03
f14	mean	3.83E+04	**1.54E+04**	1.70E+04	1.76E+04	3.24E+04	f29	9.05E+04	3.82E+04	6.26E+04	**2.20E+04**	5.79E+04
	std	2.25E+04	4.67E+03	1.93E+03	7.19E+03	1.79E+04		4.60E+04	2.04E+04	5.26E+04	1.46E+03	3.42E+04
f15	mean	5.55E+03	2.43E+03	2.38E+03	**1.77E+03**	3.29E+03						
	std	4.77E+03	3.55E+02	5.74E+02	1.95E+02	6.89E+02						

other selected algorithms are compared and analyzed in detail based on numerical statistical results, convergence behavior, and computation time.

5.2.1. Parameters analysis of elite individuals (rp)

Section 4.2.5 emphasizes that the elite individual proportion rp plays a crucial role in the elite chaotic search strategy, and its proper adjustment can significantly improve the local search capability of the algorithm. In Logistic Chaotic Elite Population, the elite probability rp is a central parameter that takes values between 0 and 1. This section aims to analyze the parameter rp by sensitivity to optimize the search efficiency of the TVESCSO algorithm. Although the value range of rp is within $(0, 1)$, too high rp may cause the algorithm to fall into a local optimum solution too early, while too low rp may limit the improvement potential of the algorithm. Therefore, this section provides an in-depth analysis of the specific impact of rp on the algorithm's performance. Table 4 shows the experimental results of the TVESCSO algorithm for different rp value settings, which are based on the data obtained from 30 independent runs performed on 29 CEC2017 benchmark functions.

By analyzing the data in Table 4, it can be seen that the TVESCSO algorithm achieves the optimal mean value for 16 out of the 29 functions tested when the proportion of elite individuals rp is set to 0.4. This outperforms the results when rp is 0.1, 0.2, 0.3, and 0.5. The worst performance of 3.45% is achieved with rp of 0.1, while the highest performance of 55.17% is achieved with rp of 0.4.

On the unimodal functions f1 and f2, increasing the value of rp enhances the convergence accuracy and stability of TVESCSO. The reason is that the unimodal functions lack local optimal solutions. Therefore, when $rp = 0.4$, the algorithm has a significant advantage in finding better potential solutions. For multimodal functions, increasing the value of rp helps to improve the algorithm's global exploration capability. For functions f4, f5, and f7, TVESCSO performs best with $rp = 0.4$; for functions f3, f6, and f8, the performance ranks second. For the mixed functions (F9–F19), TVESCSO consistently achieves the best performance at $rp = 0.4$ for functions f10, f13, f15, f17, and f18. While lower rp values may work better in some cases, a slight degradation of accuracy at higher rp values is acceptable. Among the synthetic functions (f22-f29), TVESCSO performs best at $rp = 0.4$, indicating that a moderate rp value prevents the algorithm from falling into a local optimum. Comprehensive analysis shows that TVESCSO performs best when $rp = 0.4$. Therefore, $rp = 0.4$ will be used as the parameter setting in the subsequent experiments.

5.2.2. Validation of different strategies

This study aims to evaluate the impact of different improvement strategies on the optimized search performance of the TVESCSO algorithm. To this end, we compare TVESCSO with the following three algorithms: the ESCSO, which employs exclusively the augmented SCSO population strategy; TVSCSO, which focuses on the time-varying elite population pooling strategy; and the CSCSO, which relies exclusively on the Chaotic Elite Mutation strategy. All four algorithms use the uniform parameter configurations in Table 3. To verify the effectiveness of these strategies, we evaluated the ablation performance of ESCSO, TVSCSO, CSCSO, and TVESCSO on the CEC2017 test set. Table 5 summarizes these algorithms' fitness values and variance comparison results on the CEC2017 test dataset.

According to the data analyzed in Table 5, TVESCSO, which incorporates the three strategies, performs the best in 28 test functions, while TVSCSO ranks first in 1 test function. In the unimodal function f2, TVESCSO reaches the closest solution to the theoretical value, which indicates that the adaptive time-varying elite grouping strategy plays an active role in enhancing the local search capability of SCSO. In the test of multimodal functions (f3 to f9), TVESCSO reaches the closest solution to the theoretical value in f3-f5,f8, which indicates that the adaptive time-varying elite grouping strategy effectively solves the problem of local optimal stagnation. In contrast, the Logistic Chaotic Elite Search strategy improves the algorithm's probability of finding the globally optimal solution by increasing the search depth. In the tests with hybrid and combinatorial functions (f10 to f29), TVESCSO reaches the closest solution to the theoretical value at f20 to f22 and the theoretical optimum at f27. These results show that the improved strategy effectively balances exploration and exploitation when dealing

Table 5
TVESCSO experiments with different strategies on CEC2017.

Functions	Metric	TVESCSO	ESCSO	TVSCSO	CSCSO	Functions	TVESCSO	ESCSO	TVSCSO	CSCSO
f1	mean	**1.48E+03**	6.01E+09	4.73E+08	2.76E+09	f16	7.08E+03	5.86E+04	**6.38E+03**	4.44E+04
	std	1.19E+03	1.72E+09	5.89E+08	5.56E+08		5.59E+03	1.25E+04	1.54E+03	1.75E+04
f2	mean	**2.00E+02**	3.24E+04	5.21E+03	1.76E+04	f17	**1.54E+05**	7.39E+05	3.97E+05	2.94E+05
	std	1.42E−07	1.15E+04	3.43E+03	4.40E+03		2.61E+04	1.68E+05	1.61E+05	1.69E+05
f3	mean	**3.32E+02**	1.02E+03	4.77E+02	6.16E+02	f18	**1.46E+04**	5.08E+09	9.81E+04	2.60E+09
	std	7.14E−01	6.51E+02	6.76E+01	1.01E+02		5.31E+03	2.69E+09	4.98E+04	3.51E+09
f4	mean	**4.69E+02**	7.77E+03	1.27E+03	8.42E+03	f19	**2.26E+03**	4.49E+03	2.85E+03	3.74E+03
	std	1.89E+01	3.43E+03	2.67E+02	2.18E+03		8.21E+01	2.51E+02	2.93E+01	8.92E+02
f5	mean	**5.00E+02**	5.00E+02	5.00E+02	5.00E+02	f20	**2.33E+03**	6.73E+03	2.68E+03	6.25E+03
	std	5.21E−04	6.47E−04	3.78E−04	2.06E−03		2.53E+01	9.04E+02	4.37E+02	1.31E+03
f6	mean	**7.37E+02**	1.40E+05	5.40E+04	1.57E+05	f21	**2.28E+03**	2.35E+03	2.33E+03	2.44E+03
	std	3.33E+01	6.13E+04	6.63E+04	4.12E+04		6.85E+00	3.38E+01	2.46E+01	5.16E+01
f7	mean	**8.56E+02**	1.08E+03	1.02E+03	1.13E+03	f22	**2.40E+03**	1.09E+04	2.98E+03	1.62E+04
	std	4.86E+01	3.04E+01	6.04E+01	1.45E+02		1.55E−06	4.29E+03	1.75E+02	4.05E+03
f8	mean	**8.04E+02**	8.38E+02	8.06E+02	8.10E+02	f23	**2.50E+03**	7.69E+03	5.89E+03	1.29E+04
	std	7.78E−01	7.60E+00	2.52E+00	2.46E+00		6.39E−06	2.06E+02	2.47E+03	6.03E+03
f9	mean	**5.19E+03**	6.22E+03	5.53E+03	6.00E+03	f24	**2.82E+03**	3.35E+03	2.93E+03	3.35E+03
	std	3.05E+02	2.62E+02	2.80E+02	8.59E+02		3.11E−01	1.44E+02	4.12E+01	1.43E+02
f10	mean	**1.34E+04**	6.66E+05	9.52E+04	9.44E+04	f25	**3.35E+03**	3.41E+03	3.36E+03	3.80E+03
	std	9.64E+03	4.69E+05	2.78E+04	1.83E+04		2.61E+01	5.07E+00	1.48E+01	3.18E+02
f11	mean	**1.15E+05**	9.23E+07	7.35E+06	8.09E+07	f26	**3.13E+03**	3.21E+03	3.15E+03	3.28E+03
	std	7.12E+04	4.41E+06	4.70E+06	1.71E+07		2.06E+01	3.11E+01	1.31E+01	5.46E+01
f12	mean	**1.99E+04**	4.96E+07	9.57E+05	1.02E+08	f27	**2.70E+03**	3.24E+03	3.17E+03	3.25E+03
	std	2.06E+04	5.25E+07	4.55E+05	1.21E+08		9.19E−05	5.59E+01	3.00E+01	9.43E+01
f13	mean	**1.52E+05**	2.28E+06	2.68E+05	8.00E+05	f28	**8.47E+03**	6.39E+07	9.24E+04	2.45E+08
	std	2.75E+04	2.69E+06	4.37E+04	4.58E+05		1.39E+03	5.91E+07	4.34E+04	1.70E+08
f14	mean	**1.76E+04**	1.94E+06	2.05E+05	4.82E+05	f29	**2.20E+04**	1.45E+09	2.71E+05	3.15E+08
	std	7.19E+03	7.49E+05	3.83E+04	2.04E+05		1.46E+03	1.06E+09	2.06E+04	1.08E+08
f15	mean	**1.77E+03**	1.50E+06	1.33E+06	7.83E+06					
	std	1.95E+02	1.99E+06	1.86E+06	9.12E+06					

Table 6
Results for unimodal and multimodal functions(basic algorithms) on CEC2017.

Functions	Metric	TVESCSO	SCSO	APO	SAO	RIME	INFO	BWO	AO	CHoA
f1	mean	1.48E+03	5.19E+09	1.35E+07	**7.71E+02**	7.26E+08	4.83E+03	2.66E+09	1.16E+10	3.01E+10
	std	1.19E+03	2.76E+09	8.30E+06	5.01E+02	2.81E+08	5.40E+03	6.73E+08	3.03E+09	5.39E+09
f2	mean	**2.00E+02**	1.79E+04	3.59E+02	1.42E+03	1.75E+04	2.00E+02	3.05E+04	4.68E+04	5.39E+04
	std	1.42E−07	6.83E+03	4.33E+01	6.62E+02	8.71E+03	1.53E−02	7.90E+03	1.08E+04	6.74E+03
f3	mean	**3.32E+02**	6.78E+02	4.62E+02	3.52E+02	5.97E+02	3.63E+02	6.68E+02	1.80E+03	2.69E+03
	std	7.14E−01	2.89E+02	3.49E+01	2.79E+01	1.34E+02	4.33E+01	9.36E+01	6.39E+02	4.61E+02
f4	mean	**4.69E+02**	6.90E+03	1.17E+03	5.65E+02	1.77E+03	7.52E+02	4.01E+03	1.29E+04	4.05E+04
	std	1.89E+01	3.82E+03	2.29E+02	7.57E+01	2.61E+02	1.47E+02	8.22E+02	3.55E+03	9.79E+03
f5	mean	**5.00E+02**	5.00E+02	5.00E+02	5.00E+02	5.00E+02	5.00E+02	5.00E+02	5.00E+02	5.00E+02
	std	5.21E−04	3.29E−03	2.96E−04	1.94E−05	6.62E−03	4.31E−04	3.88E−03	5.37E−03	9.19E−03
f6	mean	**7.37E+02**	1.63E+05	2.46E+03	8.43E+02	2.51E+04	1.12E+03	1.23E+05	3.63E+05	1.56E+06
	std	3.33E+01	1.35E+05	1.82E+02	2.66E+01	1.09E+04	2.52E+02	2.64E+04	1.06E+05	2.76E+05
f7	mean	**8.56E+02**	1.13E+03	9.59E+02	8.82E+02	1.44E+03	9.76E+02	1.16E+03	1.21E+03	1.21E+03
	std	4.86E+01	8.49E+01	2.97E+01	3.87E+01	1.45E+02	6.56E+01	7.45E+01	7.34E+01	5.08E+01
f8	mean	8.04E+02	8.11E+02	8.09E+02	**8.01E+02**	8.24E+02	8.06E+02	8.08E+02	8.15E+02	8.16E+02
	std	7.78E−01	3.71E+00	1.58E+00	4.35E−01	8.63E+00	3.25E+00	2.72E+00	3.29E+00	3.18E+00
f9	mean	5.19E+03	5.42E+03	3.98E+03	**3.69E+03**	6.33E+03	4.31E+03	7.33E+03	6.13E+03	8.70E+03
	std	3.05E+02	6.40E+02	8.83E+02	4.59E+02	5.89E+02	6.21E+02	7.66E+02	1.03E+03	3.28E+02

with complex optimization problems, with the adaptive time-varying elite grouping strategy playing a central role and the Logistic Chaotic Elite Search strategy significantly expanding the search range. From the analysis of the results in Table 5, it can be seen that TVESCSO integrating the three strategies significantly improves the overall performance of SCSO.

5.2.3. Assessment of the exploitative and exploratory behavior

According to the analysis in Table 6, TVESCSO outperforms all other methods in the unimodal test function f2, highlighting its superiority in optimization performance. In addition, TVESCSO also shows excellent performance in multimodal test functions f3 to f6, outperforming other methods, proving its excellence in global exploration capability and its effectiveness in complex optimization problems.

This excellent exploration capability can be attributed to the adaptive time-varying elite grouping strategy employed by TVESCSO, which allows the algorithm to adjust dynamically to the properties of different functions. Therefore, this feature of TVESCSO not only enhances its competitiveness in multimodal function optimization but also strongly supports the algorithm's overall performance.

According to the analysis of the unimodal test functions in Table 7, TVESCSO demonstrates superior performance to other methods on all test functions. This result highlights the advantages of TVESCSO in algorithm development and shows that it can converge effectively and locate the optimal solution quickly and accurately. Looking further at the results for the multimodal test functions, TVESCSO outperforms the other algorithms for the f3 to f7 functions. This excellent performance is attributed to its adaptive time-varying elite grouping strategy, significantly improving TVESCSO's global exploration ability. Thus, this feature of TVESCSO not only highlights its competitiveness in multimodal function optimization but also provides substantial proof of the algorithm's exploration capability.

The data in Table 8 reveals TVESCSO's excellent performance in balancing global search and local optimization, especially on the four

Table 7
Results for unimodal and multimodal functions (improved algorithms) on CEC2017.

Functions	Metric	TVESCSO	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f1	mean	**1.48E+03**	2.08E+10	7.37E+07	7.57E+03	5.12E+06
	std	1.19E+03	1.36E+09	3.83E+07	8.14E+03	6.60E+06
f2	mean	**2.00E+02**	7.00E+04	1.55E+04	2.00E+02	2.09E+02
	std	1.42E−07	8.21E+03	8.27E+03	2.64E−01	2.43E+00
f3	mean	**3.32E+02**	2.07E+03	6.30E+02	4.05E+02	4.66E+02
	std	7.14E−01	1.10E+03	7.19E+01	3.45E+01	1.17E+02
f4	mean	**4.69E+02**	5.32E+04	5.61E+03	6.57E+02	6.44E+02
	std	1.89E+01	1.24E+04	7.55E+02	6.42E+01	1.27E+01
f5	mean	**5.00E+02**	5.00E+02	5.00E+02	5.00E+02	5.00E+02
	std	5.21E−04	1.10E−03	1.61E−04	2.43E−03	7.37E−07
f6	mean	**7.37E+02**	1.25E+06	1.58E+05	9.23E+02	1.31E+03
	std	3.33E+01	6.13E+04	5.92E+04	1.36E+02	7.58E+01
f7	mean	**8.56E+02**	1.33E+03	1.23E+03	9.93E+02	9.13E+02
	std	4.86E+01	3.38E+01	6.54E+01	2.17E+01	1.88E+02
f8	mean	8.04E+02	8.15E+02	8.11E+02	8.02E+02	**8.01E+02**
	std	7.78E−01	3.91E+00	2.38E+00	3.74E−01	6.41E−01
f9	mean	5.19E+03	7.37E+03	5.57E+03	6.42E+03	**4.73E+03**
	std	3.05E+02	8.77E+02	1.22E+03	6.09E+02	2.10E+03

Table 8
Results for Hybrid Functions (basic algorithms) on CEC2017.

Functions	Metric	TVESCSO	SCSO	APO	SAO	RIME	INFO	BWO	AO	CHoA
f10	mean	1.34E+04	6.38E+06	2.74E+04	**1.07E+04**	2.35E+05	1.74E+04	1.37E+06	8.19E+06	9.68E+08
	std	9.64E+03	3.35E+07	4.72E+03	8.20E+03	9.36E+04	9.80E+03	8.91E+05	3.32E+07	8.68E+08
f11	mean	1.15E+05	6.71E+07	2.73E+06	6.65E+04	1.11E+08	**4.00E+04**	2.34E+08	1.14E+09	3.69E+09
	std	7.12E+04	1.20E+08	2.50E+06	3.86E+04	7.59E+07	3.76E+04	6.95E+07	1.17E+09	1.93E+09
f12	mean	1.99E+04	6.42E+07	3.49E+05	**7.62E+03**	2.11E+07	1.06E+04	7.24E+07	9.30E+08	5.79E+09
	std	2.06E+04	1.26E+08	3.82E+05	6.71E+03	1.02E+07	7.06E+03	3.50E+07	8.42E+08	3.89E+09
f13	mean	1.52E+05	5.10E+05	**8.16E+04**	2.31E+05	3.88E+06	2.23E+05	9.10E+05	2.97E+06	6.75E+06
	std	2.75E+04	4.34E+05	4.56E+04	2.68E+05	3.69E+06	1.81E+05	5.04E+05	2.39E+06	2.80E+06
f14	mean	**1.76E+04**	3.98E+07	6.17E+04	1.96E+04	8.21E+06	2.34E+04	1.83E+07	2.58E+08	1.50E+09
	std	7.19E+03	1.57E+08	1.29E+04	6.04E+03	3.96E+06	1.64E+04	8.37E+06	2.08E+08	7.75E+08
f15	mean	**1.77E+03**	1.04E+06	2.60E+03	1.85E+03	8.62E+05	2.57E+03	9.89E+05	1.09E+07	1.42E+09
	std	1.95E+02	1.45E+06	2.56E+02	2.08E+02	1.27E+06	1.62E+03	6.93E+05	1.85E+07	3.28E+09
f16	mean	7.08E+03	3.44E+04	4.64E+03	**2.20E+03**	9.73E+04	2.60E+03	9.29E+05	6.18E+08	2.31E+13
	std	5.59E+03	1.60E+04	4.80E+02	3.09E+02	4.59E+04	5.85E+02	1.30E+06	2.53E+09	1.23E+14
f17	mean	1.54E+05	6.77E+05	**4.29E+04**	7.92E+04	2.75E+06	4.94E+04	2.58E+06	1.57E+06	1.43E+07
	std	2.61E+04	6.26E+05	2.08E+04	3.45E+04	2.43E+06	2.21E+04	1.21E+06	1.12E+06	6.36E+07
f18	mean	**1.46E+04**	2.25E+09	4.36E+04	2.16E+04	5.39E+07	1.88E+04	2.62E+08	2.68E+11	8.07E+13
	std	5.31E+03	3.65E+09	5.39E+03	1.59E+04	1.46E+08	1.88E+04	1.86E+08	1.29E+12	1.01E+14
f19	mean	**2.26E+03**	3.52E+03	2.64E+03	2.26E+03	6.31E+03	2.32E+03	3.79E+03	7.01E+03	5.82E+03
	std	8.21E+01	6.48E+02	2.11E+02	1.04E+02	1.47E+03	1.90E+02	5.63E+02	2.56E+03	6.36E+02

functions f14, f15, f18, and f19. However, in function f10, TVESCSO's convergence accuracy needs to be improved, and the SAO algorithm needs to be outperformed.

Further analyzing Table 9, TVESCSO continues its excellent performance on the f10, f14, f15, and f19 functions. However, its convergence accuracy is slightly less than that of the ALSHADE algorithm on the f11, f12, f16 and f18 functions. These findings indicate that despite TVESCSO's outstanding performance on specific functions, there is still room for improvement in the balance between global search and local optimization. Therefore, future work needs to further optimize TVESCSO for more comprehensive search capabilities and more accurate local optimization.

5.2.4. Assessment of ability escaping from local optimum

The test functions f20 to f29 are dedicated to evaluating the algorithm's ability to avoid falling into local optima and to test its balance between exploration and exploitation. The results in Tables 10 and 11 show that TVESCSO exhibits significant competitiveness on functions f21 to f24 and f26 to f29. These results indicate that TVESCSO balances exploration and exploitation and effectively avoids local optimal solutions. This excellent performance of TVESCSO is attributed to the critical contributions of its adaptive time-varying elite grouping strategy and Logistic Chaotic Elite Search strategy, which facilitate the algorithm's effective coordination between global search and local exploitation. The algorithm is effectively prevented from falling into local optimal solutions.

The test results in Tables 6 to 11 show that TVESCSO can effectively avoid falling into local minima and find optimal solutions in the solution space.

5.2.5. Friedman test and nemenyi analysis

The Friedman test for the data in Tables 6 to 11 yields a p-value of only 2.16e−55, much lower than the generally accepted significance level of 0.05. This significant p-value implies that the difference in performance between the algorithms is statistically significant at the set significance level. Further analyzing the average ranking of Friedman's test in Fig. 4, the TVESCSO algorithm tops the list with a score of 2.28.

From the analysis of the Nemenyi test in Fig. 5, it can be seen that SAO is closer to TVESCSO, indicating that they have similar optimization capabilities.ALSHADE, INFO, LSHADE_cnEpSin, and APO have a specific performance gap with TVESCSO, but the gap is small. Among them, the difference between LSHADE_cnEpSin and APO is slightly more significant. DETDO, BWO, RIME, SCSO, AO, MSGWO, and CHoA have a significant gap compared to TVESCSO, especially CHoA, which has the most significant average ranking difference. Overall, regarding average ranking difference, TVESCSO performs better among these algorithms, with a smaller and more significant performance gap with some algorithms.

Table 9
Results for Hybrid Functions (improved algorithms) on CEC2017.

Functions	Metric	TVESCSO	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f10	mean	**1.34E+04**	1.34E+08	7.19E+04	1.99E+04	7.85E+04
	std	9.64E+03	1.27E+08	6.54E+03	7.57E+03	2.96E+04
f11	mean	1.15E+05	8.69E+08	5.54E+07	**1.97E+04**	8.39E+06
	std	7.12E+04	1.08E+09	5.29E+07	7.10E+03	8.19E+06
f12	mean	1.99E+04	3.56E+09	1.45E+05	**1.03E+04**	9.79E+04
	std	2.06E+04	4.64E+09	6.66E+04	3.56E+03	4.48E+04
f13	mean	1.52E+05	4.28E+06	1.39E+06	**1.73E+04**	1.47E+05
	std	2.75E+04	1.46E+06	8.91E+05	2.06E+04	2.95E+04
f14	mean	**1.76E+04**	3.91E+08	7.18E+04	1.88E+04	7.45E+04
	std	7.19E+03	5.07E+08	1.52E+04	8.63E+03	1.59E+04
f15	mean	**1.77E+03**	9.59E+05	1.70E+04	1.97E+03	5.09E+03
	std	1.95E+02	6.73E+05	2.80E+03	5.46E+02	8.07E+02
f16	mean	7.08E+03	8.57E+04	8.56E+04	**1.88E+03**	2.37E+04
	std	5.59E+03	4.10E+04	6.95E+04	1.19E+02	1.47E+04
f17	mean	1.54E+05	5.75E+05	2.66E+05	**2.24E+04**	1.44E+05
	std	2.61E+04	5.33E+04	2.09E+05	1.37E+04	6.77E+04
f18	mean	1.46E+04	2.22E+10	1.24E+08	**3.94E+03**	8.23E+04
	std	5.31E+03	2.79E+10	1.75E+08	1.76E+03	3.21E+04
f19	mean	**2.26E+03**	7.11E+03	3.47E+03	2.48E+03	2.32E+03
	std	8.21E+01	5.66E+03	3.10E+02	3.66E+02	1.03E+02

Table 10
Results for composition functions (basic algorithms) on CEC2017.

Functions	Metric	TVESCSO	SCSO	APO	SAO	RIME	INFO	BWO	AO	CHoA
f20	mean	2.33E+03	5.21E+03	2.34E+03	**2.15E+03**	3.49E+03	2.36E+03	4.10E+03	9.48E+03	2.57E+04
	std	2.53E+01	2.93E+03	1.94E+02	7.24E+01	4.08E+02	1.81E+02	7.62E+02	3.91E+03	8.02E+03
f21	mean	**2.28E+03**	2.41E+03	2.32E+03	2.28E+03	2.51E+03	2.29E+03	2.35E+03	2.67E+03	2.92E+03
	std	6.85E+00	6.44E+01	2.13E+01	1.00E+01	6.06E+02	1.26E+01	1.60E+01	2.55E+02	4.03E+02
f22	mean	**2.40E+03**	1.06E+04	2.91E+03	2.40E+03	5.29E+03	2.54E+03	8.94E+03	1.74E+04	3.60E+04
	std	1.55E−06	3.45E+03	1.31E+02	5.99E−03	8.27E+02	3.21E+02	1.13E+03	3.22E+03	8.63E+03
f23	mean	**2.50E+03**	9.52E+03	3.03E+03	2.50E+03	4.25E+03	2.50E+03	7.05E+03	1.32E+04	2.58E+04
	std	6.39E−06	4.16E+03	2.65E+02	1.24E−02	4.23E+02	9.98E−02	6.60E+02	4.82E+03	4.01E+03
f24	mean	**2.82E+03**	3.11E+03	2.88E+03	2.82E+03	3.05E+03	3.10E+03	3.01E+03	3.50E+03	4.51E+03
	std	3.11E−01	1.19E+02	3.46E+01	3.50E−01	1.07E+02	9.50E+01	4.11E+01	2.95E+02	3.88E+02
f25	mean	3.35E+03	3.50E+03	3.35E+03	3.35E+03	3.44E+03	**3.33E+03**	3.40E+03	3.96E+03	5.02E+03
	std	2.61E+01	1.42E+02	2.55E+01	1.87E+01	6.47E+01	1.40E+01	1.44E+01	6.03E+02	8.10E+02
f26	mean	**3.13E+03**	3.26E+03	3.19E+03	3.14E+03	3.37E+03	3.14E+03	3.16E+03	3.28E+03	3.58E+03
	std	2.06E+01	5.12E+01	1.85E+01	5.78E+00	9.78E+01	2.09E+01	1.51E+01	6.48E+01	9.57E+01
f27	mean	**2.70E+03**	3.26E+03	2.98E+03	2.95E+03	3.22E+03	2.86E+03	3.27E+03	3.50E+03	5.10E+03
	std	9.19E−05	7.26E+01	1.34E+02	1.81E+02	2.91E+01	1.49E+02	4.63E+01	1.63E+02	7.75E+02
f28	mean	**8.47E+03**	1.99E+08	4.27E+04	2.55E+04	2.34E+08	1.27E+04	5.75E+07	3.71E+09	3.12E+09
	std	1.39E+03	2.05E+08	1.29E+04	2.72E+04	5.95E+08	5.98E+03	6.09E+07	1.22E+10	2.81E+09
f29	mean	**2.20E+04**	3.52E+08	2.50E+05	4.99E+04	3.17E+08	3.37E+04	8.92E+07	7.95E+09	1.54E+11
	std	1.46E+03	3.33E+08	2.37E+05	5.12E+04	2.92E+08	1.88E+04	4.94E+07	3.70E+10	2.86E+11

Table 11
Results for composition functions (improved algorithms) on CEC2017.

Functions	TVESCSO	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f20	2.33E+03	2.68E+04	4.75E+03	**2.21E+03**	2.46E+03
	2.53E+01	8.83E+03	3.16E+02	1.50E+02	3.63E+01
f21	**2.28E+03**	2.66E+03	4.14E+03	2.29E+03	2.28E+03
	6.85E+00	2.99E+01	1.88E+03	1.63E+01	6.32E+00
f22	**2.40E+03**	3.10E+04	7.00E+03	2.61E+03	2.82E+03
	1.55E−06	1.04E+04	5.52E+02	3.00E+02	5.72E+02
f23	**2.50E+03**	2.59E+04	5.33E+03	2.63E+03	2.60E+03
	6.39E−06	4.84E+03	7.14E+02	1.87E+02	1.05E+01
f24	**2.82E+03**	3.97E+03	2.97E+03	2.83E+03	2.83E+03
	3.11E−01	2.67E+02	3.49E+01	1.17E+01	6.90E+00
f25	3.35E+03	5.37E+03	3.60E+03	**3.33E+03**	3.38E+03
	2.61E+01	1.92E+03	2.73E+01	4.01E+00	6.96E+01
f26	**3.13E+03**	3.19E+03	3.41E+03	3.13E+03	3.15E+03
	2.06E+01	1.05E+01	1.84E+02	1.97E+01	1.85E+01
f27	**2.70E+03**	3.61E+03	3.34E+03	2.75E+03	2.90E+03
	9.19E−05	7.94E+01	1.18E+01	1.58E+01	1.81E+02
f28	**8.47E+03**	4.74E+11	4.19E+07	1.27E+04	2.37E+05
	1.39E+03	6.69E+11	4.19E+07	7.97E+03	1.12E+05
f29	**2.20E+04**	1.18E+09	5.00E+07	3.24E+04	1.03E+06
	1.46E+03	7.82E+08	6.99E+07	1.84E+04	9.31E+05

5.2.6. Wilcoxon rank sum test

From Table 12 Wilcoxon test, it can be seen that in the comparison of TVESCSO with SCSO, RIME, BWO, AO, CHoA, MSGWO, and DETDO, the p-values are all extremely small, and much less than the 0.05 significance level, which indicates a significant difference in performance between TVESCSO and these algorithms. Regarding positive and negative rank sums, the positive rank sums are all 0 or close to 0. In contrast, the negative rank sums are more significant, indicating that TVESCSO outperforms these algorithms in most cases. In comparing TVESCSO with APO and LSHADE_cnEpSin, the same is less than 0.05, and there is a significant difference. The positive rank sum of APO is 86.00, and the negative rank sum is 349.00, while the positive rank sum of LSHADE_cnEpSin is 59.00 and the negative rank sum is 376.00, which suggests that TVESCSO also outperforms these algorithms in terms of performance. The p-values in the comparison of TVESCSO with SAO, INFO, and ALSHADE are all greater than 0.05, indicating that at the current significance level, the performance difference between TVESCSO and these algorithms is insignificant. In terms of positive and negative rank sums, these algorithms' positive and negative rank sums are relatively more balanced, with no apparent bias.

5.2.7. Analysis of convergence

Figs. 6 to 9 present the convergence curves of the TVESCSO algorithm with other algorithms on several test functions. In the graphs,

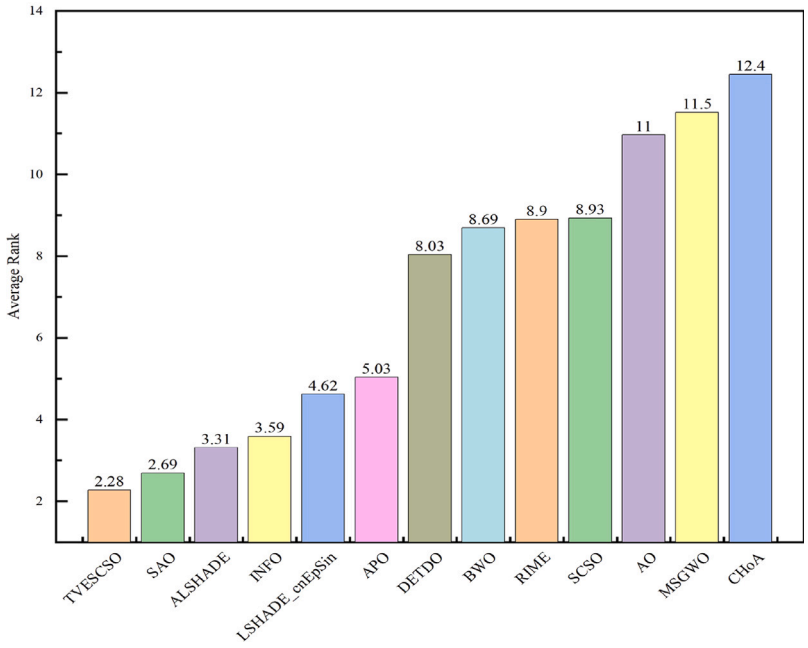


Fig. 4. Friedman mean ranking for different Algorithm on CEC2017.

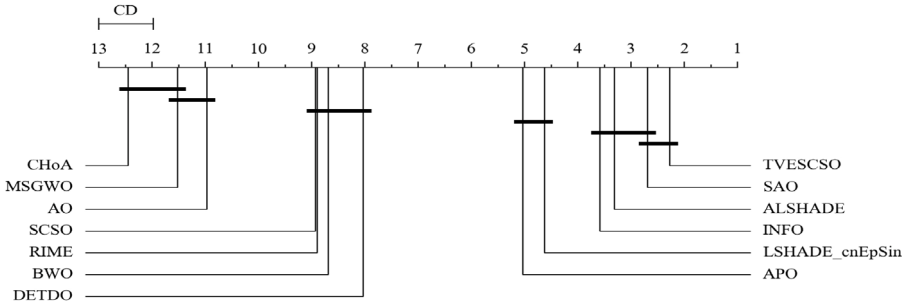


Fig. 5. Results of the Nemenyi test comparing TVESCSO with other algorithms on CEC2017.

Table 12
Wilcoxon rank-sum test on CEC2017.

Algorithm	Positive sum	Negative sum	*p*-value	Algorithm	Positive sum	Negative sum	*p*-value
TVESCSO VS SCSO	0	435	2.56E−06	TVESCSO VS AO	0	435	2.56E−06
TVESCSO VS APO	86	349	4.46E−03	TVESCSO VS CHoA	0	435	2.56E−06
TVESCSO VS SAO	181	254	4.30E−01	TVESCSO VS MSGWO	0	435	2.56E−06
TVESCSO VS RIME	0	435	2.56E−06	TVESCSO VS DETDO	1	434	2.85E−06
TVESCSO VS INFO	131	304	6.14E−02	TVESCSO VS ALSHADE	173	262	3.36E−01
TVESCSO VS LSHADE_cnEpSin	59	376	6.10E−04	TVESCSO VS BWO	0	435	2.56E−06

the horizontal axis represents the number of iterations. The vertical axis represents the logarithmic value of each algorithm's average optimal solution obtained after each iteration, with a base of 10. Regarding the single-peak functions, as shown in Fig. 6, TVESCSO quickly approximates the global optimal solution in the f2 function. Fig. 7 shows that TVESCSO also converges quickly for the f4 function, while it takes about 100 iterations for the f9 function to approach the optimal solution. In Figs. 8 and 9, TVESCSO significantly improves its performance with the adaptive time-varying elite grouping strategy and the logistic chaotic elite search strategy for complex multimodal and composite functions such as f14, f15, f27, and f29. Fig. 9 shows that TVESCSO requires approximately 400 iterations to converge to the optimal solution for the f27 function and about 600 iterations for the f29 function. In addition, the smoothness of the convergence curves further validates the efficiency of the chaotic elite search strategy in solution optimization. It highlights the excellent performance of the algorithm in dealing with composite functions. These plots show that TVESCSO performs better by configuring specific mechanisms for different phases to efficiently explore and exploit the search space. In addition, the stability of the convergence curve in the final stage demonstrates both the convergence of the algorithm and the robustness of its search process. The results show that TVESCSO's design strategy is effective in locating globally optimal solutions for both single-peak and simple multi-peak functions and that its adaptive time-varying elite grouping strategy and Logistic Chaotic Elite Search strategy perform equally well when dealing with more challenging multi-peak and composite functions, suggesting that real-world performance at each definitional stage is both observable and efficacious.

5.2.8. *Time analysis of TVESCSO*

Table 13 compares in detail the actual running times of the TVESCSO algorithm with the twelve algorithms SCSO, APO, SAO,

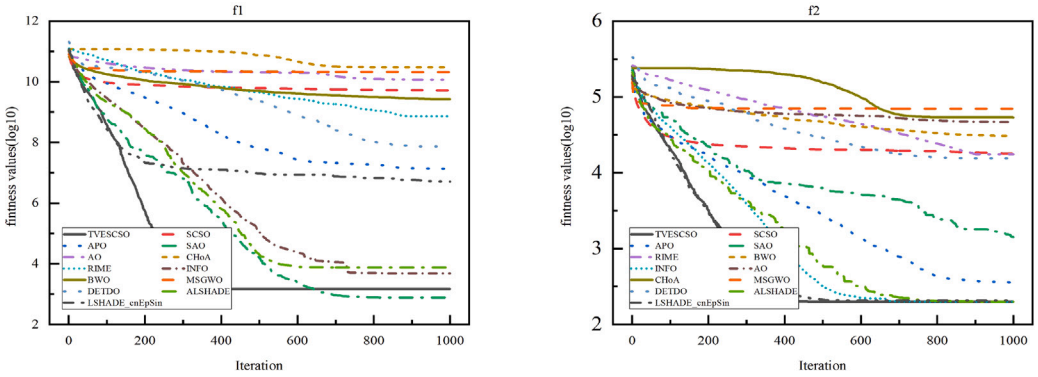


Fig. 6. TVESCSO averaged convergence curve on different Unimodal Functions on CEC2017.

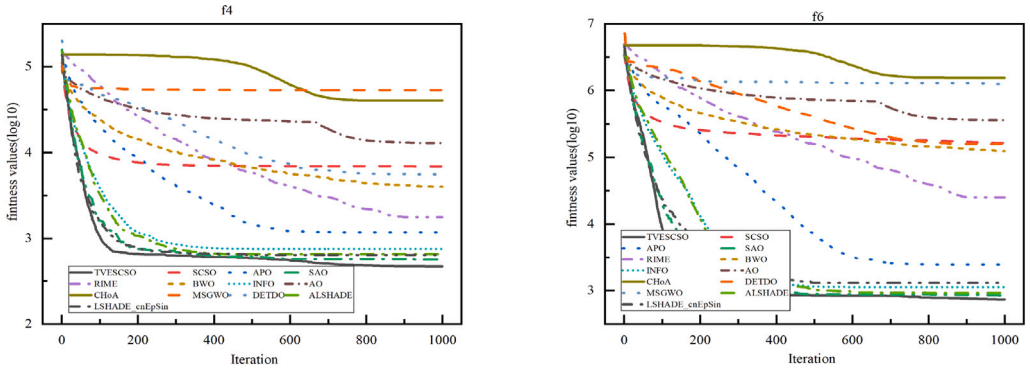


Fig. 7. TVESCSO averaged convergence curve on different Simple Multimodal Functions on CEC2017.

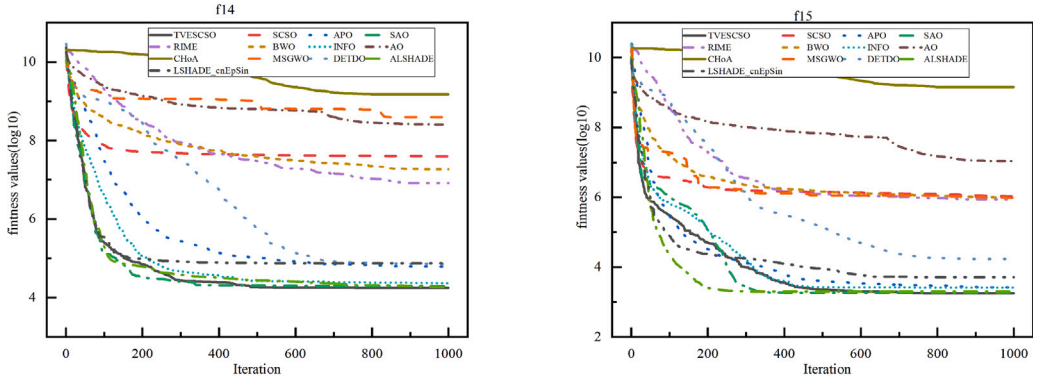


Fig. 8. TVESCSO averaged convergence curve on different Hybrid Functions on CEC2017.

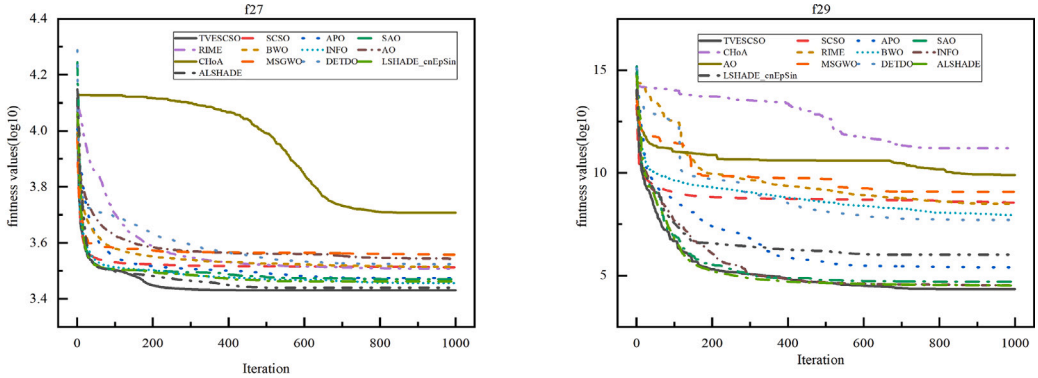


Fig. 9. TVESCSO averaged convergence curve on different Composition Functions on CEC2017.

Table 13
Running time(/s) of TVESCSO and different algorithm on CEC2017.

Functions	TVESCSO	SCSO	APO	SAO	RIME	BWO	INFO	AO	CHoA	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f1	7.64	7.38	8.94	3.89	4.68	4.92	11.85	3.44	18.45	29.36	8.12	4.64	5.15
f2	38.45	7.55	9.61	4.07	4.93	5.21	13.04	3.93	18.89	41.16	9.39	4.33	5.25
f3	27.47	8.16	10.15	4.13	5.08	5.42	13.39	4.19	18.82	31.64	7.62	5.24	5.39
f4	24.03	7.58	9.42	3.89	4.87	5.05	12.40	3.80	18.52	25.58	6.13	3.11	3.45
f5	25.62	12.44	21.14	6.93	9.04	9.67	18.73	12.23	22.67	107.92	16.58	12.46	9.28
f6	24.54	8.90	12.74	4.84	5.71	6.13	13.22	5.42	19.33	43.69	8.16	6.63	6.31
f7	26.32	10.08	15.79	5.00	7.29	8.19	15.90	8.75	20.63	83.06	13.40	6.69	5.70
f8	24.39	8.21	12.06	4.66	5.42	5.76	13.85	5.01	18.99	40.15	7.56	3.73	3.81
f9	41.63	9.71	13.83	4.63	6.42	7.02	16.36	7.07	19.41	62.75	9.59	5.13	4.49
f10	40.38	9.66	14.31	5.36	6.75	7.39	16.09	7.25	21.31	57.15	9.06	5.64	5.14
f11	39.28	11.25	20.97	6.92	7.73	8.69	19.49	9.64	20.92	86.78	11.84	6.47	5.16
f12	24.56	9.76	10.63	3.78	6.87	7.51	15.42	7.91	20.60	65.55	9.85	5.42	4.72
f13	25.14	10.92	12.10	4.21	8.09	8.92	16.82	9.97	22.26	101.92	14.52	6.28	4.99
f14	25.98	9.45	10.64	3.77	6.90	7.51	16.06	7.92	20.43	74.91	12.02	6.31	5.13
f15	48.76	16.24	37.43	11.75	13.03	14.50	25.42	20.14	26.35	207.83	25.72	12.32	7.94
f16	85.45	45.68	129.06	37.52	40.37	44.05	71.71	73.24	53.29	794.96	78.74	54.51	29.93
f17	33.30	10.62	20.58	7.00	7.23	8.06	18.99	9.11	19.88	92.03	13.68	6.18	4.97
f18	35.72	32.21	47.03	12.15	19.70	21.96	38.14	33.65	32.57	322.70	34.23	22.87	15.29
f19	53.61	26.34	100.18	29.15	25.23	26.75	72.18	44.36	33.73	747.75	73.96	46.32	24.91
f20	26.93	11.68	23.44	7.83	8.93	9.70	21.10	11.72	22.58	154.84	20.92	7.58	5.60
f21	29.04	12.83	29.84	9.57	10.63	11.44	23.39	12.90	24.63	155.47	19.80	10.00	6.70
f22	29.97	14.93	32.04	9.74	12.12	13.57	25.67	18.25	25.44	187.67	23.27	11.71	7.63
f23	28.89	13.38	29.50	9.59	10.46	11.60	22.39	15.08	23.72	145.74	19.23	9.79	6.85
f24	29.83	14.46	32.79	10.57	11.63	12.74	22.91	17.30	25.10	196.87	23.59	10.24	6.85
f25	47.47	38.61	68.25	20.73	35.15	38.82	64.48	62.76	48.55	665.23	66.40	43.74	25.22
f26	46.60	41.24	87.27	23.77	38.48	43.14	66.20	71.69	52.97	593.20	59.84	62.35	34.38
f27	49.49	40.51	73.91	22.05	37.06	41.57	61.57	68.47	51.38	608.87	74.09	36.68	20.30
f28	97.78	62.70	137.73	36.43	56.59	63.00	92.92	107.13	70.22	995.54	96.41	61.10	32.19
f29	87.07	33.20	92.64	25.11	28.96	32.52	51.32	52.20	42.24	524.81	52.48	51.41	36.95
Average	7.64	7.38	8.94	3.77	4.68	4.92	11.85	3.44	18.45	25.58	6.13	3.11	3.45
Rank	9	8	10	4	5	6	11	2	12	13	7	1	3

RIME, BWO, INFO, AO, CHoA, MSGWO, DETDO, ALSHADE and LSHADE_cnEpSin on the CEC2017 test functions. The data show that the average running times (in seconds) of these algorithms are: 7.64 s for TVESCSO, 7.38 s for SCSO, 8.94 s for APO, 3.77 s for SAO, 4.68 s for RIME, 4.92 s for BWO, 11.85 s for INFO, 3.44 s for AO, 18.45 s for CHoA, MSGWO 25.58 s, DETDO 6.13 s, ALSHADE 3.11 s, and LSHADE_cnEpSin 3.45 s. These data reveal the differences in computational efficiency among the algorithms. According to the ranking in Table 13, TVESCSO ranked 9th in terms of running time, indicating that its running time is within acceptable limits.

5.3. CEC2022 benchmark function results comparison and analysis

To further demonstrate the performance of the proposed method, we select CEC2022 in the numerical optimization competition. As can be seen from Table 2, it contains 12 functions, half of which belong to the challenging hybrid and combinatorial functions. We tested the TVESCSO algorithm against these functions against SCSO, APO, SAO, RIME, BWO, INFO, AO, CHoA, MSGWO, DETDO, ALSHADE, and LSHADE_cnEpSin. The dimensionality of the test functions for the CEC2022 was set to 10 and 20, respectively, and each method was run 30 times with a 1000 iteration limit.

5.3.1. Assessment of TVESCSO on CEC2022 test functions

According to the results in Table 14, the TVESCSO algorithm excelled in the test of 12 benchmark functions, and it achieved the best average fitness in 11 functions with a percentage of 92.67%. This percentage is significantly better than the 16.67% of the APO and SAO algorithms. It is worth noting that the other algorithms failed to achieve the best performance for any of the benchmark functions in the test with a percentage of 0%. Further analysis of Table 15 shows that the TVESCSO algorithm leads with 75%, significantly outperforming ALSHADE with 58.33%, LSHADE_cnEpSin and DETDO with 8.33%, and all other algorithms with 0% performance.

The data in Table 16 shows that the TVESCSO algorithm again performs excellently in the 12 benchmark functions, occupying the best average fitness in 9 functions with a percentage of 75%. This performance is significantly better than the SAO algorithm at 25% and the APO algorithm at 8.33%. Meanwhile, the other algorithms failed to perform best in all benchmark functions, with the percentage remaining at 0%. Finally, the results in Table 17 show that the TVESCSO algorithm again leads with an excellent performance of 75% compared to 25% for the ALSHADE algorithm. All other algorithms failed to achieve the best performance on any benchmark function with a percentage of 0%.

In summary, although the solution accuracy of all algorithms decreases when the dimension of the test function increases, TVESCSO still outperforms the other algorithms on most functions. This is mainly due to the adaptive time-varying elite grouping strategy, which improves the global search efficiency during the iteration process, and the Logistic Chaotic Elite Search Strategy mechanism, which enhances the local search accuracy and prevents the algorithm from falling into the local optimum prematurely.

5.3.2. Friedman test and Nemenyi analysis

Fig. 10 clearly shows the average rankings of TVESCSO and competing algorithms on the 10- and 20-dimensional Friedman tests. As shown in Fig. 10, the average ranking values of TVESCSO are 1.96 and 2.13 on 10 and 20 dimensions, respectively, which are lower than the other algorithms, thus highlighting its performance advantage on large-scale optimization problems. In the Nemenyi statistical analysis, TVESCSO is the best-performing algorithm when Dim = 10. ALSHADE is similar to TVESCSO in terms of performance. The rest of the algorithms significantly underperform TVESCSO, as shown in Fig. 11. TVESCSO is the best-performing algorithm when Dim = 20. ALSHADE and SAO are similar to TVESCSO in terms of performance. The remaining algorithms

Table 14
Results for CEC2022 test functions (basic algorithms, Dim = 10).

Functions	Metric	TVESCSO	SCSO	APO	SAO	RIME	INFO	BWO	AO	CHoA
f1	mean	**3.00E+02**	4.03E+02	3.00E+02	**3.00E+02**	3.12E+02	3.00E+02	3.21E+02	3.50E+03	1.39E+03
	std	0.00E+00	5.86E+01	1.90E−03	0.00E+00	8.45E+00	6.31E−14	6.79E+00	2.14E+03	9.43E+02
f2	mean	**4.04E+02**	4.08E+02	4.06E+02	4.06E+02	4.31E+02	4.15E+02	4.20E+02	4.20E+02	8.51E+02
	std	4.46E+00	3.04E+00	2.27E+00	4.13E+00	3.01E+01	2.26E+01	1.18E+01	1.24E+01	1.95E+02
f3	mean	**6.00E+02**	6.00E+02	**6.00E+02**	**6.00E+02**	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02
	std	0.00E+00	1.47E−02	7.20E−11	0.00E+00	1.96E−05	9.74E−14	6.10E−05	4.02E−03	5.29E−02
f4	mean	**8.21E+02**	8.29E+02	8.39E+02	8.30E+02	9.24E+02	8.40E+02	8.51E+02	8.91E+02	8.51E+02
	std	5.00E+00	7.36E+00	1.23E+01	7.12E+00	3.12E+01	1.32E+01	6.84E+00	2.10E+01	3.26E+00
f5	mean	**9.00E+02**	9.01E+02	9.00E+02	9.00E+02	9.02E+02	9.00E+02	9.00E+02	9.01E+02	9.00E+02
	std	0.00E+00	4.62E−01	2.99E−01	4.22E−02	1.96E+00	4.15E−01	2.20E−01	2.83E−01	5.22E−02
f6	mean	2.06E+04	5.83E+04	**2.70E+03**	3.01E+04	4.16E+04	2.20E+04	1.64E+05	5.81E+04	1.27E+06
	std	9.14E+03	2.53E+04	6.81E+02	1.10E+04	6.65E+03	1.34E+04	1.38E+05	2.47E+04	1.99E+05
f7	mean	**2.03E+03**	2.07E+03	2.04E+03	2.06E+03	2.12E+03	2.03E+03	2.06E+03	2.15E+03	2.28E+03
	std	5.58E+00	4.05E+01	1.60E+01	5.63E+01	5.17E+01	1.98E+01	1.98E+00	8.24E+01	2.40E+01
f8	mean	**2.22E+03**	2.24E+03	2.22E+03	2.22E+03	2.51E+03	2.24E+03	2.34E+03	2.82E+03	3.47E+03
	std	7.14E+00	3.37E+00	6.30E−01	2.64E+00	3.26E+02	7.46E+01	6.77E+01	8.07E+02	7.32E+02
f9	mean	**2.30E+03**	2.72E+03	2.30E+03	2.66E+03	2.67E+03	2.48E+03	2.43E+03	2.59E+03	3.25E+03
	std	3.22E−13	1.87E+01	2.28E−02	2.93E−05	6.33E+00	1.77E+02	1.72E+02	1.82E+02	2.18E+02
f10	mean	**2.62E+03**	2.77E+03	2.62E+03	2.65E+03	2.80E+03	2.73E+03	2.63E+03	2.70E+03	2.65E+03
	std	4.88E−01	1.85E+01	1.44E+00	4.37E+01	1.78E+02	1.70E+02	1.07E+00	6.46E+01	5.38E+00
f11	mean	**2.60E+03**	2.61E+03	2.60E+03	3.48E+03	2.90E+03	2.66E+03	2.60E+03	2.90E+03	3.40E+03
	std	0.00E+00	8.34E+00	4.07E−04	4.03E−01	4.08E+02	2.18E+02	2.68E−01	4.10E+02	3.34E+02
f12	mean	**2.87E+03**	2.88E+03	2.87E+03	2.87E+03	2.89E+03	2.87E+03	2.87E+03	2.93E+03	2.93E+03
	std	4.01E−01	3.32E+00	5.32E+00	3.98E+00	3.28E+01	8.45E+00	1.06E−01	2.36E+01	1.58E+01

Table 15
Results for CEC2022 test functions (improved algorithms, Dim = 10).

Functions	Metric	TVESCSO	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f1	mean	**3.00E+02**	4.81E+02	**3.00E+02**	**3.00E+02**	3.02E+02
	std	0.00E+00	1.48E+01	3.90E−07	0.00E+00	1.34E+00
f2	mean	4.04E+02	4.44E+02	4.26E+02	**4.01E+02**	4.06E+02
	std	4.46E+00	3.51E+01	2.00E+01	1.88E+00	4.20E+00
f3	mean	**6.00E+02**	6.00E+02	**6.00E+02**	**6.00E+02**	**6.00E+02**
	std	0.00E+00	6.21E−02	3.23E−11	0.00E+00	1.73E−08
f4	mean	**8.21E+02**	8.56E+02	8.43E+02	8.34E+02	8.13E+02
	std	5.00E+00	1.53E+01	3.77E+00	1.25E+01	3.40E+00
f5	mean	**9.00E+02**	9.02E+02	9.01E+02	**9.00E+02**	9.00E+02
	std	0.00E+00	7.17E−01	4.08E−01	0.00E+00	4.11E−05
f6	mean	2.06E+04	2.14E+04	2.94E+04	**1.80E+03**	7.39E+03
	std	9.14E+03	1.70E+03	8.15E+03	2.09E+00	5.13E+03
f7	mean	**2.03E+03**	2.26E+03	2.09E+03	2.03E+03	2.03E+03
	std	5.58E+00	5.43E+01	3.46E+01	4.43E+00	2.50E−01
f8	mean	**2.22E+03**	2.23E+03	2.59E+03	2.22E+03	2.22E+03
	std	7.14E+00	3.19E+00	4.44E+02	3.54E+00	6.00E−01
f9	mean	**2.30E+03**	2.65E+03	2.70E+03	2.54E+03	2.42E+03
	std	3.22E−13	8.88E+01	2.67E+01	1.69E+02	1.70E+02
f10	mean	2.62E+03	2.76E+03	2.72E+03	**2.62E+03**	2.66E+03
	std	4.88E−01	2.54E+01	6.10E+01	1.06E+00	4.96E+01
f11	mean	**2.60E+03**	2.69E+03	2.60E+03	**2.60E+03**	2.60E+03
	std	0.00E+00	1.19E+02	1.12E+00	0.00E+00	1.16E+00
f12	mean	**2.87E+03**	2.89E+03	2.89E+03	2.87E+03	2.87E+03
	std	4.01E−01	2.39E+01	1.40E+01	2.49E+00	3.38E−01

are significantly worse than TVESCSO in terms of performance, as shown in Fig. 12.

5.3.3. Wilcoxon rank sum test

As seen from Table 18, TVESCSO is significantly different from most of the other algorithms in comparison with them when Dim = 10, indicating that TVESCSO is substantially different from these algorithms in terms of performance. Significant differences between APO, ALSHADE, and LSHADE_cnEpSin are insignificant, indicating their performance is closer. When Dim = 20, TVESCSO may perform more closely than APO, SAO, ALSHADE, and LSHADE_cnEpSin. TVESCSO significantly differs from SCSO, RIME, INFO, BWO, AO, CHoA, and MSGWO.

6. TVESCSO algorithm for solving engineering design optimization problems

When solving engineering design problems, the complexity of the objective function and the number of constraints are often critical factors. To evaluate the performance of the TVESCSO algorithm, four engineering design problems with different complexities were selected for testing, and these test cases showed significant differences in the number of design variables and constraints. The complexity increases with the number of design variables and constraints, ranging from a simple problem with two design variables and three constraints to a complex problem with 48 design variables and 38 constraints. Due to only two design variables, the three bar truss problem is more straightforward among these test cases. The piston lever design problem and the welded beam problem increased to four design variables, making the problems more complex. In contrast, the optimal design problem for the propane, isobutane, n-Butane nonsharp separation not only has a large number of design variables but also contains a large number of constraints, constituting a highly complex constrained optimization problem, which poses a severe challenge to the performance and stability of the algorithm. Therefore, the performance and stability of the algorithm are of particular importance.

Table 16
Results for CEC2022 test functions (basic algorithms, Dim = 20).

Functions	Metric	TVESCSO	SCSO	APO	SAO	RIME	INFO	BWO	AO	CHoA
f1	mean	**3.00E+02**	2.69E+03	3.04E+02	3.00E+02	1.41E+03	3.00E+02	1.57E+04	1.65E+04	1.57E+04
	std	0.00E+00	1.25E+03	2.49E+00	8.11E−08	3.16E+02	4.46E−10	5.23E+03	3.00E+03	3.88E+03
f2	mean	**4.41E+02**	5.48E+02	4.54E+02	4.63E+02	5.38E+02	4.54E+02	5.51E+02	8.19E+02	9.05E+02
	std	2.87E+01	3.72E+01	2.73E+00	1.01E+01	4.63E+01	1.91E+01	8.14E+01	1.01E+02	1.18E+02
f3	mean	**6.00E+02**	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.00E+02	6.01E+02
	std	0.00E+00	6.02E−02	1.70E−05	0.00E+00	2.61E−03	1.62E−12	6.80E−03	1.30E−01	9.56E−02
f4	mean	**8.78E+02**	9.92E+02	9.05E+02	9.25E+02	1.10E+03	9.25E+02	1.04E+03	9.85E+02	1.05E+03
	std	6.53E+00	2.91E+01	3.72E+01	5.60E+01	1.68E+01	4.88E+01	2.72E+01	3.57E+01	1.92E+01
f5	mean	**9.00E+02**	9.04E+02	9.03E+02	9.01E+02	9.12E+02	9.02E+02	9.02E+02	9.04E+02	9.05E+02
	std	1.59E−01	9.99E−01	9.67E−01	2.38E−01	3.98E+00	1.71E+00	9.44E−01	1.24E+00	2.36E+00
f6	mean	4.64E+04	1.27E+05	**3.32E+04**	6.13E+04	4.87E+05	6.77E+04	1.57E+07	4.99E+07	4.25E+08
	std	1.58E+04	1.15E+05	1.54E+04	2.30E+04	2.82E+05	2.53E+04	2.90E+06	5.61E+07	2.28E+08
f7	mean	2.03E+03	2.33E+03	2.15E+03	**2.03E+03**	2.59E+03	2.05E+03	2.30E+03	2.65E+03	3.48E+03
	std	2.82E+00	1.83E+02	1.98E+01	2.67E+00	3.85E+02	2.41E+01	8.53E+01	2.48E+02	1.07E+03
f8	mean	2.65E+03	3.00E+03	2.44E+03	**2.27E+03**	5.98E+03	2.33E+03	3.90E+04	4.39E+03	6.63E+03
	std	3.81E+02	5.17E+02	6.20E+01	1.16E+01	2.48E+03	1.35E+02	4.58E+03	5.60E+02	1.67E+03
f9	mean	**2.64E+03**	2.74E+03	2.65E+03	2.64E+03	2.79E+03	2.64E+03	2.74E+03	2.94E+03	3.39E+03
	std	1.09E−01	5.97E+01	7.12E+00	2.99E−04	4.76E+01	1.14E+00	2.14E+01	1.41E+02	1.18E+02
f10	mean	**2.80E+03**	2.87E+03	2.83E+03	3.90E+03	5.69E+03	4.01E+03	2.82E+03	5.01E+03	6.51E+03
	std	1.41E+00	2.21E+01	1.02E+01	8.16E+02	6.52E+02	9.60E+02	3.28E+00	1.46E+03	2.33E+02
f11	mean	**2.60E+03**	3.19E+03	2.60E+03	**2.60E+03**	2.62E+03	2.60E+03	2.62E+03	3.01E+03	4.36E+03
	std	3.11E−12	7.04E+02	3.52E+00	8.88E−11	4.95E−01	3.78E+00	3.22E+00	2.34E+02	3.52E+02
f12	mean	**2.94E+03**	3.00E+03	2.98E+03	2.94E+03	3.09E+03	2.97E+03	2.95E+03	3.12E+03	3.44E+03
	std	3.66E+00	1.20E+01	1.05E+01	1.77E+00	1.67E+01	2.20E+01	8.34E−01	6.17E+01	1.33E+02

Table 17
Results for CEC2022 test functions (improved algorithms, Dim = 20).

Functions	Metric	TVESCSO	MSGWO	DETDO	ALSHADE	LSHADE_cnEpSin
f1	mean	**3.00E+02**	8.22E+03	4.19E+02	3.00E+02	3.73E+02
	std	0.00E+00	2.65E+03	9.70E+01	1.46E−06	6.12E+01
f2	mean	**4.41E+02**	9.11E+02	5.25E+02	4.56E+02	4.49E+02
	std	2.87E+01	1.74E+02	3.66E+01	9.78E+00	1.97E−02
f3	mean	**6.00E+02**	6.01E+02	6.00E+02	6.00E+02	6.00E+02
	std	0.00E+00	3.10E−01	2.23E−02	6.56E−14	3.53E−05
f4	mean	**8.78E+02**	9.73E+02	1.05E+03	9.95E+02	8.40E+02
	std	6.53E+00	3.56E+01	1.10E+02	3.20E+01	8.16E+00
f5	mean	**9.00E+02**	9.09E+02	9.02E+02	9.01E+02	9.00E+02
	std	1.59E−01	2.86E+00	3.99E−01	4.12E−01	2.39E−01
f6	mean	4.64E+04	4.21E+07	3.80E+04	**2.68E+04**	3.66E+04
	std	1.58E+04	5.94E+07	1.26E+04	1.14E+04	4.04E+03
f7	mean	**2.03E+03**	5.93E+03	2.18E+03	2.06E+03	2.04E+03
	std	2.82E+00	1.37E+03	3.64E+01	1.85E+01	3.61E+00
f8	mean	2.65E+03	1.06E+05	3.88E+03	**2.23E+03**	2.91E+03
	std	3.81E+02	1.44E+05	2.45E+02	1.34E+00	2.24E+02
f9	mean	**2.64E+03**	2.72E+03	2.73E+03	2.64E+03	2.64E+03
	std	1.09E−01	2.04E+01	8.96E+00	2.71E−08	2.55E+00
f10	mean	2.80E+03	4.80E+03	5.35E+03	**2.80E+03**	2.87E+03
	std	1.41E+00	1.44E+03	3.54E+02	3.71E+00	4.46E+01
f11	mean	**2.60E+03**	4.57E+03	2.63E+03	2.60E+03	2.60E+03
	std	3.11E−12	1.73E+03	1.74E+01	9.49E−07	3.42E−01
f12	mean	**2.94E+03**	3.04E+03	3.08E+03	2.96E+03	2.95E+03
	std	3.66E+00	1.41E+01	6.88E+01	1.46E+01	8.03E+00

Table 18
Wilcoxon-test results for comparing algorithms on CEC2022.

Dim = 10				Dim = 20			
TVESCSO VS	Positive sum	Negative sum	pvalue	TVESCSO VS	Positive sum	Negative sum	pvalue
SCSO	0	78	4.88E−04	SCSO	0	78	4.88E−04
APO	11	67	5.05E−02	APO	23	55	2.33E−01
SAO	0	78	5.06E−03	SAO	13	65	1.39E−01
RIME	0	78	4.88E−04	RIME	0	78	4.88E−04
INFO	0	78	5.06E−03	INFO	8	70	4.69E−02
BWO	0	78	4.88E−04	BWO	0	78	4.88E−04
AO	0	78	4.88E−04	AO	0	78	4.88E−04
CHoA	0	78	4.88E−04	CHoA	0	78	4.88E−04
MSGWO	0	78	4.88E−04	MSGWO	0	78	4.88E−04
DETDO	0	78	3.35E−03	DETDO	12	66	3.42E−02
ALSHADE	14	64	5.75E−01	ALSHADE	26	52	5.34E−01
LSHADE_cnEpSin	18	60	1.82E−01	LSHADE_cnEpSin	20	58	1.51E−01

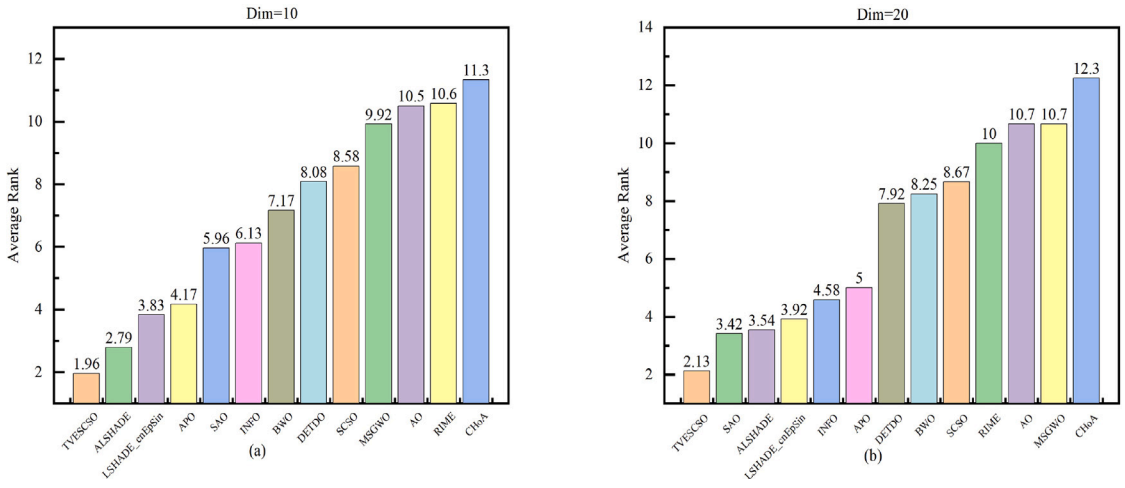


Fig. 10. Friedman mean ranking for different Algorithm on CEC2022.

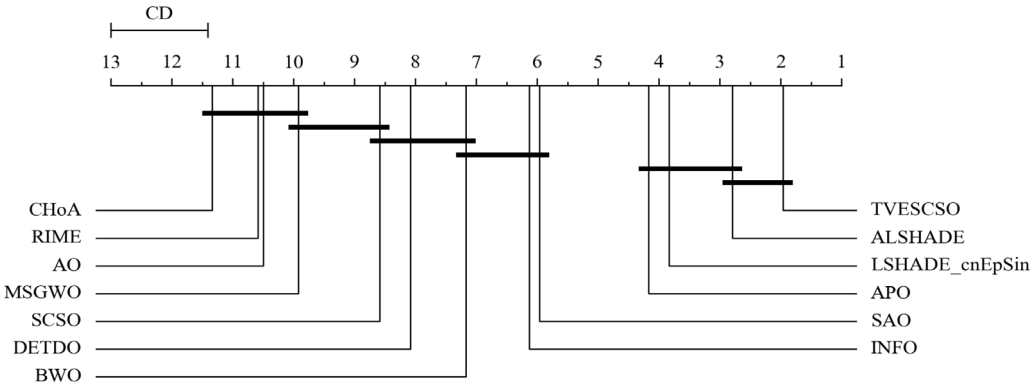


Fig. 11. Results of the Nemenyi test comparing TVESCSO with other algorithms on CEC2022(Dim = 10).

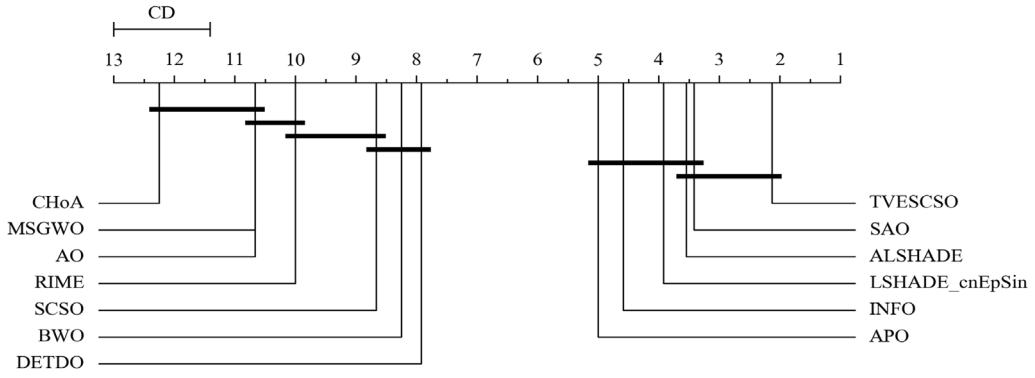


Fig. 12. Results of the Nemenyi test comparing TVESCSO with other algorithms on CEC2022(Dim = 20).

Based on these optimization problems, the TVESCSO algorithm is compared with Artificial Protozoa Optimizer (APO) (Wang et al., 2024), Snow Ablation Optimizer (SAO) (Deng & Liu, 2023), weIghted meaN oF vectOrs (INFO) (Ahmadianfar et al., 2022), Physical phenomenon of rime-ice (RIME) (Su et al., 2023), Honey Badger Algorithm (HBA) (Hashim et al., 2022) and Golden jackal optimization (GJO) (Chopra & Mohsin Ansari, 2022) are well-known optimizers for comparison. The population size of the optimizer during evaluation is set to 50, the maximum number of iterations per run is set to 1000, and the total number of optimizer runs is 30. The detailed parameter settings of all the algorithms are detailed in Table 3.

6.1. Three bar truss problem

The three bar truss problem in civil engineering is usually classified as a nonlinear fractional optimization problem. To minimize the volume of the three bar truss, multiple constraints must be imposed on each member, which mainly include buckling, deflection, and stress constraints. The core variables of the problem are x_1 and x_2, and its objective function and constraints can be succinctly formulated as follows:

(1) Design variables: $\vec{x} = [x_1, x_2]$

(2) Objective function: $f_{\min}(\vec{x}) = \left(2\sqrt{2}x_1 + x_2\right) \cdot l$

Table 19
Comparison of test results of the algorithms for solving the Three Truss Design Problem.

Algorithm	TVESCSO	HBA	APO	GJO	SAO	INFO	RIME
x1	0.788239	0.786406	0.788675	0.861024	0.790066	0.788272	0.812452
x2	0.409488	0.414724	0.408248	0.282204	0.404732	0.409449	0.381591
optimal value	263.8964	263.9015	263.8968	271.7549	263.9375	263.9018	267.9551
rank	1	3	2	7	5	4	6

Table 20
Comparison of test results of the algorithms for solving the Piston Lever Problem.

Algorithm	TVESCSO	HBA	APO	GJO	SAO	INFO	RIME
x1	0.05	200.0226	96.3135	0.051721	200.03	0.05	17.69814
x2	1.008114	200.6047	105.4362	1.01107	200.6048681	1.008114	102.3798
x3	2.016227	2.094205	2.441216	2.017889	2.094180177	2.016227	3.107591
x4	500	323.9992	292.025	499.9957	324	500	323.8925
optimal value	1.057394	67.62796	40.66174	1.065814	67.62352835	1.057394	147.4646
rank	1	5	3	2	4	1	6

(3) Constraints: $g_1(\vec{x}) = \frac{\sqrt{2}x_1+x_2}{\sqrt{2x_1^2+2x_1x_2}}P-\sigma \leqslant 0$, $g_2(\vec{x}) = \frac{x_2}{\sqrt{2x_1^2+2x_1x_2}}P-\sigma \leqslant 0$, $g_3(\vec{x}) = \frac{1}{\sqrt{2}x_2+x_1}P-\sigma \leqslant 0$
(4) Range of values: $0 \leqslant x_1, x_2 \leqslant 1$
(5) Constant variables: $l = 100$ cm, $P = 2$ km/cm^2, $\sigma = 2$ km/cm^2

In this study, we experimented with the TVESCSO algorithm on the three bar truss problem and compared and analyzed its results with those of various algorithms. Comparing the optimal solutions and decision variable values obtained by different algorithms in Table 19, we can see that the TVESCSO algorithm shows strong competitiveness. This finding highlights the efficiency and effectiveness of the TVESCSO algorithm in solving such engineering design problems.

6.2. Piston lever problem

The piston lever design problem is a typical mechanical engineering optimization problem with the primary objective of reducing the oil consumption during piston movement by optimizing the design. Specifically, the design objective is to minimize the amount of oil required as the piston rod is lifted from 0° to 45° to improve the system's energy efficiency and operational cost-effectiveness. The problem involves four key design components whose location and function are critical to achieving this optimization goal. These components include piston rod geometry (H), material selection (B), design of the lubrication system (D), and the dynamic connections that mate with the piston and cylinder (X). Configuring these components allows friction and energy losses to be effectively reduced, resulting in minimal oil consumption.
(1) Design variables: $\vec{x} = [x_1, x_2, x_3, x_4] = [H, B, D, X]$
(2) Objective function: $f_{\min}(\vec{x}) = 0.25\pi x_3^2(L_2 - L_1)$
(3) Constraints: $g_1(\vec{x}) = QL\cos\theta - R \times F \leqslant 0$, $g_2(\vec{x}) = Q(L-x_4) - M_{\max} \leqslant 0$, $g_3(\vec{x}) = 1.2(L_2 - L_1) \leqslant 0$,
$g_4(\vec{x}) = \frac{x_3}{2} - x_2 \leqslant 0$, $R = \frac{|-x_4(x_4\sin\theta+x_1)+x_1(x_2-x_4\cos\theta)|}{\sqrt{(x_4-x_2)^2+x_1^2}}$, $F = \frac{\pi P x_3^2}{4}$, $L_1 = \sqrt{(x_4-x_2)^2+x_1^2}$
$L_2 = \sqrt{(x_4\sin\theta+x_1)^2+(x_2-x_4\sin\theta)^2}$, $\theta = 45^0$, $Q = 10,000lbs$, $L = 240in$, $M_{\max} = 1.8\times10^6 ib\sin$, $P = 1500psi$
(4) Range of values: $0.05 \leqslant x_1, x_2, x_4 \leqslant 500$, $0.05 \leqslant x_3 \leqslant 120$
(5) Constant variables: $l = 100$ cm, $P = 2$ km/cm^2, $\sigma = 2$ km/cm^2

In this study, the TVESCSO algorithm was applied to optimize the piston lever design problem, and its experimental results were thoroughly compared with those of several other algorithms. Each algorithm and the specific values of the four key parameters are in Table 20. This data allows for a comparison of the performance of the different algorithms.

6.3. Welded beam problem

The welded beam problem is a practical engineering problem with complex constraints (Kumar et al., 2024), which aims to minimize the manufacturing cost of welded beams. The following constraints must be satisfied during the optimization of the welded beam problem: shear stress (τ), bending stress (σ), buckling load (Pc), deflection of the beam (δ), and side constraints. In addition, the fabrication cost of a welded beam is related to four key variables: width (t), weld thickness (h), length (l), and thickness (b). The mathematical model of the problem is formulated as follows:
(1) Design variables: $\vec{x} = [x_1, x_2, x_3, x_4] = [h, l, t, b]$
(2) Objective function: $f_{\min}(\vec{x}) = 1.1047x_1^2x_2 + 0.04811x_3x_4(14.0+x_2)$
(3) Constraints: $g_1(\vec{x}) = \tau(\vec{x}) - \tau_{\max} \leqslant 0$, $g_2(\vec{x}) = \sigma(\vec{x}) - \sigma_{\max} \leqslant 0$, $g_2(\vec{x}) = \sigma(\vec{x}) - \sigma_{\max} \leqslant 0$,
$g_3(\vec{x}) = \delta(\vec{x}) - \delta_{\max} \leqslant 0$, $g_4(\vec{x}) = x_1 - x_4 \leqslant 0$, $g_5(\vec{x}) = p - p_c(\vec{x}) \leqslant 0$, $g_6(\vec{x}) = 0.125 - x_1 \leqslant 0$,
$g_7(\vec{x}) = 0.10471x_1^2 + 0.04811x_3x_4(14.0+x_2) - 5.0 \leqslant 0$,
$\tau(\vec{x}) = \sqrt{(\tau')^2 + 2\tau'\tau''\frac{x_2}{2R} + (\tau'')^2}$, $\tau' = \frac{p}{\sqrt{2}x_1x_2}$, $\tau'' = \frac{MR}{J}$, $M = p\left(L+\frac{x_2}{2}\right)$
$R = \sqrt{\frac{x_2^2}{4} + \left(\frac{x_1+x_3}{2}\right)^2}$, $J = 2\left\{\sqrt{2}x_1x_2\left[\frac{x_2^2}{12} + \left(\frac{x_1+x_3}{2}\right)^2\right]\right\}$
$\sigma(\vec{x}) = \frac{6PL}{x_4x_3^2}$, $\delta(\vec{x}) = \frac{4PL^3}{Ex_3^3x_4}$, $P_c(\vec{x}) = \frac{4.013E\sqrt{\frac{x_3^2x_4^6}{36}}}{L^2}\left(1 - \frac{x_3}{2L}\sqrt{\frac{E}{4G}}\right)$
$p = 6000lbs$, $L = 14in$, $\delta_{\max} = 0.25in$, $E = 30\times10^6psi$, $G = 12\times10^6psi$, $\tau_{\max} = 13600psi$, $\sigma_{\max} = 30000psi$
(4) Range of values: $0.1 \leqslant x_1 \leqslant 2$, $0.1 \leqslant x_2 \leqslant 10$, $0.1 \leqslant x_2 \leqslant 10$, $0.1 \leqslant x_4 \leqslant 2$

In this study, the TVESCSO algorithm was used to optimize the welded beam problem, and its experimental results were compared with those of several other algorithms. Table 21 details the optimal solutions obtained by each algorithm and the values of the four key parameters. This data allows a clear comparison of the performance of the different algorithms.

6.4. Propane, Isobutane, n-Butane Nonsharp Separation

Propane, Isobutane, and n-butane Nonsharp Separation belong to the field of chemical engineering and process optimization (Kumar et al., 2020). It is concerned explicitly with separating a multi-component mixture to separate it into two products containing three components finely. This separation process not only requires technical precision but must also satisfy specific component ratios and quality requirements. Therefore, it is defined as a nonlinearly constrained optimization problem. The mathematical model of the problem is formulated in detail as follows:

Table 21
Comparison of test results of the algorithms for solving the Welded Beam Problem.

Algorithm	TVESCSO	HBA	APO	GJO	SAO	INFO	RIME
x1	0.205732	0.199587	0.183867	0.198035	0.221447	0.245067	0.239072
x2	3.16005	3.279042	3.791044	3.336507	3.01556	2.802093	3.782116
x3	9.036528	9.030113	8.909088	9.043963	8.766494	8.368919	6.898232
x4	0.205734	0.2061	0.211867	0.205834	0.221461	0.245068	0.398122
optimal value	1.682591	1.69132	1.75056	1.697149	1.73529	1.813931	2.478071
rank	1	2	5	3	4	6	7

Table 22
C1i values.

c	i=1	i=2
c1i	0.23947	0.75835
c2i	−0.0139904	−0.0661588
c3i	0.009351	0.033815
c4i	0.007731	0.037335
c5i	−0.0005719	0.001637
c6i	0.004266	0.0289

(1) Objective function:

$$f_{\min}(\vec{x}) = c_{11} + \left(c_{21} + c_{31}x_{24} + c_{41}x_{28} + c_{51}x_{33} + c_{61}x_{34}\right)x_5 + c_{12} + \left(c_{22} + c_{32}x_{26} + c_{42}x_{31} + c_{52}x_{38} + c_{62}x_{39}\right)x_{13}$$

where Table 22 describes C1i values.

(2) Constraints: $g_1(\vec{x}) = x_4 + x_3 + x_2 + x_1 = 300$, $g_2(\vec{x}) = x_6 - x_8 - x_7 = 0$, $g_3(\vec{x}) = x_9 - x_{12} - x_{10} - x_{11} = 0$

$g_4(\vec{x}) = x_{14} - x_{17} - x_{15} - x_{16} = 0$, $g_5(\vec{x}) = x_{18} - x_{20} - x_{19} = 0$, $g_6(\vec{x}) = x_6x_{21} - x_{24}x_{25} = 0$, $g_7(\vec{x}) = x_{14}x_{22} - x_{26}x_{27} = 0$,

$g_8(\vec{x}) = x_9x_{23} - x_{28}x_{29} = 0$, $g_9(\vec{x}) = x_{18}x_{30} - x_{31}x_{32} = 0$, $g_{10}(\vec{x}) = x_{25} - x_5x_{33} = 0$, $g_{11}(\vec{x}) = x_{29} - x_5x_{34} = 0$,

$g_{12}(\vec{x}) = x_{35} - x_5x_{36} = 0$, $g_{13}(\vec{x}) = x_{37} - x_{13}x_{38} = 0$, $g_{14}(\vec{x}) = x_{27} - x_{13}x_{39} = 0$, $g_{15}(\vec{x}) = x_{32} - x_{13}x_{40} = 0$,

$g_{16}(\vec{x}) = x_{25} - x_6x_{21} - x_9x_{41} = 0$, $g_{17}(\vec{x}) = x_{29} - x_6x_{42} - x_9x_{23} = 0$, $g_{18}(\vec{x}) = x_{35} - x_6x_{43} - x_9x_{44} = 0$,

$g_{19}(\vec{x}) = x_{37} - x_{14}x_{45} - x_{18}x_{46} = 0$, $g_{20}(\vec{x}) = x_{27} - x_{14}x_{22} - x_{18}x_{47} = 0$, $g_{21}(\vec{x}) = x_{32} - x_{14}x_{48} - x_{18}x_{30} = 0$,

$g_{22}(\vec{x}) = 0.333x_1 + x_{15}x_{45} - x_{25} = 0$, $g_{23}(\vec{x}) = 0.333x_1 + x_{15}x_{22} - x_{29} = 0$, $g_{24}(\vec{x}) = 0.333x_1 + x_{15}x_{48} - x_{35} = 0$

$g_{25}(\vec{x}) = 0.333x_2 + x_{10}x_{41} - x_{37} = 0$, $g_{26}(\vec{x}) = 0.333x_2 + x_{10}x_{23} - x_{27} = 0$, $g_{27}(\vec{x}) = 0.333x_2 + x_{10}x_{44} - x_{32} = 0$,

$g_{28}(\vec{x}) = 0.333x_3 + x_7x_{21} + x_{11}x_{41} + x_{16}x_{45} + x_{19}x_{46} = 30$, $g_{29}(\vec{x}) = 0.333x_3 + x_7x_{42} + x_{11}x_{23} + x_{16}x_{22} + x_{19}x_{47} = 50$,

$g_{30}(\vec{x}) = 0.333x_3 + x_7x_{43} + x_{11}x_{44} + x_{16}x_{48} + x_{19}x_{30} = 30$, $g_{31}(\vec{x}) = x_{33} + x_{34} + x_{36} = 1$, $g_{32}(\vec{x}) = x_{21} + x_{42} + x_{43} = 1$,

$g_{33}(\vec{x}) = x_{41} + x_{23} + x_{44} = 1$, $g_{34}(\vec{x}) = x_{38} + x_{39} + x_{40} = 1$, $g_{35}(\vec{x}) = x_{45} + x_{22} + x_{48} = 1$

$g_{36}(\vec{x}) = x_{46} + x_{47} + x_{30} = 1$, $g_{37}(\vec{x}) = x_{43} = 0$, $g_{38}(\vec{x}) = x_{46} = 0$

(3) Range of values: $0 \leqslant x_1, \ldots, x_{20} \leqslant 150$, $0 \leqslant x_{25}, x_{27}, x_{29}, x_{32}, x_{35}, x_{37} \leqslant 30$, $0 \leqslant x_{46}, x_{47}, x_{48} \leqslant 1$

$0 \leqslant x_{21}, x_{22}, x_{23}, x_{30}, x_{33}, x_{34}, x_{36}, x_{38}, x_{39}, x_{40}, x_{42}, \ldots, x_{45} \leqslant 1$

In this study, the problem of separating propane, isobutane, and n-butane has been optimized by applying the TVESCSO algorithm. Its experimental results have been analyzed exhaustively and compared with various other algorithms. The optimal solutions obtained by each algorithm are systematically listed in Table 23, along with the specific values of the forty-eight key parameters corresponding to these solutions.

7. Experiments and analysis of feature selection

The analysis of the experimental results in this study focuses on two key aspects: firstly, to assess the effectiveness of our proposed algorithm (TVESCSO) in maintaining the diversity of the population and in balancing between exploration and exploitation; and secondly, to investigate the effectiveness and efficiency of TVESCSO by comparing the classification accuracy, the feature subset size, and the computational CPU time with existing methods.

7.1. Description of the experimental data

To assess the effectiveness of our proposed feature selection algorithm (TVESCSO), we utilized datasets from several sources: two from the UCI repository (http://archive.ics.uci.edu/ml/datasets.php), two gene datasets (https://ckzixf.github.io/dataset.html), and eight high-dimensional datasets from the ASU (https://jundongl.github.io/scikit-feature/datasets.html). Table 24 provides a concise overview of these datasets. The sample sizes range from 60 to 568, with the number of features spanning from 14 to 11,225 and the number of class labels varying between 2 and 40. Specifically, datasets such as DLBCL, Brain_Tumor_1, lung_discrete, lung, lymphoma, leukemia, Carcinom, and nci9 are notable for their high-dimensional cancer microarray data.

7.2. Experimental settings

The data set displayed in Table 24 underwent a random split, assigning 70% for training and 30% for testing to enhance the experimental results' reliability and significance. Subsequently, the performance of every set of attributes was evaluated utilizing the KNN classifier (Pan et al., 2023). The experimental setup entailed standardizing the population to 50 for each test data set. The maximum number of iterations $t_{\max}$ was fixed at 100. M set to 30.

TVESCSO is compared with SCSO to validate the diversity, exploration, and exploitation capability. In addition, TVESCSO is also compared with four SCSO variants algorithms as well as four intelligent population algorithms to investigate the performance of TVESCSO in feature selection, including Sand cat arithmetic optimization algorithm (SC-AOA) (Chen & Zheng, 2023), Chaotic dynamic disturbance sand cat swarm optimization(CDD-SCSO) (Li, Xiao, Zhang et al., 2024), Enhancing Sand Cat Swarm Optimization with Dynamic Pinhole Imaging and Golden Sine Algorithm(DGS-SCSO) (Adegboye et al., 2024), Improved sand cat swarm optimization algorithm(ISCSO) (Li, Zhao, Wang, & Wen, 2024), Particle swarm optimization(PSO) (Poli et al., 2007), Moth-flame optimization algorithm(MFO) (Mirjalili, 2015), Bald Eagle Search Optimisation(BES) (Alsattar et al., 2020) and Artificial rabbits optimization(ARO) (Wang et al., 2022) and other algorithms. All parameters are set in Table 25, and the experimental framework is shown in Fig. 13.

Fig. 13 illustrates the experimental procedure, which includes the following steps: First, the TVESCSO algorithm is applied to the training data to generate a potential subset of features. Next, the original training and testing datasets are transformed into updated versions by removing unselected features. Subsequently, the classification model is trained with the refined training data. Finally, the trained classifier is used to evaluate the test data.

7.3. TVESCSO feature selection algorithm composition

The TVESCSO algorithm effectively tackles conventional SCSO algorithms' hurdles in balancing exploration and exploitation, managing time complexity, and enhancing convergence speed. It is imperative to

Table 23
Comparison of Propane, Isobutane, n-Butane Nonsharp Separation.

Algorithm	TVESCSO	HBA	APO	GJO	SAO	INFO	RIME
x1	2.65E−05	99.98546	64.31369	0	5.129943		13.97421
x2	8.75E−05	0.013278	52.03579	6.38E−05	8.622057	115.8025	2.425294
x3	149.9999	100	56.58537	150	137.2167	29.07443	98.80104
x4	149.9999	100	127.0295	150	149.0284	116.3514	72.91077
x5	3.3E−06	0	108.6701	6.58E−05	9.159062	53.40462	3.18E−14
x6	0	0	60.94036	7.94E−05	18.05089		3.84E−14
x7	0	0	15.38797	7.67E−06	10.49311		1.83E−14
x8	2.36E−05	0	73.34637	2.81E−16	7.559371		3.36E−14
x9	0	0	72.10788	0.001386	7.395244		3.87E−14
x10	0	0	9.335425	1.19E−06	1.575672	60.17868	4.58E−14
x11	0	0	16.10883	1.66E−09	3.063968		3.43E−14
x12	0	0	46.62761	5.3E−06	2.755223	113.3847	5.04E−14
x13	0	0	74.83991	7.75E−14	12.45494	58.66531	11.508
x14	0	0	81.23442	3.76E−06	11.56283	60.47068	9.87442
x15	0	0	10.87903	9.33E−05	2.266917		2.37E−14
x16	0	0	32.94343	0.001367	2.553565	7.799323	2.42E−14
x17	8.82E−08	0	37.49803	1.27E−10	6.741591	50.35295	9.76942
x18	0	0	84.29389	2.09E−06	9.426766	56.58389	5.06E−14
x19	0	0	31.55395	4.14E−05	6.250845	21.76716	4.26E−14
x20	0	0	52.56045	4.44E−05	3.171134	35.18655	2.4E−14
x21	0	0	0.239244	2.3E−10	0.054614		3.02E−16
x22	0.660207	0.333333	0.22366	4.26E−11	0.266414	0.70139	3.16E−16
x23	0.984874	0	0.26353	0.333333	0.234897	0.644193	1.19E−16
x24	0.927064	1	0.935823	0.9	0.851082	0.876378	0.85
x25	0	0	19.22004	7.62E−07	1.044225	11.60326	1.19E−14
x26	0.921454	1	0.88366	0.9	0.850184	0.960788	0.85
x27	0	0	18.71656	0	3.402999	14.21556	0.013137
x28	0.928471	1	0.911517	1	0.862076	0.901257	0.85
x29	2.31E−09	0	21.44488	0.000225	2.049831	14.37458	7.56E−15
x30	0.344572	0.333333	0.127805	0.333333	0.124349	0.152237	0.264664
x31	0.935194	1	0.900331	0.9	0.915952	0.950015	0.85
x32	2.72E−17	0	13.65514	0	0.96121		1.17E−14
x33	0.203292	0	0.290159	4.99E−05	0.134249	0.384201	1.14E−16
x34	0.458579	0	0.262808	0.333333	0.398571	0.427687	2.88E−16
x35	0	0	19.00352	2.26E−07	1.478571	22.50188	8.36E−15
x36	0.338136	0	0.153593	0.000162	0.254761	0.456046	7.99E−17
x37	0	0	19.67066	5.68E−05	2.307294	19.57896	8.57E−15
x38	0.999952	0	0.385438	0.333334	0.155162	0.503702	0.007633
x39	0	0	0.365276	0.333333	0.297192	0.297747	3.91E−16
x40	0	0	0.25178	0.333333	0.124218		3.08E−16
x41	9.1E−06	0	0.334189	0.333333	0.02087		2.68E−17
x42	0.666675	0	0.282336	0.666667	0.111608	0.368552	2.18E−16
x43	0	0	0.378381	5.68E−07	0.026294	0.440795	3.07E−16
x44	0.015136	0.333333	0.11778	4.44E−08	0.137862		0.015433
x45	0.333333	0	0.190757	2.06E−09	0.160395	0.222102	0.03169
x46	1.17E−05	0.333333	0.049356	1.62E−14	0.023342		2.74E−16
x47	0.655431	0	0.111406	0.333333	0.058104	0.750983	2.52E−16
x48	0.006436	0	0.053233	0.333333	0.007187	0.202331	0.027374
optimal value	4.03E+09	1.68E+10	1.31E+10	4.23E+09	4.37E+09	5.94E+10	1.57E+10
rank	1	6	4	2	3	7	5

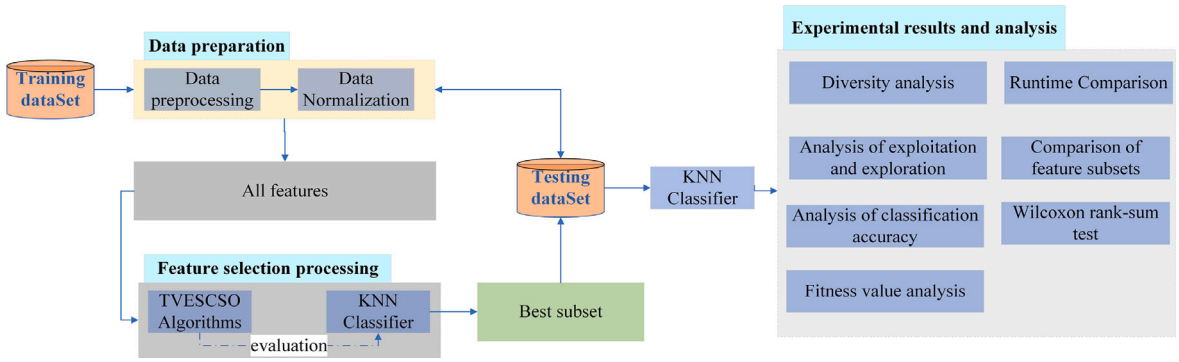


Fig. 13. Experimental framework diagram.

consider essential elements such as the binary portrayal of features and the creation of assessment functions when implementing the TVESCSO algorithm in feature selection.

7.3.1. Binary feature selection

This study focuses on binarizing TVESCSO to address the issue of feature selection. The sand cat is represented as a vector in D

Table 24
Test data set.

No.	Data set	Instances	Features	Classes	Data sources
1	Lymphography	147	19	8	UCI
2	Sonar	208	60	2	UCI
3	DLBCL	77	5469	2	Gene
4	Brain_Tumor_1	90	5920	5	Gene
5	lung_discrete	73	325	7	ASU
6	COIL20	1440	1024	20	ASU
7	ORL	400	1024	40	ASU
8	lung	203	3312	5	ASU
9	lymphoma	96	4026	9	ASU
10	leukemia	72	7070	2	ASU
11	Carcinom	174	9182	11	ASU
12	nci9	60	9712	9	ASU

Table 25
Parameter setting of the comparison algorithm.

Algorithm	Year of publication	Parameter values
SCSO	2023	$r_G \in [2,0], R \in [-2r_G, 2r_G]$
SC-AOA	2023	$r_G \in [2,0], R \in [-2r_G, 2r_G]$
ISCSO	2024	$r_G \in [2,0], R \in [-2r_G, 2r_G]$
DGS-SCSO	2024	$r_G \in [2,0], R \in [-2r_G, 2r_G]$
CDD-SCSO	2024	$r_G \in [2,0], R \in [-2r_G, 2r_G]$
PSO	2007	$w_1 = 0.9, w_2 = 0.4$
MFO	2015	$a = [-1, -2]$
BES	2020	$\alpha = 1.5, a = 10, R = 1, c_1 = c_2 = 2$
ARO	2022	$\theta \in [2,0]$

dimensions, with the number of dimensions matching the number of features. During the feature selection phase, each sand cat's position vector denotes a specific set of potential features. The values of their positions are constrained within the interval [0, 1]. It is directly depicted using a predetermined threshold. The formula is outlined as follows:

$$x_i^j = \begin{cases} 0, x_i^j < 0.5 \\ 1, x_i^j > 0.5 \end{cases} \tag{25}$$

In the ith sand cat, the jth feature is indicated by x_i^j. Eqs. (25) illustrates that if x_i^j exceeds 0.5, the feature is chosen; if it is below 0.5, it is disregarded.

7.3.2. Fitness evaluation function

In the present study, the effectiveness of the chosen feature subset is evaluated using a specific fitness evaluation function.

$$fit\left(X_i\right) = \alpha \times error + \beta \times \left(\frac{|K|}{|D|}\right) \tag{26}$$

In Eqs. (19), *error* represents the rate of classification errors. $|K|$ corresponds to the quantity of chosen feature sets, while $|D|$ represents the number of original feature sets utilized. The weight factor is denoted as $\alpha \in [0,1]$, with β being equivalent to $1-\alpha$. Assuming that α is assigned the value of 0.99, then $\beta = 0.01$ (Pan et al., 2023).

7.4. Balance between population diversity and exploration and exploitation

7.4.1. Analyses between TVESCSO and SCSO diversity

Diversity plays a crucial role in enhancing the effectiveness of swarm intelligence algorithms (Hussien, Abohany et al., 2024). It promotes an extensive exploration of the algorithm within the solution space, thereby efficiently avoiding premature convergence to suboptimal local solutions. This effectiveness can be calculated as follows.

$$Div_j = \frac{1}{N}\sum_{i=1}^{N} median\left(x^j\right) - x_i^j \tag{27}$$

$$Div = \frac{1}{D}\sum_{j=1}^{D} Div_j \tag{28}$$

Where $median\left(x^j\right)$ indicates the median value of the jth dimension. The Div_j represents the diversity measure for all individuals in the jth dimension. Div represents the overall diversity of the population across iterations. Figs. 14 and 15 show that the TVESCSO algorithm shows substantially better diversity properties when compared to the SCSO algorithm across all tested datasets.

7.4.2. Verification of the improvement in exploration and exploitation performance

The ability to explore and exploit constitutes fundamental to the success of swarm intelligence optimization algorithms (Hussien, Abohany et al., 2024). This study aims to quantitatively assess the difference in exploration and exploitation performance between the TVESCSO and SCSO algorithms. By analyzing the proportion of exploration and exploitation activities (Li et al., 2021) performed by the algorithms in each iteration, we aim to reveal the balance between these two activities. A detailed description of the computational methodology is below.

- Exploration Rate: Refers to the percentage of exploration per individual iteration within the algorithm, as follows.

$$Xpr\% = \frac{Div}{Div_{\max}} \times 100 \tag{29}$$

- Exploitation Rate: This represents the proportion of exploitation for each iteration within the algorithm, as follows.

$$Xpt\% = \frac{\left|Div - Div_{\max}\right|}{Div_{\max}} \times 100 \tag{30}$$

Div represents the diversity of the population for the specific iteration, while $Div_{\max}$ denotes the highest diversity observed across all iterations.

Based on the data in Table 26, the TVESCSO algorithm demonstrates a more even distribution between exploration and exploitation. For instance, when considering the Lymphography dataset, the exploration and exploitation ratios for TVESCSO are 66.46% and 33.54%, respectively. In contrast, the SCSO algorithm exhibits ratios of 77.03% and 22.97% for exploration and exploitation, respectively. A similar trend can be observed in the lung_discrete dataset, with TVESCSO showing percentages of 70.33% for exploration and 29.67% for exploitation, while SCSO presents percentages of 85.72% and 14.28%, respectively. The Carcinom dataset also follows this pattern, with TVESCSO at 73.41% for exploration and 26.59% for exploitation and SCSO at 89.01% for exploration and 10.99% for exploitation. This equilibrium is crucial for maintaining diversity throughout the search process, preventing a rapid decline in diversity and enhancing the algorithm's overall performance.

The TVESCSO can achieve this equilibrium by incorporating mechanisms for an elite layer that varies with time and a chaotic elite layer. These mechanisms significantly improve the diversity of the algorithm and provide it with enhanced abilities for exploration and exploitation, effectively preventing a sudden decrease in the diversity of the search population. Moreover, this strategy for optimization significantly boosts both the global search and local exploitation capabilities of the TVESCSO algorithm, allowing it to demonstrate exceptional performance in tackling intricate optimization challenges.

By examining Figs. 16 and 17, we can gain deeper insights into the exploration and exploitation abilities demonstrated by the TVESCSO algorithm across all test datasets. Initially, the TVESCSO algorithm swiftly expands its coverage of the solution space, prioritizing exploration in the early search phases. As the algorithm progresses, it shifts its focus towards identifying and optimizing specific solution regions that have the potential to enhance the overall solution quality. Analysis

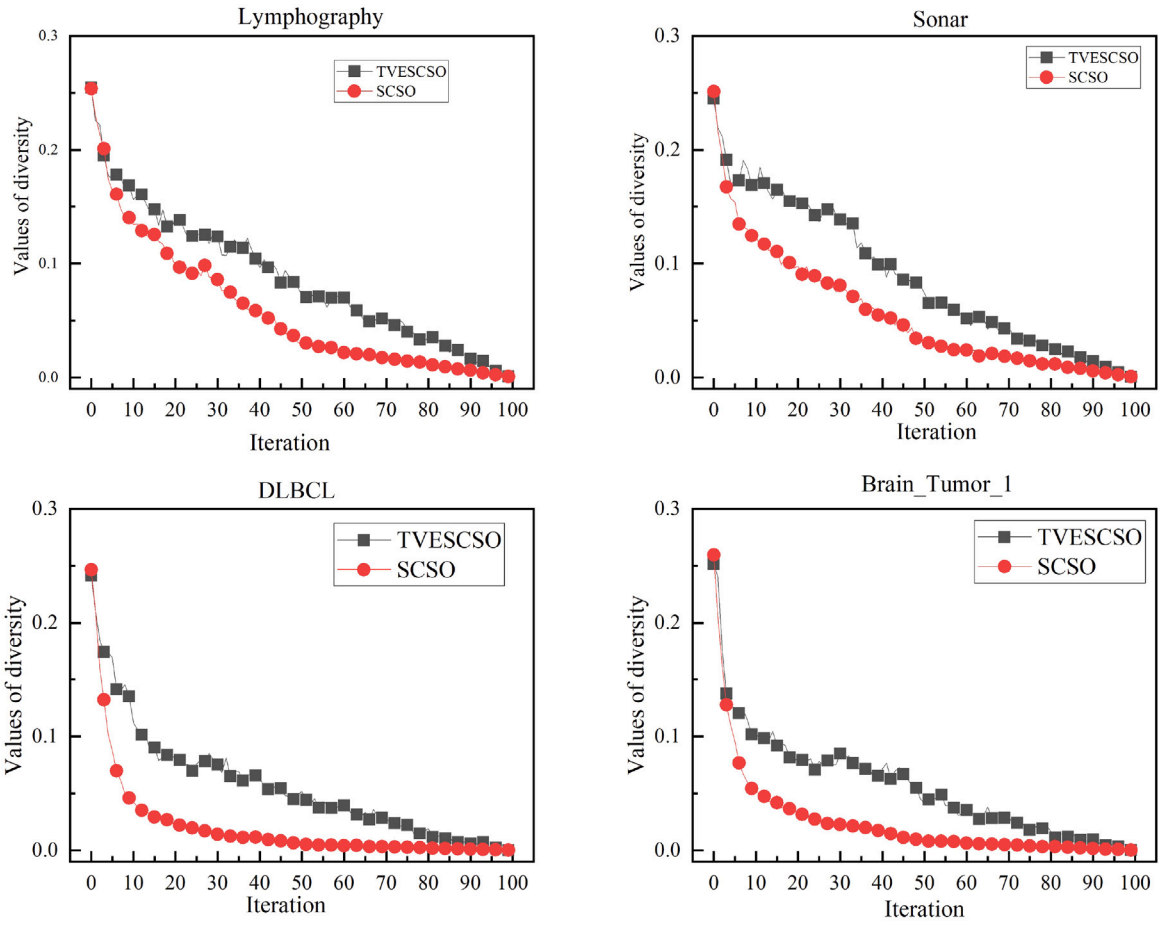


Fig. 14. Diversity of TVESCSO and SCSO on UCI and Gene dataset.

Table 26
Average Xpr%:Xpt% of TVESCSO and SCSO algorithms on 12 datasets.

Data set	TVESCSO-Expt: TVESCSO-Expr	SCSO-Expt: SCSO-Expr	Data set	TVESCSO-Expt: TVESCSO-Expr	SCSO-Expt: SCSO-Expr
Lymphography	66.46%:33.54%	77.03%:22.97%	ORL	56.46%:43.54%	79.16%:20.84%
Sonar	66.53%:33.47%	78.24%:21.76%	lung	81.33%:18.67%	90.48%:9.52%
DLBCL	76.85%:23.15%	91.85%:8.15%	lymphoma	82.10%:17.90%	93.24%:6.76%
Brain_Tumor_1	78.17%:21.83%	90.60%:9.40%	leukemia	90.66%:9.34%	94.38%:5.62%
lung_discrete	70.33%:29.67%	85.72%:14.28%	Carcinom	73.41%:26.59%	89.01%:10.99%
COIL20	66.45%:33.55%	85.81%:14.19%	nci9	80.39%:19.61%	91.42%:8.58%

of the DLBCL dataset displayed in Fig. 16 reveals that the TVESCSO algorithm rapidly extends its search efforts during the initial iterations, reaching peak exploration efficiency by the 60th iteration. This efficient exploration behavior is consistently maintained throughout the iterative process. Conversely, the exploitation efficiency of the algorithm exhibits a gradual decline at the onset of the iterations, stabilizing around the 60th iteration.

While examining the lung dataset as depicted in Fig. 17, the TVESCSO algorithm's superior exploration capability was promptly evident from the initial iterations, reaching its peak around the 60th iteration. Subsequently, this exploration prowess was sustained throughout the iteration process. Concurrently, the algorithm's exploitative ability gradually declines from the onset of the iterations until the 60th iteration, at which point its exploitative behavior stabilizes.

The study findings suggest that the TVESCSO outperforms the SCSO regarding exploration capacity. Additionally, regarding exploitation capacity, the TVESCSO demonstrates strong performance and superior outcomes compared to the SCSO algorithm.

7.5. Comparison of classification accuracy and selected feature subsets

7.5.1. Evaluation of quantitative indicators

The TVESCSO proposed in this paper is evaluated by six metrics: accuracy, sensitivity, specificity, selected feature subset, fitness value, and runtime, which are in Eqs. (31) to (33). In addition, TVESCSO is compared with some well-known algorithms, including SC-AOA, CDD-SCSO, DGS-SCSO, ISCSO, PSO, MFO, BES, and ARO algorithms.

- The formula for the specific method of calculating classification accuracy is as follows:

$$acc = \frac{TN + TP}{TN + TP + FN + FP} \tag{31}$$

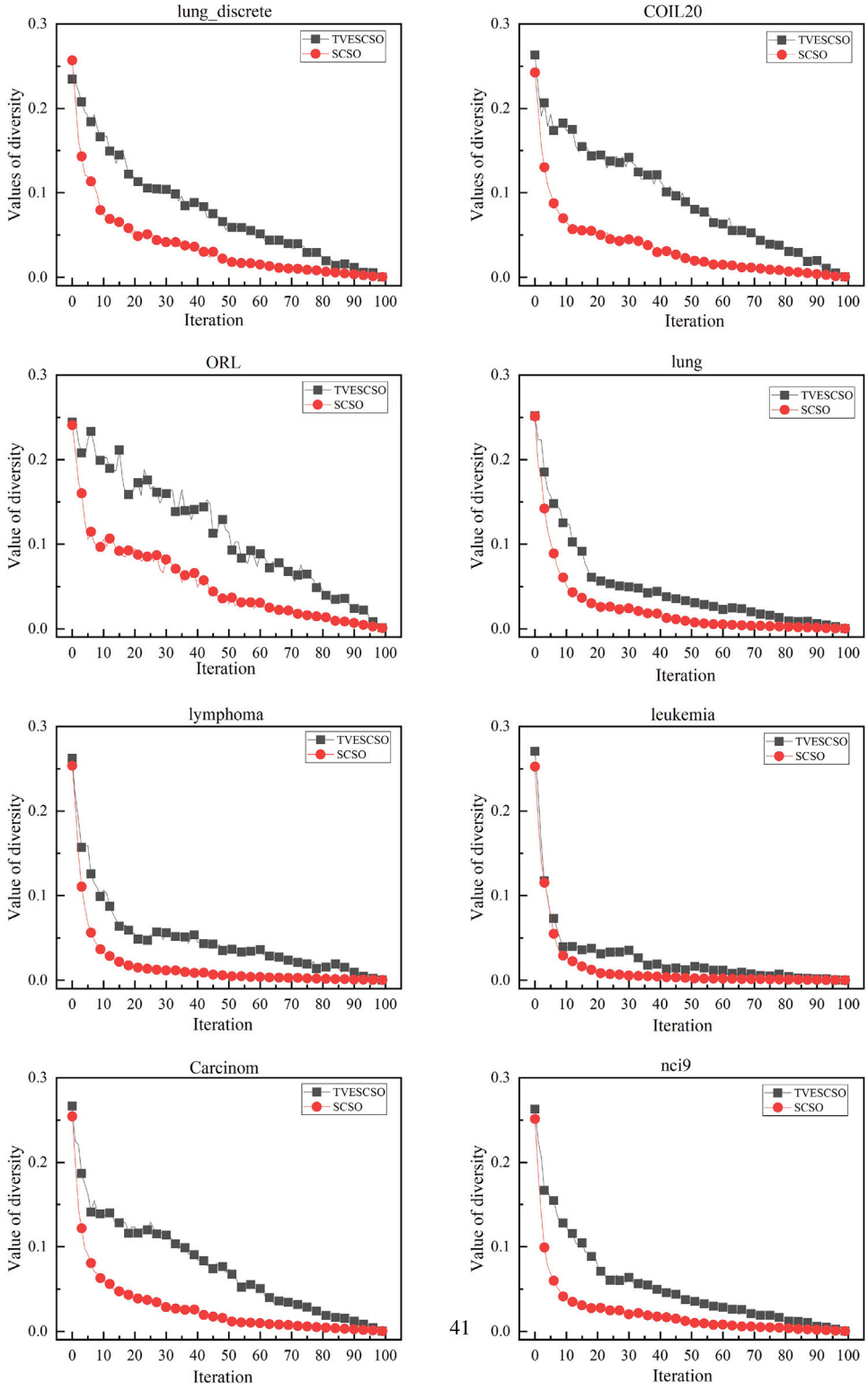


Fig. 15. Diversity of TVESCSO and SCSO on ASU dataset.

- The formula for the specific method of calculating the classification sensitivity is as follows:

$$Sensitivity = \frac{TP}{TP + FN} \tag{32}$$

- The formula for the specific calculation method of classification specificity is as follows:

$$Specificity = \frac{TN}{TN + FP} \tag{33}$$

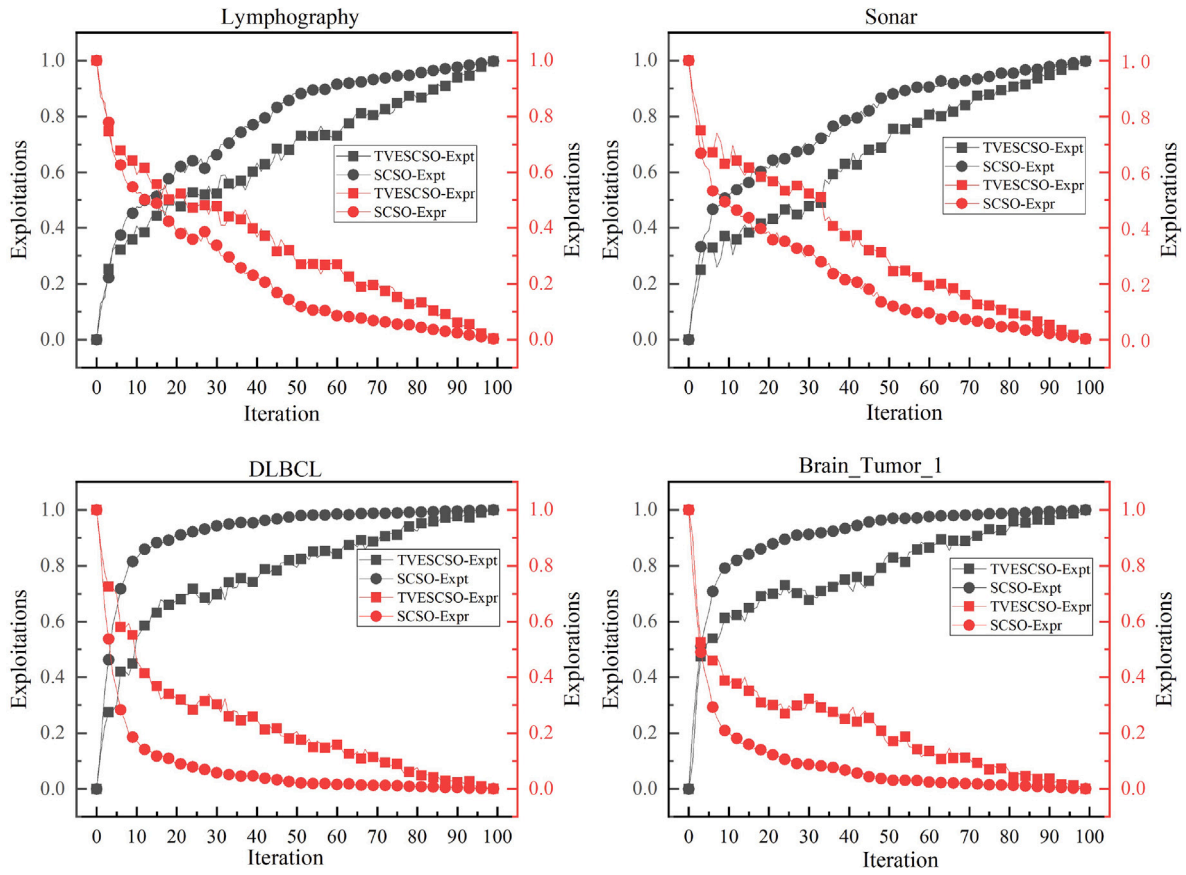


Fig. 16. Exploration and exploitation of TVESCSO and SCSO on UCI and Gene dataset.

Among these metrics are the counts of true positives (TP) and true negatives (TN), which indicate accurate classifications. Furthermore, misclassifications are measured by the counts of false positives (FP) and false negatives (FN).

7.5.2. Comparison of TVESCSO and SCSO variants

To validate the effectiveness of the TVESCSO, this section selects several enhanced SCSO algorithms for comparison based on fitness value, classification accuracy, sensitivity, specificity, and selected feature subset. Table 27 presents the average fitness value for each algorithm. Table 28 shows each algorithm's average and maximum classification accuracy. Table 29 outlines the average sensitivity values for the algorithms. Table 30 indicates the average specificity values for the algorithms. Table 31 reveals each algorithm's average number of selected feature subsets. These comparisons effectively demonstrate the superiority of the TVESCSO algorithm in tackling feature selection problems.

Based on Tables 27, 28, 29, 30, and 31, the TVESCSO algorithm exhibits noteworthy competitiveness across several crucial performance metrics. The TVESCSO algorithm shows a substantial lead in 11 datasets regarding fitness value. It holds its premier position consistently across all datasets except for the nci9 dataset, where it ranks second, showing a slight disadvantage. In terms of average classification accuracy, the TVESCSO algorithm not only performs superbly but also maintains a top ranking in 11 datasets. Despite placing third on the nci9 dataset, it demonstrates robust capability. Concerning sensitivity, the performance of the TVESCSO algorithm is suboptimal in the Lymphography and nci9 datasets. Regarding specificity, the algorithm does not achieve peak performance on the Lymphography dataset. The TVESCSO algorithm excels in the DLBCL and lung datasets for feature subset selection, though its performance is weaker on the Sonar, ORL, and Carcinom datasets.

Table 27
Comparison of SCSO variants fitness value.

Data set	TVESCSO	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	**4.61E−01**	5.15E−01	4.95E−01	5.25E−01	4.79E−01
Sonar	**7.26E−02**	1.45E−01	1.13E−01	1.51E−01	8.04E−02
DLBCL	**4.01E−05**	3.72E−02	4.32E−03	2.90E−02	8.58E−05
Brain_Tumor_1	**4.42E−02**	9.18E−02	5.52E−02	9.20E−02	6.26E−02
lung_discrete	**7.62E−04**	1.18E−01	3.64E−02	8.62E−02	7.27E−02
COIL20	**3.02E−03**	1.57E−02	6.24E−03	1.43E−02	6.79E−03
ORL	**9.41E−02**	1.50E−01	1.09E−01	1.33E−01	1.12E−01
lung	**1.39E−04**	1.69E−02	2.21E−04	1.18E−02	5.24E−04
lymphoma	**3.42E−02**	3.43E−02	3.42E−02	3.43E−02	3.42E−02
leukemia	**2.77E−05**	1.36E−02	3.71E−05	4.57E−03	7.64E−05
Carcinom	**3.23E−02**	1.16E−01	5.83E−02	8.83E−02	5.84E−02
nci9	2.26E−01	3.08E−01	**1.76E−01**	2.59E−01	2.09E−01

7.5.3. TVESCSO and some well-known swarm intelligence algorithms

To assess the impact of the TVESCSO method, we have chosen to examine the SCSO, BES, PSO, ARO, and MFO algorithms in this section, focusing on comparisons of fitness value, classification accuracy, and selected feature subsets. The average fitness values of TVESCSO-based algorithms in conjunction with SCSO, BES, PSO, ARO, and MFO algorithms are illustrated in Table 32. Additionally, Table 33 displays the average and maximum classification accuracy of each algorithm, while Table 34 presents the average number of selected feature subsets for each algorithm. These analyses offer compelling support for the superiority and efficacy of the TVESCSO algorithm when addressing feature selection challenges.

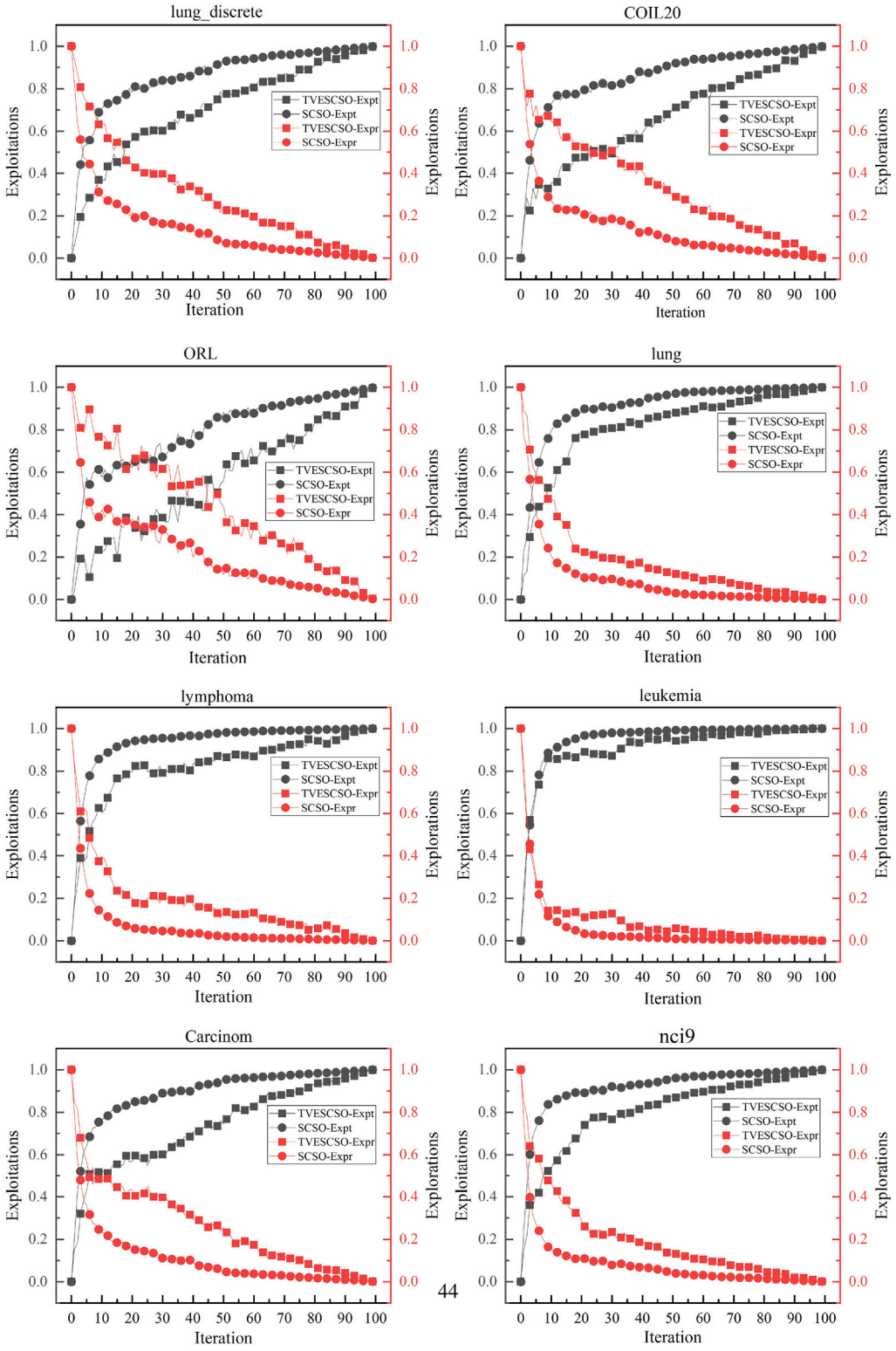


Fig. 17. Exploration and exploitation of TVESCSO and SCSO on ASU dataset.

Based on the information presented in Table 32, the TVESCSO algorithm demonstrates superior fitness values across all 12 and 10 datasets. TVESCSO falls slightly behind the ARO algorithm on the Sonar dataset. Moreover, on the nci9 dataset, TVESCSO is second behind the SCSO algorithm by a narrow difference.

Table 33 presents the mean classification accuracy rates for six population-based intelligence optimization algorithms: TVESCSO, SCSO, BES, PSO, ARO, and MFO. Their respective accuracies are 91.95%, 90.61%, 88.29%, 87.18%, 90.67%, and 87.26%. According to the data, the TVESCSO algorithm achieves the highest average classification accuracy, emphasizing its robustness and adaptability. This efficiency in solving intricate optimization problems indicates TVESCSO's superior performance.

Table 28
Comparison of SCSO variants classification accuracy.

Data set	TVESCSO		ISCSO		DGS-SCSO		CDD-SCSO		SC-AOA	
	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max
Lymphography	53.78%	60.00%	48.22%	51.11%	50.22%	51.11%	47.11%	51.11%	52.00%	55.56%
Sonar	92.86%	95.24%	85.56%	90.48%	88.73%	92.06%	84.92%	87.30%	92.06%	95.24%
DLBCL	100%	100%	96.25%	100%	99.58%	100%	97.08%	100%	100%	100%
Brain_Tumor_1	95.56%	100%	90.74%	92.59%	94.44%	96.30%	90.74%	96.30%	93.70%	96.30%
lung_discrete	100%	100%	88.18%	90.91%	96.36%	100%	91.36%	95.45%	92.73%	100%
COIL20	99.79%	100%	98.47%	99.07%	99.44%	100%	98.63%	99.07%	99.42%	100%
ORL	90.83%	91.67%	85%	86.67%	89.17%	90%	86.67%	88.33%	88.89%	90%
lung	100%	100%	98.36%	100%	100%	100%	98.85%	100%	100%	100%
lymphoma	96.55%	96.55%	96.55%	96.55%	96.55%	96.55%	96.55%	96.55%	96.55%	96.55%
leukemia	100%	100%	98.64%	100%	100%	100%	99.55%	100%	100%	100%
Carcinom	96.79%	100%	88.30%	92.45%	94.15%	98.11%	91.13%	96.23%	94.15%	98.11%
nci9	77.22%	83.33%	68.89%	72.22%	82.22%	88.89%	73.89%	83.33%	78.89%	83.33%

Table 29
Comparison of SCSO variants classification Sensitivity.

Data set	TVESCSO	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	50.74%	45.73%	51.83%	48.57%	54.85%
Sonar	96.47%	90.88%	93.82%	88.24%	95.59%
DLBCL	100.00%	97.78%	100.00%	98.33%	100.00%
Brain_Tumor_1	100.00%	100.00%	100.00%	100.00%	100.00%
lung_discrete	100.00%	100.00%	100.00%	100.00%	100.00%
COIL20	100.00%	100.00%	100.00%	100.00%	100.00%
ORL	100.00%	100.00%	100.00%	100.00%	100.00%
lung	100.00%	94.00%	100.00%	92.00%	100.00%
lymphoma	100.00%	100.00%	100.00%	100.00%	100.00%
leukemia	100.00%	96.25%	100.00%	100.00%	100.00%
Carcinom	100.00%	95.00%	100.00%	100.00%	100.00%
nci9	80.00%	80.00%	90.00%	73.33%	86.67%

Table 30
Comparison of SCSO variants classification Specificity.

Data set	TVESCSO	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	95.11%	95.00%	93.33%	100.00%	96.67%
Sonar	88.62%	79.31%	82.76%	81.03%	87.93%
DLBCL	100.00%	91.67%	98.33%	93.33%	100.00%
Brain_Tumor_1	100.00%	95.00%	93.33%	100.00%	96.67%
lung_discrete	100.00%	100.00%	100.00%	100.00%	100.00%
COIL20	100.00%	100.00%	100.00%	100.00%	100.00%
ORL	100.00%	100.00%	100.00%	100.00%	100.00%
lung	100.00%	100.00%	100.00%	100.00%	100.00%
lymphoma	100.00%	100.00%	100.00%	100.00%	100.00%
leukemia	100.00%	100.00%	100.00%	99.29%	100.00%
Carcinom	100.00%	100.00%	100.00%	100.00%	100.00%
nci9	100.00%	95.00%	100.00%	100.00%	100.00%

Table 31
Comparison of selected features based on TVESCSO and SCSO variants.

Data set	TVESCSO	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	5.3	4.3	4	2.7	6.8
Sonar	11.1	9.9	10.6	8	10.8
DLBCL	21.9	60.7	107.8	83.2	46.9
Brain_Tumor_1	130.2	87.1	142.3	188.3	130.7
lung_discrete	24.7	25.8	13.1	23.8	22.5
COIL20	98.4	59.8	75.9	74.7	109
ORL	339.7	107.3	203.3	141	243.3
lung	46	223.3	73.3	134.9	173.5
lymphoma	38.1	77.2	32.8	76.7	37.1
leukemia	19.6	45.2	26.2	47.4	54
Carcinom	465.7	496.3	391.6	428.8	450.6
nci9	218.5	176.3	256.9	210.5	243.6

Based on the data presented in Table 34, the TVESCSO algorithm excelled in feature subset selection, outperforming other methods in all six datasets: DLBCL, Brain_Tumor_1, lung, lymphoma, leukemia, and nci9. For datasets like Lymphography, Sonar, and COIL20, the TVESCSO algorithm secured the second position. It consistently achieved third place in the remaining datasets, demonstrating its broad applicability and stability across various datasets.

Fig. 18 compares the TVESCSO algorithm against other algorithms on 12 high-dimensional datasets. This includes the average Friedman ranking tests for classification accuracy and several of the best feature subsets. These figures visually evaluate the TVESCSO algorithm's performance, emphasizing its strengths in feature selection and classification tasks relative to the other algorithms.

Specifically, in Fig. 18(a), the TVESCSO algorithm excels and ranks first in classification accuracy. In Fig. 18(b), the algorithm ranks second in the ranking of the number of selected feature subsets. This result shows that the TVESCSO algorithm is competitive in improving classification accuracy and performs well in the efficiency of feature selection.

7.6. Convergence curve analysis

To further evaluate and compare the convergence performance of the TVESCSO algorithm, Figs. 19 and 20 display the fitness convergence curves of TVESCSO alongside other intelligent population feature selection algorithms, including ISCSO, DGS-SCSO, CDD-SCSO, SC-AOA, SCSO, BES, PSO, ARO, and MFO, across 12 datasets. This section aims to analyze and assess the strengths and weaknesses of the algorithms by examining their convergence curves and evaluating their convergence speed.

By examining Figs. 19 and 20, we see that the TVESCSO algorithm slightly underperforms on just two datasets, namely sonar and nci9, from a total of 12 test datasets. It is suggested that it can efficiently approximate the optimal solution in most cases, showing greater competitiveness than the other algorithms.

7.7. Running time

Tables 35 and 36 provide an exhaustive comparison of the actual running times of the ten algorithms TVESCSO, ISCSO, DGS-SCSO, CDD-SCSO, SC-AOA, SCSO, BES, PSO, ARO, and MFO on the 12 datasets. Based on these data, we can learn that the average running times (in seconds) of these algorithms are 354.53, 363.97, 369.56, 399.29, 356.81, 381.00, 13.41, 38.20, 32.21 and 55.55 s, respectively. These results reveal the differences in computational efficiency between the different algorithms. Further, Fig. 21 illustrates the ranking of the TVESCSO algorithm among all the compared algorithms, and it is ranked 6th in terms of running time. This ranking demonstrates that the execution time of the TVESCSO algorithm falls within an acceptable range.

7.8. Wilcoxon rank sum test

This section employs the Wilcoxon rank sum test to evaluate notable distinctions between the outcomes derived from this algorithm and

Table 32
Comparison of well-known swarm intelligence algorithms the fitness values.

Data set	TVESCSO	SCSO	BES	PSO	ARO	MFO
Lymphography	**4.61E−01**	4.95E−01	5.08E−01	4.95E−01	4.86E−01	4.90E−01
Sonar	7.26E−02	1.07E−01	9.52E−02	6.49E−02	**6.32E−02**	6.51E−02
DLBCL	**4.01E−05**	1.21E−04	1.31E−02	4.58E−02	4.40E−03	4.19E−02
Brain_Tumor_1	**4.42E−02**	4.79E−02	7.45E−02	8.53E−02	4.45E−02	8.20E−02
lung_discrete	**7.62E−04**	6.36E−02	1.04E−01	1.25E−01	6.36E−02	1.38E−01
COIL20	**3.02E−03**	5.35E−03	1.45E−02	1.55E−02	9.11E−03	1.70E−02
ORL	**9.41E−02**	1.14E−01	1.28E−01	1.09E−01	1.09E−01	1.07E−01
lung	**1.39E−04**	5.05E−03	2.61E−03	1.26E−02	3.94E−04	1.28E−02
lymphoma	**3.42E−02**	3.42E−02	4.14E−02	7.28E−02	3.43E−02	7.29E−02
leukemia	**2.77E−05**	3.56E−05	6.76E−02	1.53E−01	4.59E−03	1.44E−01
Carcinom	**3.23E−02**	5.84E−02	8.44E−02	1.17E−01	5.64E−02	1.17E−01
nci9	2.26E−01	1.93E−01	2.76E−01	2.80E−01	2.48E−01	2.80E−01

Table 33
Comparison of well-known swarm intelligence algorithms classification accuracy.

Data set	TVESCSO		SCSO		BES		PSO		ARO		MFO	
	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max	Avg	Max
Lymphography	53.78%	60.00%	50.22%	51.11%	49.11%	55.56%	50.44%	53.33%	51.33%	53.33%	50.89%	53.33%
Sonar	92.86%	95.24%	89.37%	92.06%	90.63%	93.65%	93.81%	96.83%	93.97%	95.24%	93.81%	95.24%
DLBCL	100%	100%	100%	100%	98.75%	100.00%	95.83%	95.83%	99.58%	100.00%	96.25%	100.00%
Brain_Tumor_1	95.56%	100%	95.19%	96.30%	92.59%	92.59%	91.85%	92.59%	95.56%	96.30%	92.22%	92.59%
lung_discrete	100%	100%	93.64%	95.45%	89.55%	90.91%	87.73%	90.91%	93.64%	95.45%	86.36%	86.36%
COIL20	99.79%	100%	99.54%	99.77%	98.70%	99.54%	98.89%	99.07%	99.24%	99.54%	98.77%	99.07%
ORL	90.83%	91.67%	88.61%	90%	87.50%	89.17%	89.42%	91.67%	89.33%	90.00%	89.67%	90.83%
lung	100%	100%	99.51%	100%	99.84%	100.00%	99.18%	100.00%	100.00%	100.00%	99.18%	100.00%
lymphoma	96.55%	96.55%	96.55%	96.55%	95.86%	96.55%	93.10%	93.10%	96.55%	96.55%	93.10%	93.10%
leukemia	100%	100%	100%	100%	93.18%	95.45%	85.00%	95.45%	99.55%	100.00%	85.91%	90.91%
Carcinom	96.79%	100%	94.15%	98.11%	91.51%	96.23%	88.68%	88.68%	94.34%	94.34%	88.68%	88.68%
nci9	77.22%	83.33%	80.56%	83.33%	72.22%	72.22%	72.22%	72.22%	75.00%	77.78%	72.22%	72.22%

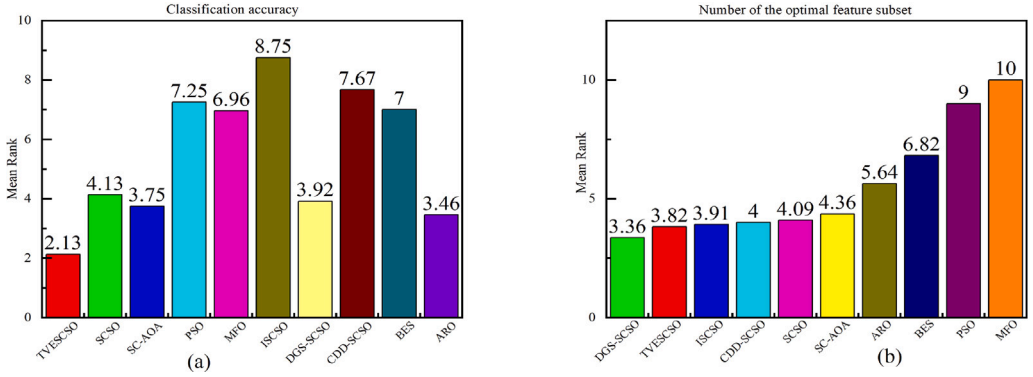


Fig. 18. Mean Friedman test ranks of ten algorithms on twelve high-dimensional datasets.

Table 34
Comparison of selected features based on TVESCSO and well-known swarm intelligence algorithms.

Data set	TVESCSO	SCSO	BES	PSO	ARO	MFO
Lymphography	5.3	4.2	7.5	7.4	7.0	7.3
Sonar	11.1	10.9	15.2	21.8	21.2	22.9
DLBCL	21.9	66.3	397.6	2505.6	147.8	2594.4
Brain_Tumor_1	130.2	133.2	718.7	2760.1	307.6	2983.1
lung_discrete	24.7	19.2	18.7	97.3	18.8	111.8
COIL20	98.4	78.6	167.8	455.9	157.9	499.2
ORL	339.7	168.0	418.1	479.4	336.5	518.4
lung	46.0	61.5	325.3	1490.8	130.3	1558.9
lymphoma	38.1	40.1	166.1	1800.9	70.3	1855.1
leukemia	19.6	25.2	83.8	3264.6	66.4	3449.1
Carcinom	465.7	481.5	340.9	4233.3	312.2	4403.1
nci9	218.5	268.9	887.1	4507.9	632.9	4694.0

Table 35
Running time(/s) of TVESCSO and SCSO variants.

Data set	TVESCSO	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	8.78	10.15	6.85	6.49	16.35
Sonar	14.69	18.03	12.68	11.9	24.43
DLBCL	499.55	515.58	571.08	499.42	517.4
Brain_Tumor_1	493.97	519.81	572.77	508.62	527.47
lung_discrete	32.07	36.13	36.32	32.33	38.5
COIL20	161.88	186.04	155.39	142.4	245.85
ORL	116.89	129.9	130.29	116.62	142.56
lung	290.46	311.42	336.91	299.59	319.99
lymphoma	354.75	377.62	415.32	373.12	388.38
leukemia	615	648.26	709.67	661.17	660.89
Carcinom	800.09	808.61	893.59	789.4	824.68
nci9	866.26	873.11	950.55	840.64	865.53

those generated by other algorithm executions. A p-value below 5% signifies a substantial divergence between the compared algorithms. Conversely, a p-value equal to or exceeding 5% signifies comparable findings in terms of optimality between the two algorithms under review. Table 37, the computed p-values from the rank sum test for TVESCSO versus ISCSO, DGS-SCSO, CDD-SCSO, and SC-AOA across the 12 test datasets are documented. The analysis presented in Table 37

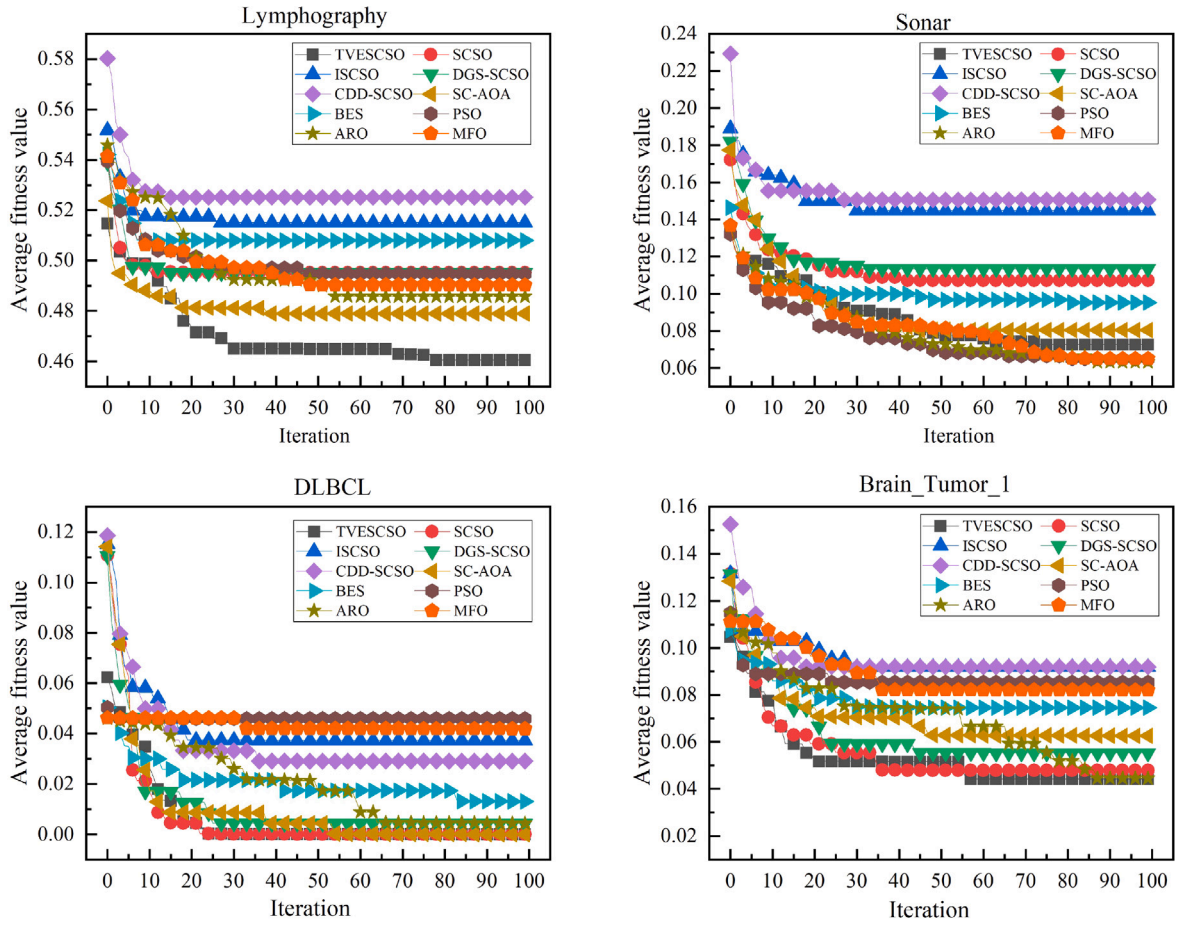


Fig. 19. Convergence curves of all algorithms on UCI and Gene datasets.

Table 36
Running time(/s) of TVESCSO and well-known swarm intelligence algorithms.

Data set	TVESCSO	SCSO	BES	PSO	ARO	MFO
Lymphography	8.78	6.71	4.4	5.01	9.43	9.66
Sonar	14.69	12.13	3.98	4.74	9.16	9
DLBCL	499.55	492.09	13.44	45.95	30.64	59.95
Brain_Tumor_1	493.97	531.79	13.74	48.59	33.18	63.37
lung_discrete	32.07	32.44	2.73	5.73	7.28	9.09
COIL20	161.88	151.99	33.36	52.22	87.68	102.69
ORL	116.89	117.05	10.02	16.95	23.01	28.86
lung	290.46	298.06	11.52	33.12	26.16	47.78
lymphoma	354.75	388.17	9.33	32.48	22.32	44.17
leukemia	615	634.39	14.2	53.54	33.69	74.84
Carcinom	800.09	818.36	23.38	84.83	57.39	120.43
nci9	866.26	884.41	20.79	75.19	46.59	96.74

Table 37
Wilcoxon rank-sum test of TVESCSO versus SCSO variants.

Data set	ISCSO	DGS-SCSO	CDD-SCSO	SC-AOA
Lymphography	3.01E−18	4.60E−18	3.01E−18	9.72E−16
Sonar	3.77E−18	3.77E−18	3.77E−18	7.29E−07
DLBCL	3.90E−18	1.20E−13	3.89E−18	7.69E−10
Brain_Tumor_1	3.90E−18	3.89E−18	3.90E−18	3.90E−18
lung_discrete	3.89E−18	3.89E−18	3.89E−18	3.87E−18
COIL20	3.90E−18	3.90E−18	3.90E−18	3.90E−18
ORL	3.88E−18	3.89E−18	3.89E−18	3.89E−18
lung	3.89E−18	2.33E−16	3.89E−18	3.90E−18
lymphoma	2.00E−01	3.89E−18	3.25E−01	6.04E−04
leukemia	8.31E−04	5.05E−12	7.62E−01	5.29E−01
Carcinom	3.90E−18	3.90E−18	3.90E−18	3.90E−18
nci9	2.14E−16	9.31E−18	1.06E−01	7.88E−16

indicates that the *p*-values were predominantly under the 5% threshold in most datasets analyzed. TVESCSO exhibits parallel optimality findings in the lymphoma dataset with ISCSO and CDD-SCSO algorithms. Similarly, in the nci9 dataset, the optimization outputs from TVESCSO and the CDD-SCSO algorithm coincide.

Table 38 presents the *p* values obtained from rank sum tests comparing TVESCSO with SCSO, BES, PSO, ARO, and MFO across 12 test datasets. The examination of Table 38 reveals that the *p* values are predominantly below 5% for most test datasets. Particularly noteworthy is the congruence in optimality findings between TVESCSO and SCSO algorithms for the DLBCL dataset.

Based on the information provided in Tables 37 and 38, a notable significant difference exists between TVESCSO and the other nine algorithms. This result suggests that TVESCSO demonstrates superior effectiveness compared to the different algorithms.

8. Conclusion

This study delves into the SCSO algorithm's inherent limitations, especially the inadequacy of its elite strategy in maintaining population diversity and balancing exploration and exploitation. To this end, our proposed TVESCSO algorithm introduces time-varying dynamics and chaotic elite strategies, which improve search efficiency and enhance the exploration of globally optimal solutions. Compared with algorithms such as APO, SAO, RIME, BWO, INFO, AO, CHOA, MSGWO, DETDO, ALSHADE, LSHADE_cnEpSin, HBA, GJO, SC-AOA, CDD-SCSO,

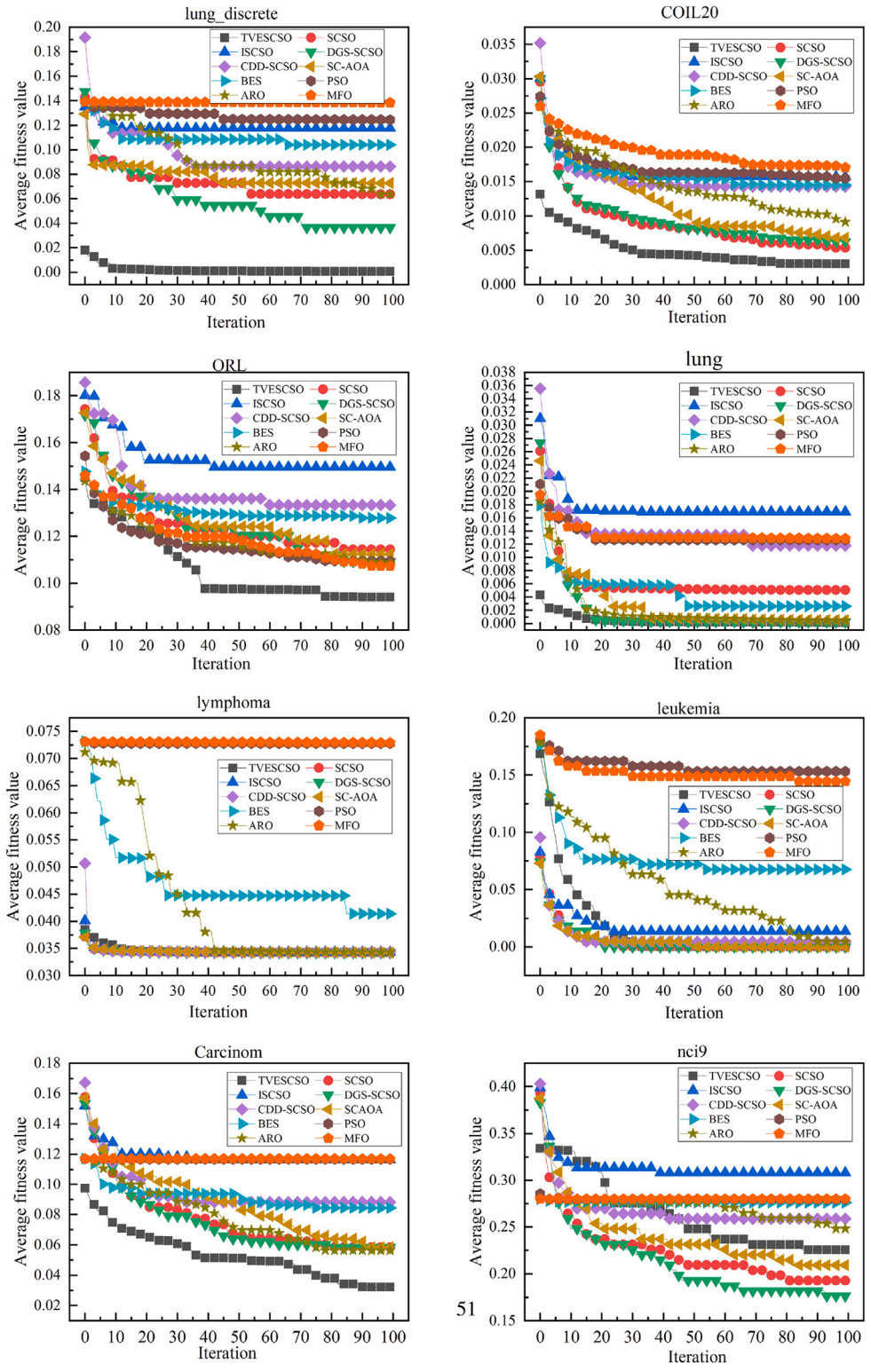


Fig. 20. Convergence curves of all algorithms on ASU datasets.

DGS-SCSO, ISCSO, PSO, MFO, BES, and ARO, the TVESCSO algorithm has obvious advantages in terms of overall optimization performance. Specifically, TVESCSO performs excellently on the standard test functions of CEC2017 and CEC2022 and proves its outstanding capability on four engineering design problems and 12 feature selection datasets. This demonstrates that TVESCSO's performance maintains performance stability in increasingly complex optimization challenges, proving that TVESCSO's ability to identify high-quality solutions robustly stems from its ability to balance exploration and exploitation while preserving population diversity. The summary results are as follows.

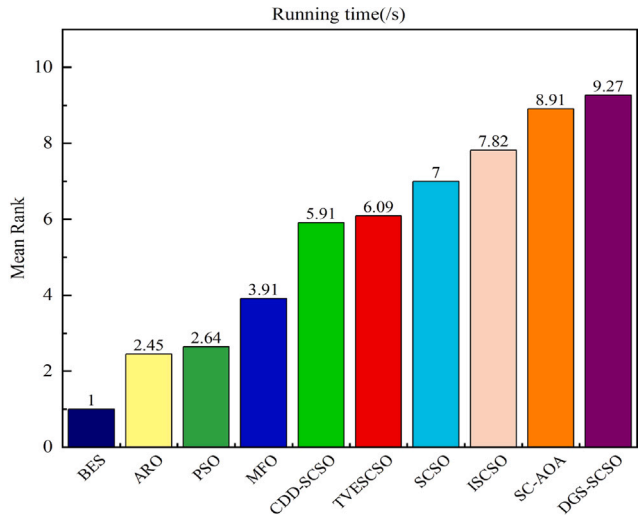


Fig. 21. Running time(/s) of ten algorithms on twelve high-dimensional datasets.

Table 38
Wilcoxon rank-sum test of TVESCSO versus well-known swarm intelligence algorithms.

Data set	SCSO	BES	PSO	ARO	MFO
Lymphography	7.02E−18	3.01E−18	2.97E−18	3.62E−18	3.56E−18
Sonar	3.77E−18	4.36E−14	3.98E−18	1.73E−17	3.02E−12
DLBCL	4.24E−01	4.88E−17	9.32E−18	1.01E−16	9.04E−18
Brain_Tumor_1	4.85E−09	4.96E−18	5.60E−18	3.90E−18	3.90E−18
lung_discrete	3.88E−18	3.89E−18	3.89E−18	3.89E−18	3.88E−18
COIL20	3.90E−18	3.90E−18	3.89E−18	3.90E−18	3.90E−18
ORL	3.89E−18	3.89E−18	4.65E−15	4.52E−18	3.90E−18
lung	3.89E−18	3.89E−18	3.89E−18	3.90E−18	3.89E−18
lymphoma	1.62E−17	3.89E−18	3.89E−18	3.89E−18	3.90E−18
leukemia	4.50E−13	3.89E−18	3.89E−18	3.90E−18	3.90E−18
Carcinom	3.90E−18	3.90E−18	3.90E−18	3.90E−18	3.90E−18
nci9	5.18E−17	1.52E−03	3.40E−05	2.90E−02	4.30E−05

- In the CEC2017 benchmark test function, TVESCSO exhibits stable performance in most test scenarios, especially when the ratio of elite individuals (rp) is set to 0.4 in the chaotic search strategy. The ablation experiments confirm that a reasonable combination of strategies can enhance the TVESCSO algorithm's performance. TVESCSO obtained 16 optimal solutions out of 29 tested functions, ranking the highest among all compared algorithms.
- In the CEC2022 benchmark functions, TVESCSO provides 8 and 7 optimal solutions out of 12 10-dimensional and 20-dimensional CEC2022 functions, respectively, and achieves the optimal algorithmic ranking among all compared algorithms.
- In the actual engineering optimization problem, the best function value obtained by TVESCSO is 263.8964 for the design problem of a three-bar truss, 1.057394 for the piston rod problem, 1.682591 for the design problem of welded beams, 1.682591 for the Propane Isobutane NButane Non-sharp Separation Problem, TVESCSO derives an optimal function value of 4.03E+09.
- In feature selection, TVESCSO consistently surpasses other methods in terms of classification accuracy while retaining a small scale of the selected feature subset, which implies the algorithm's balance in optimizing the trade-off between feature selection and classification accuracy. TVESCSO also exhibits remarkable computational efficiency.
- Statistical analyses, including Friedman, Nemenyi, and Wilcoxon rank sum tests, further validate TVESCSO's effectiveness and efficiency, highlighting its significant differences from other methods.

In conclusion, the TVESCSO algorithm effectively overcomes the limitations of the SCSO algorithm in terms of exploration-exploitation balance and population diversity. By introducing strategies such as dynamic sub-population partitioning, elite individual and elite pool steering, and Logistic Chaotic Elite Search, TVESCSO prevents the algorithm from falling into local optimal solutions. Experimental results on CEC2017 and CEC2022 benchmark test functions, real-world engineering problems, and feature selection datasets underline the superiority and robustness of TVESCSO when dealing with complex optimization problems.

Although significant results have been achieved in this study, several areas still need to be explored in further depth. Future work will develop novel, dynamic strategies for SCSO algorithms that optimize population diversity and exploration-exploitation trade-offs to improve algorithm performance. Developing variants of SCSO algorithms that can effectively deal with these complex problems is urgent. This will expand the range of SCSO applications.

Ethical approval

This study does not involve any ethical issues.

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Declaration of competing interest

No conflict of interest exists in the submission of this manuscript, and the manuscript is approved by all authors for publication. I would like to declare on behalf of my co-authors that the work described was original research that has not been published previously, and not under consideration for publication elsewhere, in whole or in part.

Data availability

The experimental cancer microarray dataset set selects the world-famous data set (https://ckzixf.github.io/dataset.html, http://archive.ics.uci.edu/ and https://jundongl.github.io/scikit-feature/datasets.html).

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